## spec_extractor

# Test-case library

Tool version 1.4.0 | 149 cases in 8 areas, 13 of them known limitations

**What this is.** Every situation the tool has to get right - a common layout, or a detail that went wrong once - is a **case** with a fixed ID (TAB-01, LST-07 ...). Each case is one small page, saved as a scanned document of its own, together with the exact rows the tool must produce from it. Use the IDs to talk about a situation: in a bug report, a review or a commit message ("LST-10 fails again").

**This catalog** shows each case on its own page: the situation, what the tool must do and why (and what went wrong before, where it did), the page itself - drawn again here as text, where the scan has it - and the rows expected, with their checks. A **known limitation** is a case the tool still gets wrong: it is kept, with what happens instead, so that it is tracked and so that a fix shows up at once.

**The scanned documents** (image PDFs, as the tool sees them) are drawn on your computer by `python tests/cases/build_cases.py`, into `tests/cases/documents/` - one file per case, named by its ID, e.g. `TAB-01_letters_labels_and_two_line_values_on_wide_tab_stops.pdf`. Open one to see exactly what the tool reads, or run the tool on it yourself.

**Running the cases.** `python tests/cases/run_cases.py` runs the tool on every document (offline; a few minutes) and checks each case: PASS, FAIL, KNOWN (a known limitation, still wrong) or KNOWN-PASSES (a known limitation that now passes - check it and remove the mark). It lists every case whose result changed since the last run, so the effect of an edit is seen at once. `--only TAB` runs one area, `--only TAB-07` one case. The full comparison is written to `tests/cases/_results/case_report.txt` and `case_results.xlsx`.

**How a case is checked.** The Requirement rows of the Extracted Items sheet must match the expected rows one for one: the same number of rows, each Doc Ref exactly as expected and in order, and each Content at least 90 % similar to the expected text - both as a whole and after the lead-in, so an item that lost its value cannot pass on the strength of its lead-in. Checks written after a row (a review note, the Section, the Page(s)) and the number of tables and captions are checked too.

**Adding a case.** Write the page as text in the right tests/cases/area_*.py file (the format is explained at the top of tests/cases/casekit.py), give it the next free ID of its area and the rows the tool must produce, then run `python tests/cases/build_cases.py` to draw its document and rewrite this catalog. Never reuse an ID for a different situation.

| Area | Cases | IDs | Known limitations |
|---|---|---|---|
| Lists on tab stops | 54 | TAB-01 to TAB-54 | TAB-38 |
| List markers and list structure | 25 | LST-01 to LST-25 | LST-20, LST-24 |
| Lead-ins | 9 | LEAD-01 to LEAD-09 | LEAD-05, LEAD-06 |
| Paragraph numbers and headings | 13 | NUM-01 to NUM-13 | NUM-03 |
| Paragraphs around lists | 9 | PARA-01 to PARA-09 | PARA-08, PARA-09 |
| Two-column pages | 13 | COL-01 to COL-13 | COL-13 |
| Tables and lists | 21 | TBL-01 to TBL-21 | TBL-15, TBL-16, TBL-17, TBL-21 |
| Page breaks | 5 | PAGE-01 to PAGE-05 | - |

# Index of the cases

| ID | Situation | Area | Known limitation |
|---|---|---|---|
| TAB-01 | Letters, labels and two-line values on wide tab stops | Lists on tab stops | |
| TAB-02 | Numbered list under a lead-in without a colon, then the next paragraph | Lists on tab stops | |
| TAB-03 | (a)-(d), labels without colons, very large tab, blank lines between items | Lists on tab stops | |
| TAB-04 | (1)-(4) with small tabs and no gap between items | Lists on tab stops | |
| TAB-05 | 1) to 4) markers, lead-in ending 'the following.' | Lists on tab stops | |
| TAB-06 | Roman numerals i. to iv. | Lists on tab stops | |
| TAB-07 | Roman numerals in brackets (i) to (iii) | Lists on tab stops | |
| TAB-08 | Capital letters A. to D., then a sub-paragraph 3.4.3.1 | Lists on tab stops | |
| TAB-09 | Bullets with upper-case labels | Lists on tab stops | |
| TAB-10 | No label column: markers and values of one to four lines | Lists on tab stops | |
| TAB-11 | A label that wraps onto a second line beside its value | Lists on tab stops | |
| TAB-12 | A nested (1)/(2) list under an item that has no value | Lists on tab stops | |
| TAB-13 | Decimal values that look like paragraph numbers, then an all-capitals heading 4. | Lists on tab stops | |
| TAB-14 | Proportional font, markers at the margin, three-line value | Lists on tab stops | |
| TAB-15 | Proportional font, numbered list, wrapped lead-in without a colon | Lists on tab stops | |
| TAB-16 | Proportional font, (a)-(c), a four-line value with numbers inside | Lists on tab stops | |
| TAB-17 | Proportional font, every label wraps onto two lines | Lists on tab stops | |
| TAB-18 | Proportional font, bullets, then the next paragraph | Lists on tab stops | |
| TAB-19 | TAB-01 on a photocopy | Lists on tab stops | |
| TAB-20 | TAB-02 on a photocopy | Lists on tab stops | |
| TAB-21 | A NOTE right under the list | Lists on tab stops | |
| TAB-22 | A heading right under the list | Lists on tab stops | |
| TAB-23 | The lead-in's last line sits beside the markers and labels | Lists on tab stops | |
| TAB-24 | One label of five words in a six-item list | Lists on tab stops | |
| TAB-25 | Values in lower case without a full stop | Lists on tab stops | |
| TAB-26 | Values that end with commas, one sentence over all items | Lists on tab stops | |
| TAB-27 | Values that end with commas, proportional font | Lists on tab stops | |
| TAB-28 | A value that starts with a quantity and a sentence ('5.0 V. Ripple shall ...') | Lists on tab stops | |
| TAB-29 | A sub-list (1), (2) inside the value column | Lists on tab stops | |
| TAB-30 | A sub-list printed without its own indent | Lists on tab stops | |
| TAB-31 | A label that wraps and is longer than its value | Lists on tab stops | |
| TAB-32 | A label that wraps and is longer than its value, proportional font | Lists on tab stops | |
| TAB-33 | A wrapped label whose second line has four words | Lists on tab stops | |
| TAB-34 | A wrapped label whose first line ends with an abbreviation | Lists on tab stops | |
| TAB-35 | Values that run on in lower case, each ending 'and' | Lists on tab stops | |
| TAB-36 | Long labels ('Mean time between failures:') | Lists on tab stops | |
| TAB-37 | A value that wraps back under the label column | Lists on tab stops | |
| TAB-38 | A value that wraps back under the label column, proportional font | Lists on tab stops | yes |
| TAB-39 | A sentence under the label column after a two-line value | Lists on tab stops | |
| TAB-40 | A 'Note:' line inside a tabbed item | Lists on tab stops | |
| TAB-41 | A 'Note:' line inside a tabbed item, proportional font | Lists on tab stops | |
| TAB-42 | Sub-labels without markers under 'a. Vibration: as follows:' | Lists on tab stops | |
| TAB-43 | A value that wraps back under a label without a colon, then a sub-list | Lists on tab stops | |
| TAB-44 | A value '1.5 V peak to peak. Measured at the' ... | Lists on tab stops | |
| TAB-45 | A value line '2.5 G rms overall. Duration 3 hours per axis.' | Lists on tab stops | |
| TAB-46 | A nested (1)/(2) list inside a value that ends with ':' | Lists on tab stops | |
| TAB-47 | Each label on its own line, its text on the lines below | Lists on tab stops | |
| TAB-48 | A whole page typed double-spaced | Lists on tab stops | |
| TAB-49 | A second paragraph of a value, after a blank line, in the value column | Lists on tab stops | |
| TAB-50 | A value that ends with a colon of its own | Lists on tab stops | |
| TAB-51 | Contents of a report: 'Title:', 'Procedure:', 'Remarks:', 'Signature:' | Lists on tab stops | |
| TAB-52 | An item whose value is empty | Lists on tab stops | |
| TAB-53 | Labels without colons | Lists on tab stops | |
| TAB-54 | A value line '12.5 V. The auxiliary output shall be' after a finished value line | Lists on tab stops | |
| LST-01 | An ordinary tight lettered list (control) | List markers and list structure | |

| ID | Situation | Area | Known limitation |
|---|---|---|---|
| LST-02 | An ordinary tight numbered list whose last item wraps (control) | List markers and list structure | |
| LST-03 | A lost marker: the item starts in the marker column | List markers and list structure | |
| LST-04 | A lost marker: the item starts in the text column | List markers and list structure | |
| LST-05 | A lost marker in the marker column, between long paragraphs | List markers and list structure | |
| LST-06 | A lost marker in the text column, between long paragraphs | List markers and list structure | |
| LST-07 | The first two markers lost in a tab list | List markers and list structure | |
| LST-08 | The first two markers lost in a tab list, proportional font | List markers and list structure | |
| LST-09 | A figure caption between two items, and the next item's marker lost | List markers and list structure | |
| LST-10 | A lost marker in a single-spaced list | List markers and list structure | |
| LST-11 | A lost marker after an item of three lines | List markers and list structure | |
| LST-12 | A marker misread as the next one ('c.', 'c.') | List markers and list structure | |
| LST-13 | A numbered item whose second sentence starts on its own line | List markers and list structure | |
| LST-14 | A numbered list that skips on purpose (1, 3, 5) | List markers and list structure | |
| LST-15 | Capital letters with a small-letter list under each | List markers and list structure | |
| LST-16 | Lettered items that wrap back to the left margin | List markers and list structure | |
| LST-17 | Capital-lettered items on tab stops whose value wraps back to the margin | List markers and list structure | |
| LST-18 | Sub-items without markers under 'a. ... which shall include:' | List markers and list structure | |
| LST-19 | A requirement sentence between items, after a sub-list | List markers and list structure | |
| LST-20 | A second lead-in inside the same paragraph | List markers and list structure | yes |
| LST-21 | Two requirement sentences between a. and b. | List markers and list structure | |
| LST-22 | A NOTE between two items of a tab list | List markers and list structure | |
| LST-23 | A capital 'C.' printed in a small-letter list | List markers and list structure | |
| LST-24 | The last item's marker lost in a single-spaced list | List markers and list structure | yes |
| LST-25 | Lettered items without full stops, then a paragraph | List markers and list structure | |
| LEAD-01 | No colon; items separated by blank lines; then a requirement at the same indent | Lead-ins | |
| LEAD-02 | No colon; four single-spaced items right under the lead-in | Lead-ins | |
| LEAD-03 | No colon; three single-spaced items after a blank line; then a requirement | Lead-ins | |
| LEAD-04 | '... listed below.' with plain one-line items | Lead-ins | |
| LEAD-05 | No colon, at the top of the page; three single-spaced items after a blank line | Lead-ins | yes |
| LEAD-06 | No colon, at the top of the page; items separated by blank lines | Lead-ins | yes |
| LEAD-07 | Lead-in words in the middle of a paragraph | Lead-ins | |
| LEAD-08 | A colon with nothing below it | Lead-ins | |
| LEAD-09 | Lead-in words, then the next numbered paragraph | Lead-ins | |
| NUM-01 | Numbered lists in sections 2 and 3, each followed by the next paragraph | Paragraph numbers and headings | |
| NUM-02 | A title-case heading with a full stop after a list at the margin | Paragraph numbers and headings | |
| NUM-03 | A heading with a full stop after a list of documents without full stops | Paragraph numbers and headings | yes |
| NUM-04 | Empty sections in a row ('5. Packaging.', '6. Notes.') | Paragraph numbers and headings | |
| NUM-05 | Wrapped lines that start with a cross-reference or a quantity | Paragraph numbers and headings | |
| NUM-06 | Paragraph numbers that skip after a list (3.2.4, then 3.2.8) | Paragraph numbers and headings | |
| NUM-07 | A numbered list straight before '4. Verification' and 4.1 | Paragraph numbers and headings | |
| NUM-08 | A wrapped line that starts with the next paragraph's number | Paragraph numbers and headings | |
| NUM-09 | Two numbered paragraphs with no blank line between them | Paragraph numbers and headings | |
| NUM-10 | A numbered list of documents at the margin | Paragraph numbers and headings | |
| NUM-11 | A title page with a document-control block | Paragraph numbers and headings | |
| NUM-12 | A table of contents with dot leaders | Paragraph numbers and headings | |
| NUM-13 | A paragraph number run into its title ('4.2Extended verification.') | Paragraph numbers and headings | |
| PARA-01 | Paragraphs with a first-line indent after a list at the margin | Paragraphs around lists | |
| PARA-02 | A numbered sub-paragraph indented under a list | Paragraphs around lists | |
| PARA-03 | A one-line indented paragraph after a list at the margin | Paragraphs around lists | |
| PARA-04 | A paragraph after a last item of three lines | Paragraphs around lists | |
| PARA-05 | A NOTE on tab stops after a list | Paragraphs around lists | |
| PARA-06 | 'Note.' level with the list's markers | Paragraphs around lists | |
| PARA-07 | Numbered items without full stops, then a paragraph | Paragraphs around lists | |
| PARA-08 | Paragraph numbers set apart from their text (hanging indent) | Paragraphs around lists | yes |
| PARA-09 | Lettered paragraphs 'A. General:' under a heading | Paragraphs around lists | yes |
| COL-01 | Two columns of prose with a tight list in the left column (control) | Two-column pages | |
| COL-02 | A list of documents in the left column, prose in the right | Two-column pages | |

| ID | Situation | Area | Known limitation |
|---|---|---|---|
| COL-03 | A list of documents in the left column, prose in the right, proportional font | Two-column pages | |
| COL-04 | A lead-in and short labelled items in the left column, prose in the right | Two-column pages | |
| COL-05 | A list set in two columns (a-c left, d-f right) | Two-column pages | |
| COL-06 | A list set in two columns, proportional font | Two-column pages | |
| COL-07 | A short list beside a narrow column of prose | Two-column pages | |
| COL-08 | A short list beside a ragged column of typed prose | Two-column pages | |
| COL-09 | Two lists side by side | Two-column pages | |
| COL-10 | A list beside prose that has a small list of its own | Two-column pages | |
| COL-11 | A short right column beside a list and prose | Two-column pages | |
| COL-12 | A short list beside prose with one numbered paragraph | Two-column pages | |
| COL-13 | A list on tab stops straight above a two-column section | Two-column pages | yes |
| TBL-01 | A borderless table with real columns (control) | Tables and lists | |
| TBL-02 | A requirement table with a header row and labels with colons | Tables and lists | |
| TBL-03 | A requirement table with a header row and labels without colons | Tables and lists | |
| TBL-04 | A requirement table without a header row, labels with colons | Tables and lists | |
| TBL-05 | A requirement table without a header row, labels without colons | Tables and lists | |
| TBL-06 | A three-column table with a header under a lead-in | Tables and lists | |
| TBL-07 | A three-column table without a header under a lead-in | Tables and lists | |
| TBL-08 | A three-column table with a header, labels without colons | Tables and lists | |
| TBL-09 | A three-column table without a header, labels without colons | Tables and lists | |
| TBL-10 | A table of documents under a lead-in | Tables and lists | |
| TBL-11 | A table of documents with a header row | Tables and lists | |
| TBL-12 | Labels with a requirement sentence each, under a lead-in | Tables and lists | |
| TBL-13 | A table whose first column holds long labels | Tables and lists | |
| TBL-14 | A table under a lead-in with its caption BELOW it | Tables and lists | |
| TBL-15 | A table whose labels are abbreviations with full stops ('Min.', 'Nom.', 'Max.') | Tables and lists | yes |
| TBL-16 | A table whose labels are abbreviations ('Temp.', 'Alt.', 'Vib.', 'Hum.') | Tables and lists | yes |
| TBL-17 | Requirement IDs with a colon on a tab stop ('SRS-101:') | Tables and lists | yes |
| TBL-18 | A list of labels, then the next table's caption and a ruled table | Tables and lists | |
| TBL-19 | A list of labels, then the next table's caption and a ruled table, proportional font | Tables and lists | |
| TBL-20 | A list of labels, then a sentence that starts 'Table 4 ...' | Tables and lists | |
| TBL-21 | A lead-in whose short last line sits right above a table's header | Tables and lists | yes |
| PAGE-01 | A tab-list value that carries on at the top of the next page | Page breaks | |
| PAGE-02 | A value carried onto the next page that starts with a decimal quantity | Page breaks | |
| PAGE-03 | A value carried onto the next page that starts with a cross-referenced paragraph | Page breaks | |
| PAGE-04 | A paragraph that carries on at the top of the next page | Page breaks | |
| PAGE-05 | A list that carries on at the top of the next page | Page breaks | |

# Lists on tab stops

Lists on wide tab stops: a marker, a large tab, a label ("Acceleration:"), another large tab, then a value that may run over several lines, its continuation lines aligned under the value column. TAB-01 to TAB-20 are the pages of the earlier 'list suite', kept exactly as they were drawn (their Doc Ref and Content convention is the tool's: '<paragraph number> <marker>' and '<lead-in> <marker> <label> <value>'). The later cases come from the adversarial reviews of the tab-list rules.

## TAB-01  Letters, labels and two-line values on wide tab stops

Lists on tab stops | document: documents/TAB-01_letters_labels_and_two_line_values_on_wide_tab_stops.pdf | typewriter 11 pt, 300 dpi

**Situation**

a. / b. / c. / d. at 0.4 in, a label ('Acceleration:') at 0.9 in and a value at 2.6 in - tabs of about 0.4 and 1 inch. Every value has two lines that start with numbers ('5 g', '20 Hz', '40 g', '-40 C', '2.5 C per minute'); the lead-in ends 'each of the following:'; half a line between items.

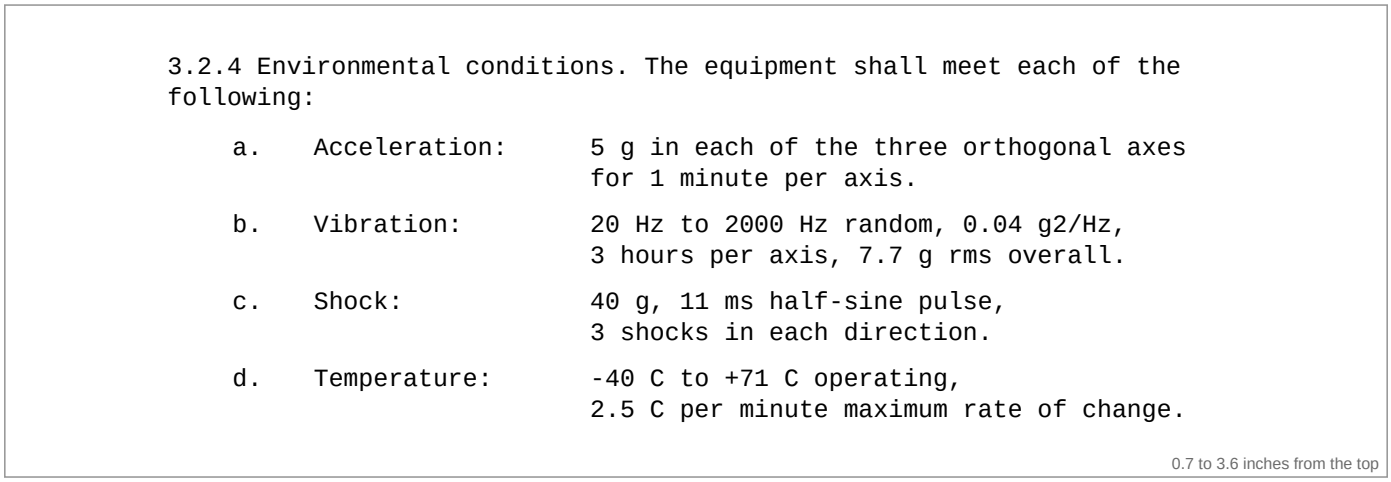
**What the tool must do**

Each item is ONE row with its marker, its label and its whole value, joined to the lead-in. The wide tabs are not the gutter between two text columns, the list is not a table, and a value line that starts with a number is not a new paragraph. The OCR tends to read the lone marker 'c.' as 'C.'; the Doc Ref must still be '<para> c', because the a, b, ?, d sequence says it is c.

**Source**

The user's report (the layout that started the tab-list work); list suite p01.

**The page (as the scan document has it)**



3.2.4 Environmental conditions. The equipment shall meet each of the following:

a. Acceleration: 5 g in each of the three orthogonal axes for 1 minute per axis.

b. Vibration: 20 Hz to 2000 Hz random, 0.04 g2/Hz, 3 hours per axis, 7.7 g rms overall.

c. Shock: 40 g, 11 ms half-sine pulse, 3 shocks in each direction.

d. Temperature: -40 C to +71 C operating, 2.5 C per minute maximum rate of change.

0.7 to 3.6 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The equipment shall meet each of the following: a. Acceleration: 5 g in each of the three orthogonal axes for 1 minute per axis. | |
| 3.2.4 b | The equipment shall meet each of the following: b. Vibration: 20 Hz to 2000 Hz random, 0.04 g2/Hz, 3 hours per axis, 7.7 g rms overall. | |
| 3.2.4 c | The equipment shall meet each of the following: c. Shock: 40 g, 11 ms half-sine pulse, 3 shocks in each direction. | |
| 3.2.4 d | The equipment shall meet each of the following: d. Temperature: -40 C to +71 C operating, 2.5 C per minute maximum rate of change. | |

# TAB-02  Numbered list under a lead-in without a colon, then the next paragraph

Lists on tab stops | document: documents/TAB-02_numbered_list_under_a_lead_in_without_a_colon_then_the_next_.pdf | typewriter 11 pt, 300 dpi

### Situation

1. to 4. on tab stops; the lead-in ends '... within the following limits.' (no colon); value lines start with '24 to', '1.5 V peak-to-peak', '3.2 A continuous', '150 W'; paragraph 3.2.6 follows.

### What the tool must do

A lead-in phrase ('the following limits.') leads into the list even without a colon; the values that start with '1.5' or '3.2' do not start new sections; 3.2.6 is a paragraph of its own.

### Source

List suite p01.

### The page (as the scan document has it)

```
3.2.5 Electrical conditions. The equipment shall operate within the
following limits.

    1.   Input voltage:     24 to 32 VDC, 28 VDC nominal.

    2.   Ripple:           1.5 V peak-to-peak maximum,
                           10 Hz to 50 kHz.

    3.   Current:          3.2 A continuous, 4.0 A peak.

    4.   Power:            150 W maximum at 28 VDC.

3.2.6 Weight. The equipment shall not exceed 45 kilograms.
```

0.7 to 3.3 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.5 1 | The equipment shall operate within the following limits. 1. Input voltage: 24 to 32 VDC, 28 VDC nominal. | |
| 3.2.5 2 | The equipment shall operate within the following limits. 2. Ripple: 1.5 V peak-to-peak maximum, 10 Hz to 50 kHz. | |
| 3.2.5 3 | The equipment shall operate within the following limits. 3. Current: 3.2 A continuous, 4.0 A peak. | |
| 3.2.5 4 | The equipment shall operate within the following limits. 4. Power: 150 W maximum at 28 VDC. | |
| 3.2.6 | The equipment shall not exceed 45 kilograms. | |

# TAB-03 (a)-(d), labels without colons, very large tab, blank lines between items

Lists on tab stops | document: documents/TAB-03_a_d_labels_without_colons_very_large_tab_blank_lines_between.pdf | typewriter 11 pt, 300 dpi

### Situation

(a)-(d) at 0.4 in, labels WITHOUT ':' at 1.0 in, values at 3.2 in (a tab of 1.4-2 inches); lead-in 'as follows:'; a full blank line between items; values start with '9.3 GHz', '25 kW', '0.5 us', '0.1 percent'.

### What the tool must do

Each item one row; the blank lines between items do not end the list.

### Source

List suite p02.

### The page (as the scan document has it)

```
3.3.1 Transmitter. The transmitter output shall be as follows:

    (a)    Frequency              9.3 GHz to 9.5 GHz, tunable
                                  in 10 MHz steps.

    (b)    Peak power             25 kW minimum.

    (c)    Pulse width            0.5 us, 1.0 us or 2.0 us,
                                  selectable by the operator.

    (d)    Duty cycle             0.1 percent maximum.
```

0.7 to 3.3 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.3.1 a | The transmitter output shall be as follows: (a) Frequency 9.3 GHz to 9.5 GHz, tunable in 10 MHz steps. | |
| 3.3.1 b | The transmitter output shall be as follows: (b) Peak power 25 kW minimum. | |
| 3.3.1 c | The transmitter output shall be as follows: (c) Pulse width 0.5 us, 1.0 us or 2.0 us, selectable by the operator. | |
| 3.3.1 d | The transmitter output shall be as follows: (d) Duty cycle 0.1 percent maximum. | |

# TAB-04  (1)-(4) with small tabs and no gap between items

Lists on tab stops | document: documents/TAB-04_1_4_with_small_tabs_and_no_gap_between_items.pdf | typewriter 11 pt, 300 dpi

### Situation

(1)-(4) at 0.4 in, labels at 0.8 in (a small tab after the marker), values at 2.3 in (only 2-4 characters after the longest label); lead-in 'as specified below.' (no colon); no gap between items.

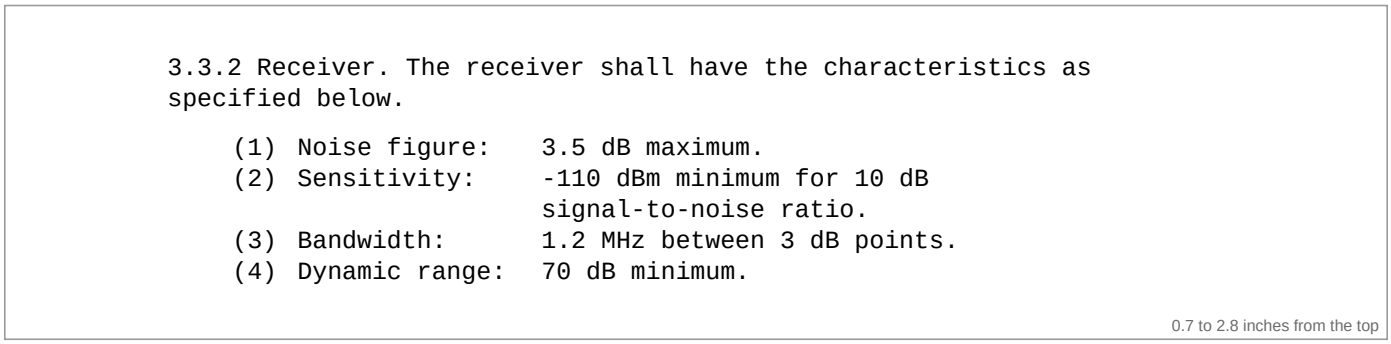
### What the tool must do

Each item one row even though the tabs are small and the items are single-spaced.

### Source

List suite p02.

### The page (as the scan document has it)



3.3.2 Receiver. The receiver shall have the characteristics as specified below.

(1) Noise figure: 3.5 dB maximum.
(2) Sensitivity: -110 dBm minimum for 10 dB signal-to-noise ratio.
(3) Bandwidth: 1.2 MHz between 3 dB points.
(4) Dynamic range: 70 dB minimum.

0.7 to 2.8 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.3.2 1 | The receiver shall have the characteristics as specified below. (1) Noise figure: 3.5 dB maximum. | |
| 3.3.2 2 | The receiver shall have the characteristics as specified below. (2) Sensitivity: -110 dBm minimum for 10 dB signal-to-noise ratio. | |
| 3.3.2 3 | The receiver shall have the characteristics as specified below. (3) Bandwidth: 1.2 MHz between 3 dB points. | |
| 3.3.2 4 | The receiver shall have the characteristics as specified below. (4) Dynamic range: 70 dB minimum. | |

# TAB-05  1) to 4) markers, lead-in ending 'the following.'

Lists on tab stops | document: documents/TAB-05_1_to_4_markers_lead_in_ending_the_following.pdf | typewriter 11 pt, 300 dpi

### Situation

1)-4) at 0.4 in, labels at 0.9 in, values at 2.6 in; lead-in ends 'the following.' (no colon); value lines start with '32 dBi', '2.5 degrees', '-25 dB', '1.5:1'; paragraph 3.3.4 follows.

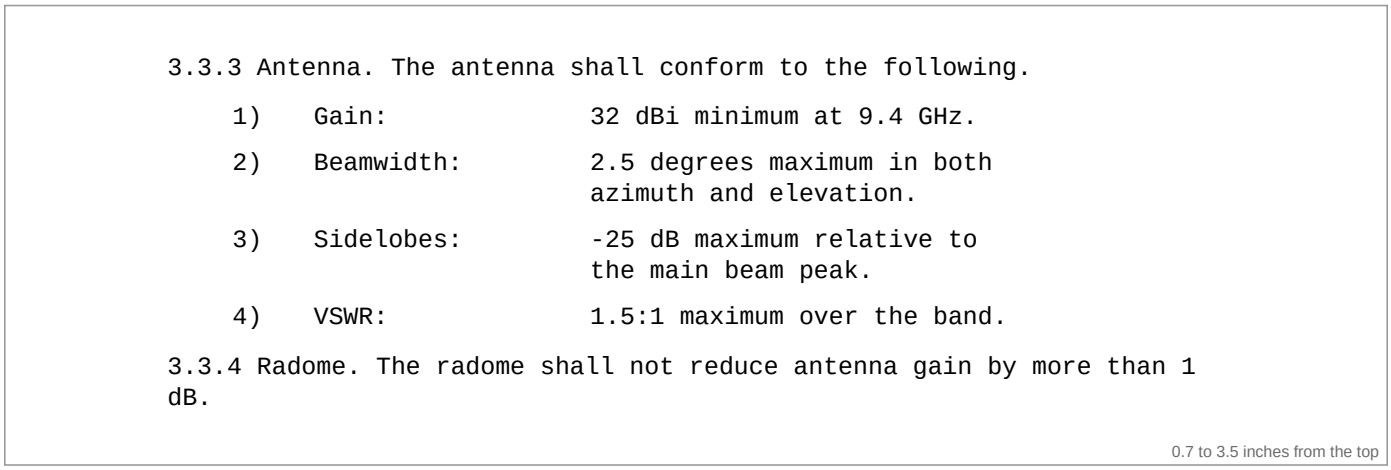
### What the tool must do

Each item one row; 3.3.4 is a paragraph of its own.

### Source

List suite p02.

### The page (as the scan document has it)



```
3.3.3 Antenna. The antenna shall conform to the following.

    1)   Gain:              32 dBi minimum at 9.4 GHz.

    2)   Beamwidth:         2.5 degrees maximum in both
                            azimuth and elevation.

    3)   Sidelobes:         -25 dB maximum relative to
                            the main beam peak.

    4)   VSWR:              1.5:1 maximum over the band.

3.3.4 Radome. The radome shall not reduce antenna gain by more than 1
dB.
```

0.7 to 3.5 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.3.3 1 | The antenna shall conform to the following. 1) Gain: 32 dBi minimum at 9.4 GHz. | |
| 3.3.3 2 | The antenna shall conform to the following. 2) Beamwidth: 2.5 degrees maximum in both azimuth and elevation. | |
| 3.3.3 3 | The antenna shall conform to the following. 3) Sidelobes: -25 dB maximum relative to the main beam peak. | |
| 3.3.3 4 | The antenna shall conform to the following. 4) VSWR: 1.5:1 maximum over the band. | |
| 3.3.4 | The radome shall not reduce antenna gain by more than 1 dB. | |

# TAB-06  Roman numerals i. to iv.

Lists on tab stops | document: documents/TAB-06_roman_numerals_i_to_iv.pdf | typewriter 11 pt, 300 dpi

### Situation

i./ii./iii./iv. at 0.4 in, labels at 1.0 in, values at 2.8 in; lead-in 'the following.' (no colon); two-line values.

### What the tool must do

Roman markers count i, ii, iii, iv - 'i.' is not the ninth letter here.

### Source

List suite p03.

### The page (as the scan document has it)

```
3.4.1 Built-in test. The BIT function shall perform the following.

    i.    Power-up:          Full test at power-on, completed
                             within 30 seconds.

    ii.   Periodic:          Non-interfering test every
                             60 seconds in operation.

    iii.  Initiated:         Full test started by the operator.

    iv.   Isolation:         To one LRU with 95 percent
                             probability of correct isolation.
```

0.7 to 3.2 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.4.1 i | The BIT function shall perform the following. i. Power-up: Full test at power-on, completed within 30 seconds. | |
| 3.4.1 ii | The BIT function shall perform the following. ii. Periodic: Non-interfering test every 60 seconds in operation. | |
| 3.4.1 iii | The BIT function shall perform the following. iii. Initiated: Full test started by the operator. | |
| 3.4.1 iv | The BIT function shall perform the following. iv. Isolation: To one LRU with 95 percent probability of correct isolation. | |

# TAB-07  Roman numerals in brackets (i) to (iii)

Lists on tab stops | document: documents/TAB-07_roman_numerals_in_brackets_i_to_iii.pdf | typewriter 11 pt, 300 dpi

**Situation**

(i)/(ii)/(iii) at 0.4 in, labels at 1.0 in, values at 2.6 in; the lead-in ends ':'.

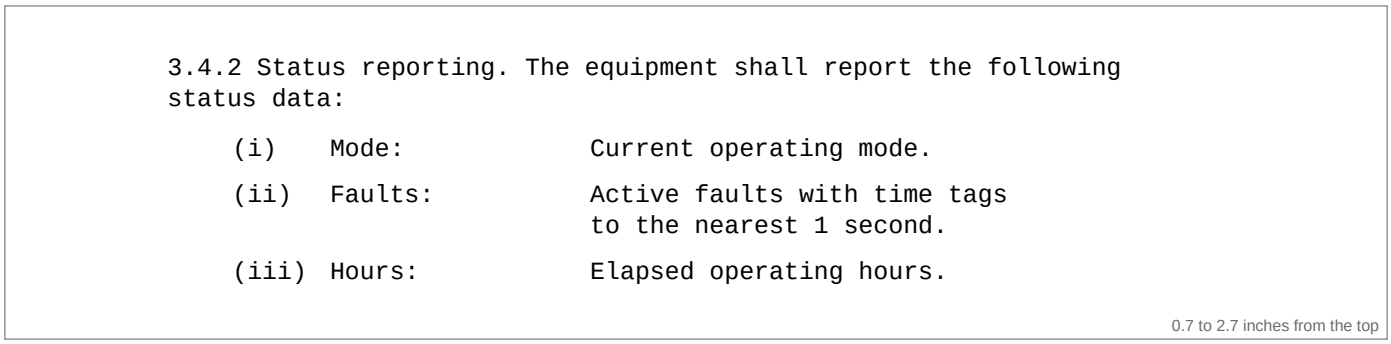
**What the tool must do**

Each item one row, Doc Refs '3.4.2 i', '3.4.2 ii', '3.4.2 iii'.

**Source**

List suite p03.

**The page (as the scan document has it)**



```
3.4.2 Status reporting. The equipment shall report the following
status data:

    (i)   Mode:           Current operating mode.

    (ii)  Faults:         Active faults with time tags
                          to the nearest 1 second.

    (iii) Hours:          Elapsed operating hours.
```

0.7 to 2.7 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.4.2 i | The equipment shall report the following status data: (i) Mode: Current operating mode. | |
| 3.4.2 ii | The equipment shall report the following status data: (ii) Faults: Active faults with time tags to the nearest 1 second. | |
| 3.4.2 iii | The equipment shall report the following status data: (iii) Hours: Elapsed operating hours. | |

# TAB-08  Capital letters A. to D., then a sub-paragraph 3.4.3.1

Lists on tab stops | document: documents/TAB-08_capital_letters_a_to_d_then_a_sub_paragraph_3_4_3_1.pdf | typewriter 11 pt, 300 dpi

### Situation

A./B./C./D. at 0.4 in, labels at 1.0 in, values at 3.0 in; the lead-in 'within the following limits:' wraps over two lines; values start with '2.5 kg/min', '15 C', '1.2 kPa', '5 to 95'; the child paragraph 3.4.3.1 follows.

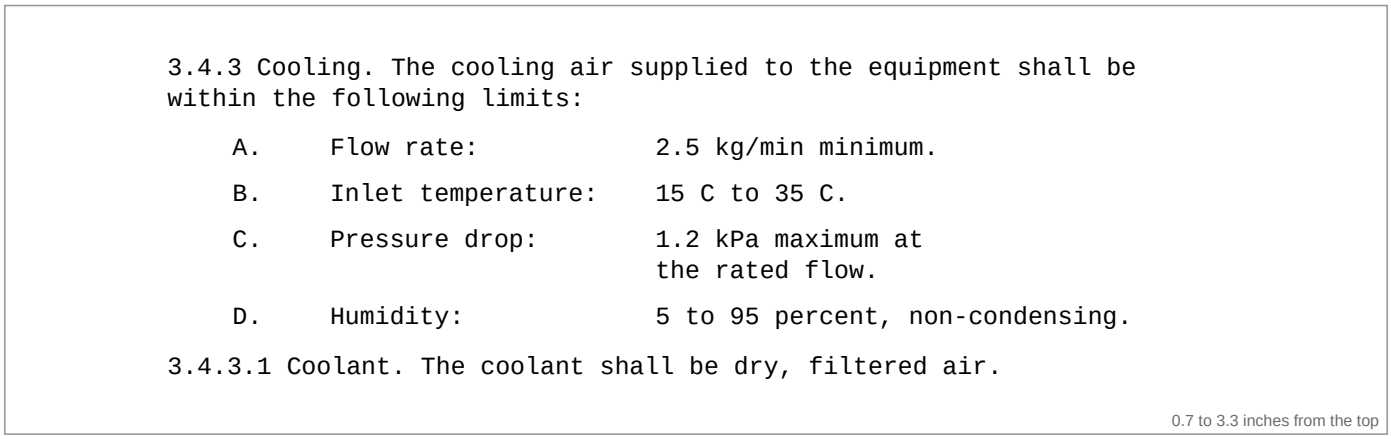
### What the tool must do

A list lettered in capitals keeps its capitals ('3.4.3 A' ... '3.4.3 D'); 3.4.3.1 (a valid next number after 3.4.3) is a paragraph of its own.

### Source

List suite p03.

### The page (as the scan document has it)



```
3.4.3 Cooling. The cooling air supplied to the equipment shall be
within the following limits:

    A.    Flow rate:           2.5 kg/min minimum.

    B.    Inlet temperature:   15 C to 35 C.

    C.    Pressure drop:       1.2 kPa maximum at
                               the rated flow.

    D.    Humidity:            5 to 95 percent, non-condensing.

3.4.3.1 Coolant. The coolant shall be dry, filtered air.
```

0.7 to 3.3 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.4.3 A | The cooling air supplied to the equipment shall be within the following limits: A. Flow rate: 2.5 kg/min minimum. | |
| 3.4.3 B | The cooling air supplied to the equipment shall be within the following limits: B. Inlet temperature: 15 C to 35 C. | |
| 3.4.3 C | The cooling air supplied to the equipment shall be within the following limits: C. Pressure drop: 1.2 kPa maximum at the rated flow. | |
| 3.4.3 D | The cooling air supplied to the equipment shall be within the following limits: D. Humidity: 5 to 95 percent, non-condensing. | |
| 3.4.3.1 | The coolant shall be dry, filtered air. | |

# TAB-09  Bullets with upper-case labels

Lists on tab stops | document: documents/TAB-09_bullets_with_upper_case_labels.pdf | typewriter 11 pt, 300 dpi

### Situation

Bullets at 0.4 in, upper-case labels ('POWER:') at 0.8 in, values at 2.4 in; the lead-in ends ':'.

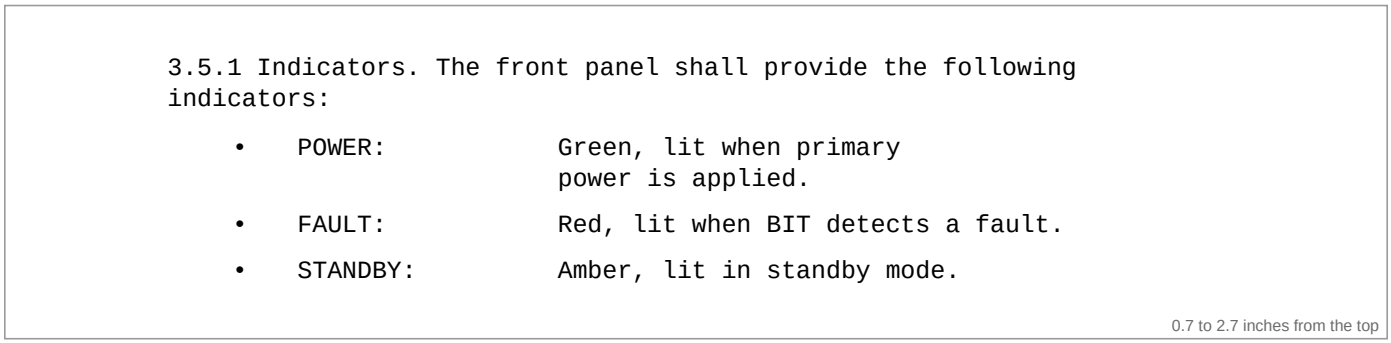
### What the tool must do

Bullet items have no marker reference: every row has Doc Ref '3.5.1'. The OCR returns no glyph at all for a bullet that stands alone, so these items have no visible marker and must be found from their label column under the lead-in.

### Source

List suite p04.

### The page (as the scan document has it)



3.5.1 Indicators. The front panel shall provide the following indicators:

- POWER: Green, lit when primary power is applied.
- FAULT: Red, lit when BIT detects a fault.
- STANDBY: Amber, lit in standby mode.

0.7 to 2.7 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.5.1 | The front panel shall provide the following indicators: • POWER: Green, lit when primary power is applied. | |
| 3.5.1 | The front panel shall provide the following indicators: • FAULT: Red, lit when BIT detects a fault. | |
| 3.5.1 | The front panel shall provide the following indicators: • STANDBY: Amber, lit in standby mode. | |

# TAB-10  No label column: markers and values of one to four lines

Lists on tab stops | document: documents/TAB-10_no_label_column_markers_and_values_of_one_to_four_lines.pdf | typewriter 11 pt, 300 dpi

### Situation

a.-d. at 0.4 in, no label, values at 1.2 in (a large tab); values of 2, 1, 4 and 2 lines; continuation lines start with '10 Mbit/s' and 'system time.'; item c holds a second 'shall' sentence.

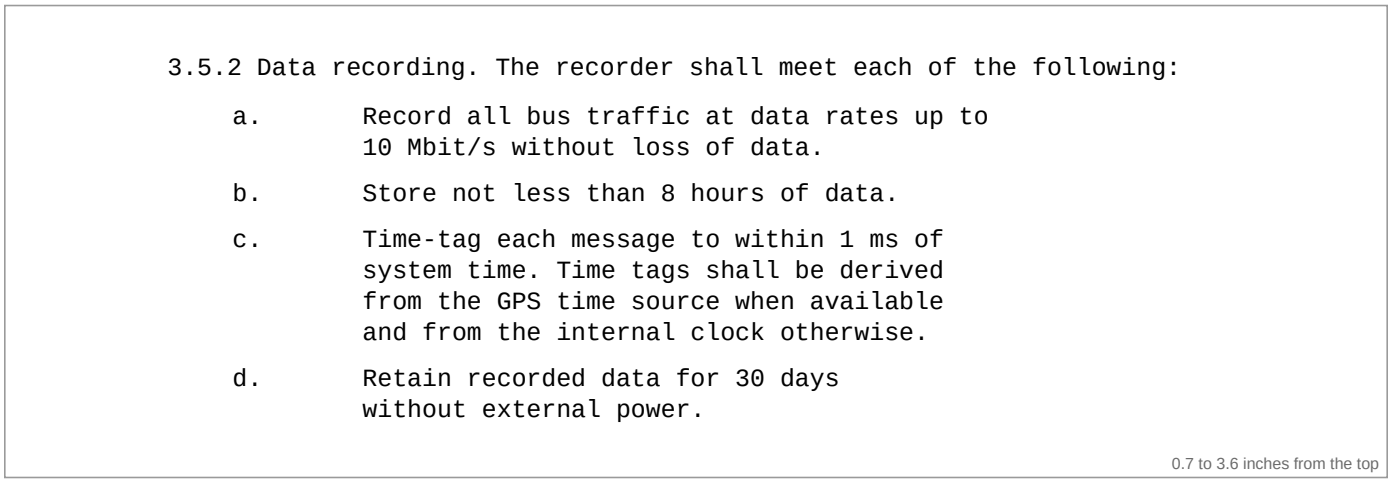
### What the tool must do

Each item one row with its whole value, whatever its continuation lines start with.

### Source

List suite p04.

### The page (as the scan document has it)



3.5.2 Data recording. The recorder shall meet each of the following:

a. Record all bus traffic at data rates up to 10 Mbit/s without loss of data.

b. Store not less than 8 hours of data.

c. Time-tag each message to within 1 ms of system time. Time tags shall be derived from the GPS time source when available and from the internal clock otherwise.

d. Retain recorded data for 30 days without external power.

0.7 to 3.6 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.5.2 a | The recorder shall meet each of the following: a. Record all bus traffic at data rates up to 10 Mbit/s without loss of data. | |
| 3.5.2 b | The recorder shall meet each of the following: b. Store not less than 8 hours of data. | |
| 3.5.2 c | The recorder shall meet each of the following: c. Time-tag each message to within 1 ms of system time. Time tags shall be derived from the GPS time source when available and from the internal clock otherwise. | |
| 3.5.2 d | The recorder shall meet each of the following: d. Retain recorded data for 30 days without external power. | |

# TAB-11 A label that wraps onto a second line beside its value

Lists on tab stops | document: documents/TAB-11_a_label_that_wraps_onto_a_second_line_beside_its_value.pdf | typewriter 11 pt, 300 dpi

### Situation

a.-c. at 0.4 in; the label at 0.9 in wraps ('Storage' / 'temperature:') beside a two-line value at 2.6 in; paragraph 3.5.4 follows.

### What the tool must do

The label is read first, then the value: 'a. Storage temperature: -55 C to +85 C for up to 5 years.' - not 'Storage -55 C to +85 C for temperature: up to 5 years.'

### Source

List suite p04.

### The page (as the scan document has it)

```
3.5.3 Storage. The equipment shall not be damaged by storage in the
following conditions:

    a.   Storage           -55 C to +85 C for
         temperature:      up to 5 years.

    b.   Relative          Up to 100 percent,
         humidity:         including condensation.

    c.   Altitude:         Up to 12,200 m (40,000 ft).

3.5.4 Transport. The packaged equipment shall withstand transport by
road, rail, sea and air.
```

0.7 to 3.4 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.5.3 a | The equipment shall not be damaged by storage in the following conditions: a. Storage temperature: -55 C to +85 C for up to 5 years. | |
| 3.5.3 b | The equipment shall not be damaged by storage in the following conditions: b. Relative humidity: Up to 100 percent, including condensation. | |
| 3.5.3 c | The equipment shall not be damaged by storage in the following conditions: c. Altitude: Up to 12,200 m (40,000 ft). | |
| 3.5.4 | The packaged equipment shall withstand transport by road, rail, sea and air. | |

# TAB-12  A nested (1)/(2) list under an item that has no value

Lists on tab stops | document: documents/TAB-12_a_nested_1_2_list_under_an_item_that_has_no_value.pdf | typewriter 11 pt, 300 dpi

### Situation

a./b./c. on tab stops; item b. 'Random:' has no value of its own and a nested (1)/(2) list under its label (markers 0.9 in, labels 1.4 in, values 3.0 in).

### What the tool must do

The sub-items are filed under b: '3.6.1 b.1' and '3.6.1 b.2', each with 'b. Random:' in front.

### Source

List suite p05.

### The page (as the scan document has it)

```
3.6.1 Vibration. The equipment shall withstand the following vibration
environments:

    a.   Sinusoidal:       5 Hz to 500 Hz, 2 g peak,
                           1 octave per minute.

    b.   Random:

         (1)  Functional:      20 Hz to 2000 Hz, 0.04 g2/Hz,
                               1 hour per axis.

         (2)  Endurance:       20 Hz to 2000 Hz, 0.1 g2/Hz,
                               3 hours per axis.

    c.   Transport:        As specified in MIL-STD-810H,
                           Method 514.8, Procedure II.
```

0.7 to 3.9 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.6.1 a | The equipment shall withstand the following vibration environments: a. Sinusoidal: 5 Hz to 500 Hz, 2 g peak, 1 octave per minute. | |
| 3.6.1 b.1 | The equipment shall withstand the following vibration environments: b. Random: (1) Functional: 20 Hz to 2000 Hz, 0.04 g2/Hz, 1 hour per axis. | |
| 3.6.1 b.2 | The equipment shall withstand the following vibration environments: b. Random: (2) Endurance: 20 Hz to 2000 Hz, 0.1 g2/Hz, 3 hours per axis. | |
| 3.6.1 c | The equipment shall withstand the following vibration environments: c. Transport: As specified in MIL-STD-810H, Method 514.8, Procedure II. | |

# TAB-13 Decimal values that look like paragraph numbers, then an all-capitals heading 4.

Lists on tab stops | document: documents/TAB-13_decimal_values_that_look_like_paragraph_numbers_then_an_all_.pdf | typewriter 11 pt, 300 dpi

### Situation

1./2./3. on tab stops, no gap between items; every value line starts with a decimal and a capital unit ('9.0 G', '2.5 G', '3.2 G'); the first value line ends with '.' and the next starts with a number; then the heading '4. QUALITY ASSURANCE PROVISIONS' (4 would continue 1, 2, 3!) and paragraph 4.1.

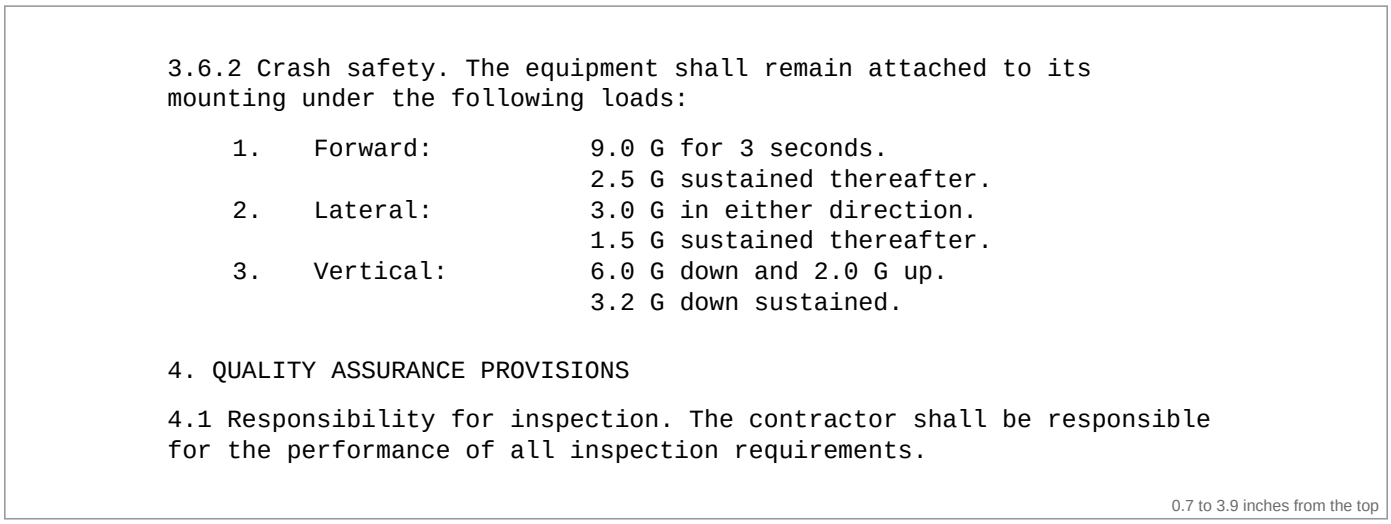
### What the tool must do

'2.5 G sustained thereafter.' stays in its item (it is not paragraph 2.5); '4. QUALITY ASSURANCE PROVISIONS' is a heading, not item 4; 4.1 is a paragraph.

### Source

List suite p05.

### The page (as the scan document has it)



```
3.6.2 Crash safety. The equipment shall remain attached to its
mounting under the following loads:

    1.   Forward:          9.0 G for 3 seconds.
                           2.5 G sustained thereafter.
    2.   Lateral:          3.0 G in either direction.
                           1.5 G sustained thereafter.
    3.   Vertical:         6.0 G down and 2.0 G up.
                           3.2 G down sustained.


4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The contractor shall be responsible
for the performance of all inspection requirements.
```

0.7 to 3.9 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.6.2 1 | The equipment shall remain attached to its mounting under the following loads: 1. Forward: 9.0 G for 3 seconds. 2.5 G sustained thereafter. | |
| 3.6.2 2 | The equipment shall remain attached to its mounting under the following loads: 2. Lateral: 3.0 G in either direction. 1.5 G sustained thereafter. | |
| 3.6.2 3 | The equipment shall remain attached to its mounting under the following loads: 3. Vertical: 6.0 G down and 2.0 G up. 3.2 G down sustained. | |
| 4.1 | The contractor shall be responsible for the performance of all inspection requirements. | |

# TAB-14 Proportional font, markers at the margin, three-line value

Lists on tab stops | document: documents/TAB-14_proportional_font_markers_at_the_margin_three_line_value.pdf | proportional 12 pt, 200 dpi

### Situation

Serif, 200 dpi. a.-d. at the LEFT MARGIN, labels at 0.5 in, values at 2.2 in; the lead-in ':' wraps over two lines; item a. has a three-line value whose lines start '-40 C', '-55 C', '5 C per minute'.

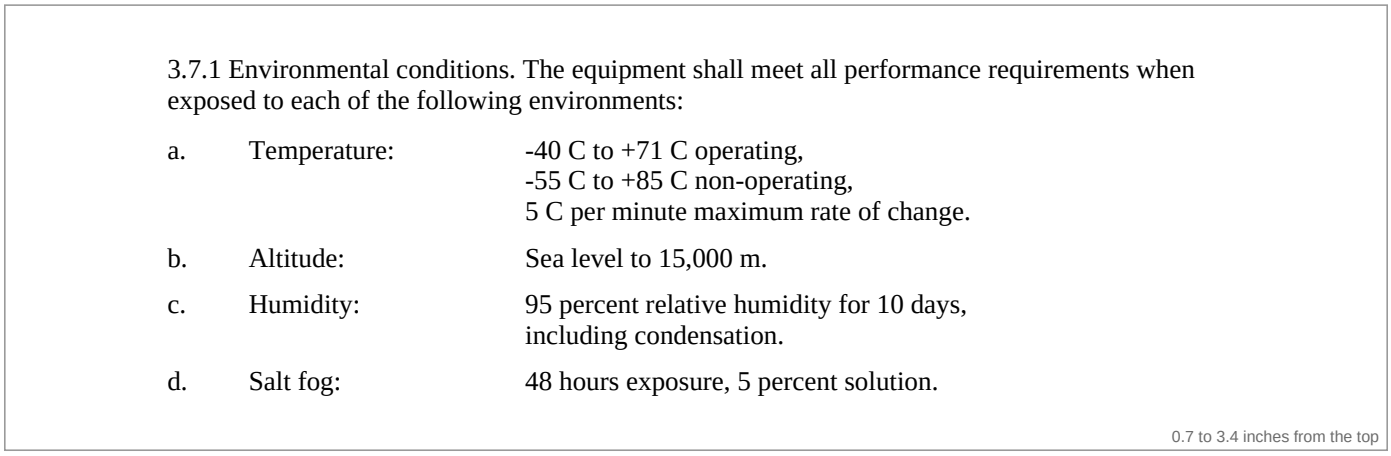
### What the tool must do

Each item one row with its whole value. The OCR tends to read the lone marker 'c.' as 'C.'; the Doc Ref must still be '<para> c', because the a, b, ?, d sequence says it is c.

### Source

List suite p06.

### The page (as the scan document has it)



3.7.1 Environmental conditions. The equipment shall meet all performance requirements when exposed to each of the following environments:

| | | |
|---|---|---|
| a. | Temperature: | -40 C to +71 C operating,<br>-55 C to +85 C non-operating,<br>5 C per minute maximum rate of change. |
| b. | Altitude: | Sea level to 15,000 m. |
| c. | Humidity: | 95 percent relative humidity for 10 days,<br>including condensation. |
| d. | Salt fog: | 48 hours exposure, 5 percent solution. |

0.7 to 3.4 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.7.1 a | The equipment shall meet all performance requirements when exposed to each of the following environments: a. Temperature: -40 C to +71 C operating, -55 C to +85 C non-operating, 5 C per minute maximum rate of change. | |
| 3.7.1 b | The equipment shall meet all performance requirements when exposed to each of the following environments: b. Altitude: Sea level to 15,000 m. | |
| 3.7.1 c | The equipment shall meet all performance requirements when exposed to each of the following environments: c. Humidity: 95 percent relative humidity for 10 days, including condensation. | |
| 3.7.1 d | The equipment shall meet all performance requirements when exposed to each of the following environments: d. Salt fog: 48 hours exposure, 5 percent solution. | |

# TAB-15 Proportional font, numbered list, wrapped lead-in without a colon

Lists on tab stops | document: documents/TAB-15_proportional_font_numbered_list_wrapped_lead_in_without_a_co.pdf | proportional 12 pt, 200 dpi

### Situation

Serif, 200 dpi. 1.-4. at 0.3 in, labels at 0.7 in, values at 2.2 in; the lead-in '... within the following limits.' wraps over two lines; a value line ending '.' is followed by one starting '4.5 A'; paragraph 3.7.3 follows.

### What the tool must do

Each item one row; '4.5 A peak' is not paragraph 4.5. The OCR does not return the lone marker '4.' of 'Inrush' at all: the Doc Ref '3.7.2 4' can only come from the 1, 2, 3 sequence and the label column.

### Source

List suite p06.

### The page (as the scan document has it)

3.7.2 Electrical interface. The equipment shall accept primary power from the aircraft bus within the following limits.

| | | |
|---|---|---|
| 1. | Voltage: | 22 VDC to 29 VDC,<br>28 VDC nominal. |
| 2. | Ripple: | 1.5 V peak-to-peak maximum. |
| 3. | Current: | 3.2 A continuous.<br>4.5 A peak for 100 ms. |
| 4. | Inrush: | 12 A for 10 ms maximum. |

3.7.3 Grounding. The equipment case shall be bonded to structure with a resistance not exceeding 2.5 milliohms.

0.7 to 3.7 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.7.2 1 | The equipment shall accept primary power from the aircraft bus within the following limits. 1. Voltage: 22 VDC to 29 VDC, 28 VDC nominal. | |
| 3.7.2 2 | The equipment shall accept primary power from the aircraft bus within the following limits. 2. Ripple: 1.5 V peak-to-peak maximum. | |
| 3.7.2 3 | The equipment shall accept primary power from the aircraft bus within the following limits. 3. Current: 3.2 A continuous. 4.5 A peak for 100 ms. | |
| 3.7.2 4 | The equipment shall accept primary power from the aircraft bus within the following limits. 4. Inrush: 12 A for 10 ms maximum. | |
| 3.7.3 | The equipment case shall be bonded to structure with a resistance not exceeding 2.5 milliohms. | |

# TAB-16 Proportional font, (a)-(c), a four-line value with numbers inside

Lists on tab stops | document: documents/TAB-16_proportional_font_a_c_a_four_line_value_with_numbers_inside.pdf | proportional 12 pt, 200 dpi

**Situation**

Serif, 200 dpi. (a)-(c) on tab stops; lead-in 'within the following limits.' (no colon); item (b) has a four-line value whose second line ends with '.' and whose third and fourth lines start '3.0 kg/min' and '3,000 m'.

**What the tool must do**

(b) is one row with all four value lines.

**Source**

List suite p07.

**The page (as the scan document has it)**

3.7.4 Cooling air. The equipment shall operate with cooling air within the following limits.

| | | |
|---|---|---|
| (a) | Temperature: | 15 C to 35 C at the inlet. |
| (b) | Flow: | 2.5 kg/min minimum per kW of dissipated power at sea level. 3.0 kg/min per kW above 3,000 m altitude. |
| (c) | Pressure: | 1.2 kPa maximum drop across the equipment at rated flow. |

0.7 to 3.2 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.7.4 a | The equipment shall operate with cooling air within the following limits. (a) Temperature: 15 C to 35 C at the inlet. | |
| 3.7.4 b | The equipment shall operate with cooling air within the following limits. (b) Flow: 2.5 kg/min minimum per kW of dissipated power at sea level. 3.0 kg/min per kW above 3,000 m altitude. | |
| 3.7.4 c | The equipment shall operate with cooling air within the following limits. (c) Pressure: 1.2 kPa maximum drop across the equipment at rated flow. | |

# TAB-17 Proportional font, every label wraps onto two lines

Lists on tab stops | document: documents/TAB-17_proportional_font_every_label_wraps_onto_two_lines.pdf | proportional 12 pt, 200 dpi

**Situation**

Serif, 200 dpi. a.-c.; two-line labels ('Maximum' / 'weight:') at 0.9 in beside two-line values at 2.4 in; the lead-in ends ':'.

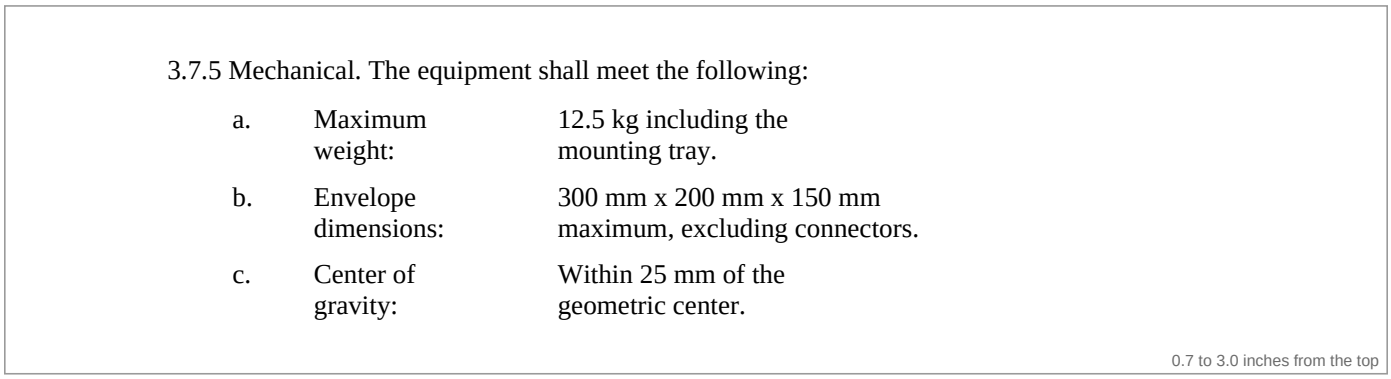
**What the tool must do**

Each item reads label first, then value: 'a. Maximum weight: 12.5 kg including the mounting tray.'

**Source**

List suite p07.

**The page (as the scan document has it)**



3.7.5 Mechanical. The equipment shall meet the following:

| | | |
|---|---|---|
| a. | Maximum weight: | 12.5 kg including the mounting tray. |
| b. | Envelope dimensions: | 300 mm x 200 mm x 150 mm maximum, excluding connectors. |
| c. | Center of gravity: | Within 25 mm of the geometric center. |

0.7 to 3.0 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.7.5 a | The equipment shall meet the following: a. Maximum weight: 12.5 kg including the mounting tray. | |
| 3.7.5 b | The equipment shall meet the following: b. Envelope dimensions: 300 mm x 200 mm x 150 mm maximum, excluding connectors. | |
| 3.7.5 c | The equipment shall meet the following: c. Center of gravity: Within 25 mm of the geometric center. | |

# TAB-18  Proportional font, bullets, then the next paragraph

Lists on tab stops | document: documents/TAB-18_proportional_font_bullets_then_the_next_paragraph.pdf | proportional 12 pt, 200 dpi

**Situation**

Serif, 200 dpi. Bullets at 0.4 in, labels at 0.7 in, values at 2.2 in; the lead-in ends ':'; paragraph 3.7.7 follows.

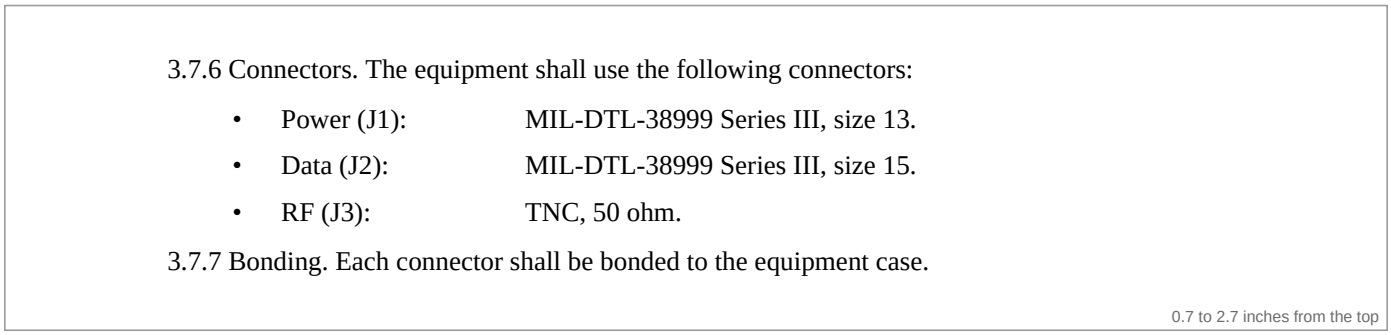
**What the tool must do**

Bullet items: every row has Doc Ref '3.7.6'. The OCR returns no glyph at all for a bullet that stands alone, so these items have no visible marker and must be found from their label column under the lead-in.

**Source**

List suite p07.

**The page (as the scan document has it)**



3.7.6 Connectors. The equipment shall use the following connectors:

- Power (J1): MIL-DTL-38999 Series III, size 13.
- Data (J2): MIL-DTL-38999 Series III, size 15.
- RF (J3): TNC, 50 ohm.

3.7.7 Bonding. Each connector shall be bonded to the equipment case.

0.7 to 2.7 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.7.6 | The equipment shall use the following connectors: • Power (J1): MIL-DTL-38999 Series III, size 13. | |
| 3.7.6 | The equipment shall use the following connectors: • Data (J2): MIL-DTL-38999 Series III, size 15. | |
| 3.7.6 | The equipment shall use the following connectors: • RF (J3): TNC, 50 ohm. | |
| 3.7.7 | Each connector shall be bonded to the equipment case. | |

# TAB-19  TAB-01 on a photocopy

Lists on tab stops | document: documents/TAB-19_tab_01_on_a_photocopy.pdf | typewriter 11 pt, 300 dpi, photocopy (drawn here without the damage)

### Situation

The page of TAB-01 run through photocopier-style damage: faded ink, speckle, a 1.3 degree skew, punch holes.

### What the tool must do

The same rows as TAB-01.

### Source

List suite p10.

### The page (as the scan document has it)

```
3.2.4 Environmental conditions. The equipment shall meet each of the
following:

    a.   Acceleration:     5 g in each of the three orthogonal axes
                           for 1 minute per axis.

    b.   Vibration:        20 Hz to 2000 Hz random, 0.04 g2/Hz,
                           3 hours per axis, 7.7 g rms overall.

    c.   Shock:            40 g, 11 ms half-sine pulse,
                           3 shocks in each direction.

    d.   Temperature:      -40 C to +71 C operating,
                           2.5 C per minute maximum rate of change.
```

0.7 to 3.6 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The equipment shall meet each of the following: a. Acceleration: 5 g in each of the three orthogonal axes for 1 minute per axis. | |
| 3.2.4 b | The equipment shall meet each of the following: b. Vibration: 20 Hz to 2000 Hz random, 0.04 g2/Hz, 3 hours per axis, 7.7 g rms overall. | |
| 3.2.4 c | The equipment shall meet each of the following: c. Shock: 40 g, 11 ms half-sine pulse, 3 shocks in each direction. | |
| 3.2.4 d | The equipment shall meet each of the following: d. Temperature: -40 C to +71 C operating, 2.5 C per minute maximum rate of change. | |

# TAB-20  TAB-02 on a photocopy

Lists on tab stops | document: documents/TAB-20_tab_02_on_a_photocopy.pdf | typewriter 11 pt, 300 dpi, photocopy (drawn here without the damage)

### Situation

The page of TAB-02 run through photocopier-style damage: faded ink, speckle, a 1.3 degree skew, punch holes.

### What the tool must do

The same rows as TAB-02.

### Source

List suite p10.

### The page (as the scan document has it)

```
3.2.5 Electrical conditions. The equipment shall operate within the
following limits.

    1.   Input voltage:     24 to 32 VDC, 28 VDC nominal.

    2.   Ripple:           1.5 V peak-to-peak maximum,
                           10 Hz to 50 kHz.

    3.   Current:          3.2 A continuous, 4.0 A peak.

    4.   Power:            150 W maximum at 28 VDC.

3.2.6 Weight. The equipment shall not exceed 45 kilograms.
```

0.7 to 3.3 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.5 1 | The equipment shall operate within the following limits. 1. Input voltage: 24 to 32 VDC, 28 VDC nominal. | |
| 3.2.5 2 | The equipment shall operate within the following limits. 2. Ripple: 1.5 V peak-to-peak maximum, 10 Hz to 50 kHz. | |
| 3.2.5 3 | The equipment shall operate within the following limits. 3. Current: 3.2 A continuous, 4.0 A peak. | |
| 3.2.5 4 | The equipment shall operate within the following limits. 4. Power: 150 W maximum at 28 VDC. | |
| 3.2.6 | The equipment shall not exceed 45 kilograms. | |

# TAB-21  A NOTE right under the list

Lists on tab stops | document: documents/TAB-21_a_note_right_under_the_list.pdf | typewriter 11 pt, 300 dpi

### Situation

The list of TAB-01's shape (values on tab stops), then, after a blank line, 'NOTE: Values apply at 25 C.' at the margin, then paragraph 3.2.5.

### What the tool must do

The NOTE sits in the same band of the page as the list, but it is not part of the list: it must not make the tab look like the gutter between two text columns (round 2: the values were lost, or all given to the last item). The NOTE has no 'shall', so it is not a row.

### Source

Review round 2, finding layout/F1.

### The page (as the scan document has it)

```
3.2.4 Environmental conditions. The equipment shall meet
each of the following:

    a.     Acceleration:    5 g in each of the three orthogonal axes
                            for 1 minute per axis.

    b.     Vibration:       20 Hz to 2000 Hz random, 0.04 g2/Hz,
                            3 hours per axis, 7.7 g rms overall.

    c.     Shock:           40 g, 11 ms half-sine pulse,
                            3 shocks in each direction.

    d.     Temperature:     -40 C to +71 C operating.

NOTE: Values apply at 25 C.

3.2.5 Weight. The equipment shall not exceed 45 kilograms.
```

0.7 to 4.2 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The equipment shall meet each of the following: a. Acceleration: 5 g in each of the three orthogonal axes for 1 minute per axis. | |
| 3.2.4 b | The equipment shall meet each of the following: b. Vibration: 20 Hz to 2000 Hz random, 0.04 g2/Hz, 3 hours per axis, 7.7 g rms overall. | |
| 3.2.4 c | The equipment shall meet each of the following: c. Shock: 40 g, 11 ms half-sine pulse, 3 shocks in each direction. | |
| 3.2.4 d | The equipment shall meet each of the following: d. Temperature: -40 C to +71 C operating. | |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

# TAB-22  A heading right under the list

Lists on tab stops | document: documents/TAB-22_a_heading_right_under_the_list.pdf | typewriter 11 pt, 300 dpi

**Situation**

The list, then the heading '3.2.5 Size, weight and finish.' and its first paragraph 3.2.5.1.

**What the tool must do**

The heading does not cut the list into columns and does not become an item; 3.2.5.1 is a paragraph.

**Source**

Review round 2, finding layout/F1.

**The page (as the scan document has it)**

```
3.2.4 Environmental conditions. The equipment shall meet
each of the following:

    a.     Acceleration:        5 g in each of the three orthogonal
                                axes for 1 minute per axis.

    b.     Vibration:           20 Hz to 2000 Hz random, 0.04 g2/Hz,
                                3 hours per axis, 7.7 g rms overall.

    c.     Shock:               40 g, 11 ms half-sine pulse,
                                3 shocks in each direction.

    d.     Temperature:         -40 C to +71 C operating.

3.2.5 Size, weight and finish.

3.2.5.1 Weight. The equipment shall not exceed 45 kilograms in any
configuration.
```

0.7 to 4.3 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The equipment shall meet each of the following: a. Acceleration: 5 g in each of the three orthogonal axes for 1 minute per axis. | |
| 3.2.4 b | The equipment shall meet each of the following: b. Vibration: 20 Hz to 2000 Hz random, 0.04 g2/Hz, 3 hours per axis, 7.7 g rms overall. | |
| 3.2.4 c | The equipment shall meet each of the following: c. Shock: 40 g, 11 ms half-sine pulse, 3 shocks in each direction. | |
| 3.2.4 d | The equipment shall meet each of the following: d. Temperature: -40 C to +71 C operating. | |
| 3.2.5.1 | The equipment shall not exceed 45 kilograms in any configuration. | |

# TAB-23  The lead-in's last line sits beside the markers and labels

Lists on tab stops | document: documents/TAB-23_the_lead_in_s_last_line_sits_beside_the_markers_and_labels.pdf | typewriter 11 pt, 300 dpi

### Situation

A two-line lead-in whose second line, 'to meet each of the following:' (six words), ends just before the value column, so it sits in the same band as the markers and labels of the list below it.

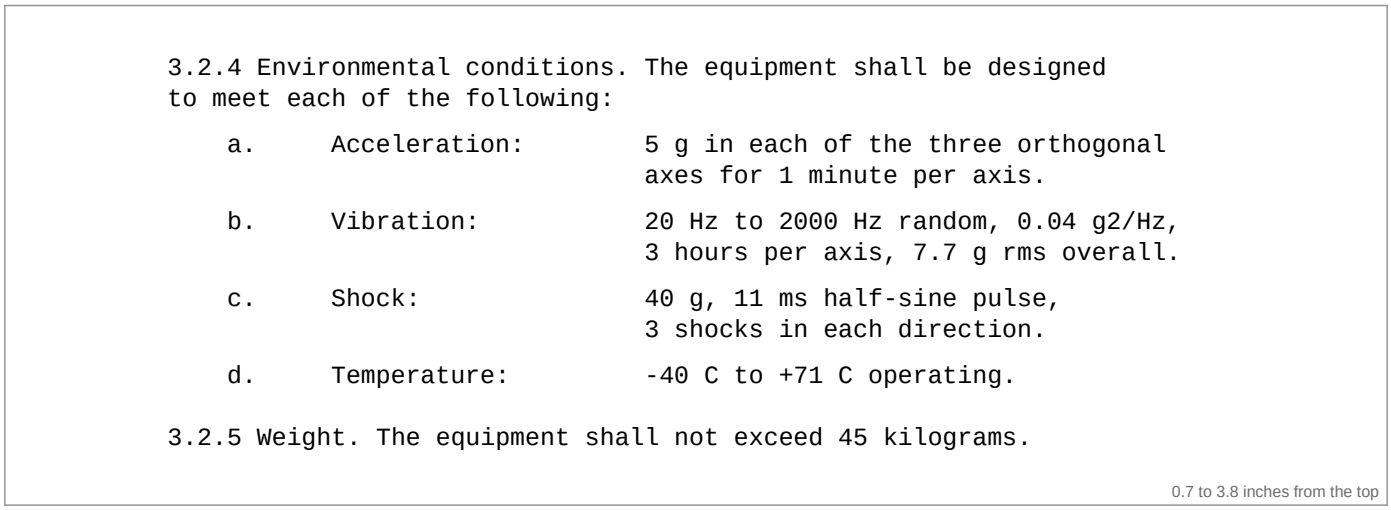
### What the tool must do

The lead-in's tail is part of the lead-in, not a label of the list, and it must not make the list look like two columns of text.

### Source

Review round 2, finding layout/F1.

### The page (as the scan document has it)



```
3.2.4 Environmental conditions. The equipment shall be designed
to meet each of the following:

    a.     Acceleration:        5 g in each of the three orthogonal
                                axes for 1 minute per axis.

    b.     Vibration:           20 Hz to 2000 Hz random, 0.04 g2/Hz,
                                3 hours per axis, 7.7 g rms overall.

    c.     Shock:               40 g, 11 ms half-sine pulse,
                                3 shocks in each direction.

    d.     Temperature:         -40 C to +71 C operating.

3.2.5 Weight. The equipment shall not exceed 45 kilograms.
```

0.7 to 3.8 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The equipment shall be designed to meet each of the following: a. Acceleration: 5 g in each of the three orthogonal axes for 1 minute per axis. | |
| 3.2.4 b | The equipment shall be designed to meet each of the following: b. Vibration: 20 Hz to 2000 Hz random, 0.04 g2/Hz, 3 hours per axis, 7.7 g rms overall. | |
| 3.2.4 c | The equipment shall be designed to meet each of the following: c. Shock: 40 g, 11 ms half-sine pulse, 3 shocks in each direction. | |
| 3.2.4 d | The equipment shall be designed to meet each of the following: d. Temperature: -40 C to +71 C operating. | |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

# TAB-24 One label of five words in a six-item list

Lists on tab stops | document: documents/TAB-24_one_label_of_five_words_in_a_six_item_list.pdf | typewriter 11 pt, 300 dpi

### Situation

Six items; one label is 'Peak inrush current (cold):' (five words with the marker).

### What the tool must do

A long label is still a label: the list is not cut into columns (round 2: the five-word row made the whole band look like running text, and every value was given to the last item).

### Source

Review round 2, finding layout/F1.

### The page (as the scan document has it)

```
3.2.6 Input power. The unit shall meet the following:
    a.     Input voltage:              22 to 29 VDC.
    b.     Ripple:                     1 V peak to peak maximum.
    c.     Peak inrush current (cold): 20 A for 10 ms maximum.
    d.     Power:                      150 W maximum.
    e.     Transients:                 as in MIL-STD-704F.
    f.     Hold-up:                    50 ms after loss of input.

3.2.7 Weight. The unit shall weigh less than 5 kg.
```

0.7 to 3.6 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.6 a | The unit shall meet the following: a. Input voltage: 22 to 29 VDC. | |
| 3.2.6 b | The unit shall meet the following: b. Ripple: 1 V peak to peak maximum. | |
| 3.2.6 c | The unit shall meet the following: c. Peak inrush current (cold): 20 A for 10 ms maximum. | |
| 3.2.6 d | The unit shall meet the following: d. Power: 150 W maximum. | |
| 3.2.6 e | The unit shall meet the following: e. Transients: as in MIL-STD-704F. | |
| 3.2.6 f | The unit shall meet the following: f. Hold-up: 50 ms after loss of input. | |
| 3.2.7 | The unit shall weigh less than 5 kg. | |

# TAB-25  Values in lower case without a full stop

Lists on tab stops | document: documents/TAB-25_values_in_lower_case_without_a_full_stop.pdf | typewriter 11 pt, 300 dpi

### Situation

Values such as 'within 22 to 29 VDC', 'less than 1 V peak to peak', 'not more than 150 W', 'per MIL-STD-704F' - lower case, no final full stop.

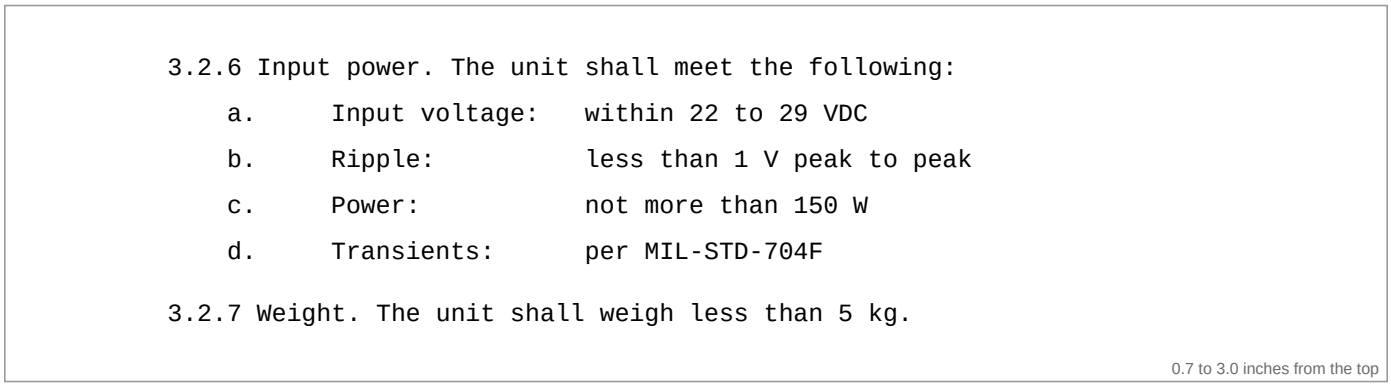
### What the tool must do

Values that start in lower case are not a column of prose running on from line to line: each stays with its label (round 2: every value went into the last item).

### Source

Review round 2, finding layout/F2.

### The page (as the scan document has it)



```
3.2.6 Input power. The unit shall meet the following:
    a.     Input voltage:   within 22 to 29 VDC
    b.     Ripple:          less than 1 V peak to peak
    c.     Power:           not more than 150 W
    d.     Transients:      per MIL-STD-704F

3.2.7 Weight. The unit shall weigh less than 5 kg.
```

0.7 to 3.0 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.6 a | The unit shall meet the following: a. Input voltage: within 22 to 29 VDC | |
| 3.2.6 b | The unit shall meet the following: b. Ripple: less than 1 V peak to peak | |
| 3.2.6 c | The unit shall meet the following: c. Power: not more than 150 W | |
| 3.2.6 d | The unit shall meet the following: d. Transients: per MIL-STD-704F | |
| 3.2.7 | The unit shall weigh less than 5 kg. | |

# TAB-26  Values that end with commas, one sentence over all items

Lists on tab stops | document: documents/TAB-26_values_that_end_with_commas_one_sentence_over_all_items.pdf | typewriter 11 pt, 300 dpi

### Situation

'within 22 to 29 VDC,' / 'less than 1 V peak to peak,' / 'not more than 150 W, and' / 'per MIL-STD-704F.' - the items read as one sentence; the values are short and all about as long as each other.

### What the tool must do

Each value stays with its label. Short values of equal length look like 'full' lines of a narrow text column, so the column test must not be fooled (round 3: on the scan the list was cut into two columns).

### Source

Review round 2, finding layout/F2; real-scan failure found in round 3.

### The page (as the scan document has it)

```
3.2.6 Input power. The unit shall operate with an input:
    a.     Voltage:         within 22 to 29 VDC,
    b.     Ripple:          less than 1 V peak to peak,
    c.     Power:           not more than 150 W, and
    d.     Transients:      per MIL-STD-704F.

3.2.7 Weight. The unit shall weigh less than 5 kg.
```

0.7 to 3.0 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.6 a | The unit shall operate with an input: a. Voltage: within 22 to 29 VDC, | |
| 3.2.6 b | The unit shall operate with an input: b. Ripple: less than 1 V peak to peak, | |
| 3.2.6 c | The unit shall operate with an input: c. Power: not more than 150 W, and | |
| 3.2.6 d | The unit shall operate with an input: d. Transients: per MIL-STD-704F. | |
| 3.2.7 | The unit shall weigh less than 5 kg. | |

# TAB-27 Values that end with commas, proportional font

Lists on tab stops | document: documents/TAB-27_values_that_end_with_commas_proportional_font.pdf | proportional 11 pt, 300 dpi

**Situation**

TAB-26 in a proportional (serif) font, where the OCR also tends to read the marker 'c.' as 'C.'.

**What the tool must do**

As TAB-26: four rows, each with its own value.

**Source**

Review round 2, finding layout/F2; real-scan failure found in round 3.

**The page (as the scan document has it)**

3.2.6 Input power. The unit shall operate with an input:

| | | |
|---|---|---|
| a. | Voltage: | within 22 to 29 VDC, |
| b. | Ripple: | less than 1 V peak to peak, |
| c. | Power: | not more than 150 W, and |
| d. | Transients: | per MIL-STD-704F. |

3.2.7 Weight. The unit shall weigh less than 5 kg.

0.7 to 3.0 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.6 a | The unit shall operate with an input: a. Voltage: within 22 to 29 VDC, | |
| 3.2.6 b | The unit shall operate with an input: b. Ripple: less than 1 V peak to peak, | |
| 3.2.6 c | The unit shall operate with an input: c. Power: not more than 150 W, and | |
| 3.2.6 d | The unit shall operate with an input: d. Transients: per MIL-STD-704F. | |
| 3.2.7 | The unit shall weigh less than 5 kg. | |

# TAB-28  A value that starts with a quantity and a sentence ('5.0 V. Ripple shall ...')

Lists on tab stops | document: documents/TAB-28_a_value_that_starts_with_a_quantity_and_a_sentence_5_0_v_rip.pdf | proportional 11 pt, 300 dpi

### Situation

'a. Output voltage: 5.0 V. Ripple shall not exceed 50 mV.' and 'b. Output current: 2.5 A. The output shall be protected' / 'against short circuits.'

### What the tool must do

'5.0 V.' is a quantity, not the start of a numbered paragraph 5.0 with the title 'V': no paragraphs '5.0' or '2.5' are invented, and the values stay in their items (round 2).

### Source

Review round 2, finding layout/F3.

### The page (as the scan document has it)

3.2.9 Power supply. The power supply shall provide the following:

| | | |
|---|---|---|
| a. | Output voltage: | 5.0 V. Ripple shall not exceed 50 mV. |
| b. | Output current: | 2.5 A. The output shall be protected against short circuits. |
| c. | Efficiency: | 85 percent minimum. |

3.2.10 Weight. The power supply shall weigh less than 5 kg.

0.7 to 2.9 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.9 a | The power supply shall provide the following: a. Output voltage: 5.0 V. Ripple shall not exceed 50 mV. | |
| 3.2.9 b | The power supply shall provide the following: b. Output current: 2.5 A. The output shall be protected against short circuits. | |
| 3.2.9 c | The power supply shall provide the following: c. Efficiency: 85 percent minimum. | |
| 3.2.10 | The power supply shall weigh less than 5 kg. | |

# TAB-29  A sub-list (1), (2) inside the value column

Lists on tab stops | document: documents/TAB-29_a_sub_list_1_2_inside_the_value_column.pdf | typewriter 11 pt, 300 dpi

### Situation

'a. Search mode: (1) Range of 100 km.' with '(2) Azimuth of 360 degrees.' under it in the value column; b. has its own (1), (2); c. has a plain value.

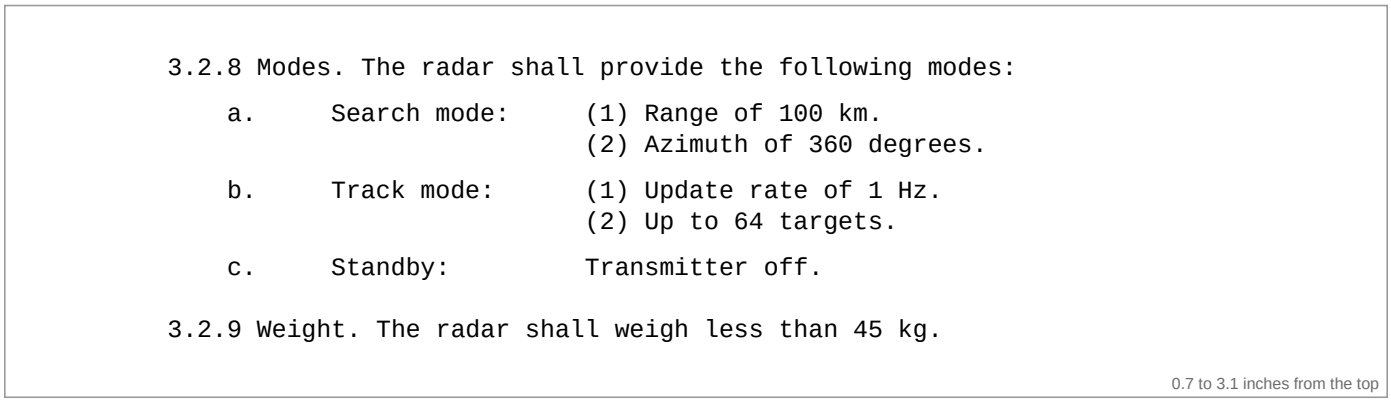
### What the tool must do

The sub-items are filed under their own item: 3.2.8 a.1, a.2, b.1, b.2 - not a second column of the list (round 2: all sub-items were given to c), and the (2) rows keep their parent.

### Source

Review round 2, finding layout/F4; the (2) rows lost their parent until round 3.

### The page (as the scan document has it)



```
3.2.8 Modes. The radar shall provide the following modes:

    a.     Search mode:     (1) Range of 100 km.
                            (2) Azimuth of 360 degrees.

    b.     Track mode:      (1) Update rate of 1 Hz.
                            (2) Up to 64 targets.

    c.     Standby:         Transmitter off.

3.2.9 Weight. The radar shall weigh less than 45 kg.
```

0.7 to 3.1 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.8 a.1 | The radar shall provide the following modes: a. Search mode: (1) Range of 100 km. | |
| 3.2.8 a.2 | The radar shall provide the following modes: a. Search mode: (2) Azimuth of 360 degrees. | |
| 3.2.8 b.1 | The radar shall provide the following modes: b. Track mode: (1) Update rate of 1 Hz. | |
| 3.2.8 b.2 | The radar shall provide the following modes: b. Track mode: (2) Up to 64 targets. | |
| 3.2.8 c | The radar shall provide the following modes: c. Standby: Transmitter off. | |
| 3.2.9 | The radar shall weigh less than 45 kg. | |

# TAB-30  A sub-list printed without its own indent

Lists on tab stops | document: documents/TAB-30_a_sub_list_printed_without_its_own_indent.pdf | typewriter 11 pt, 300 dpi

**Situation**

'b. Vibration:' with no value, then '(1) Random: ...' and '(2) Sine: ...' at the SAME indent as a. and b., then 'c. Shock: ...'.

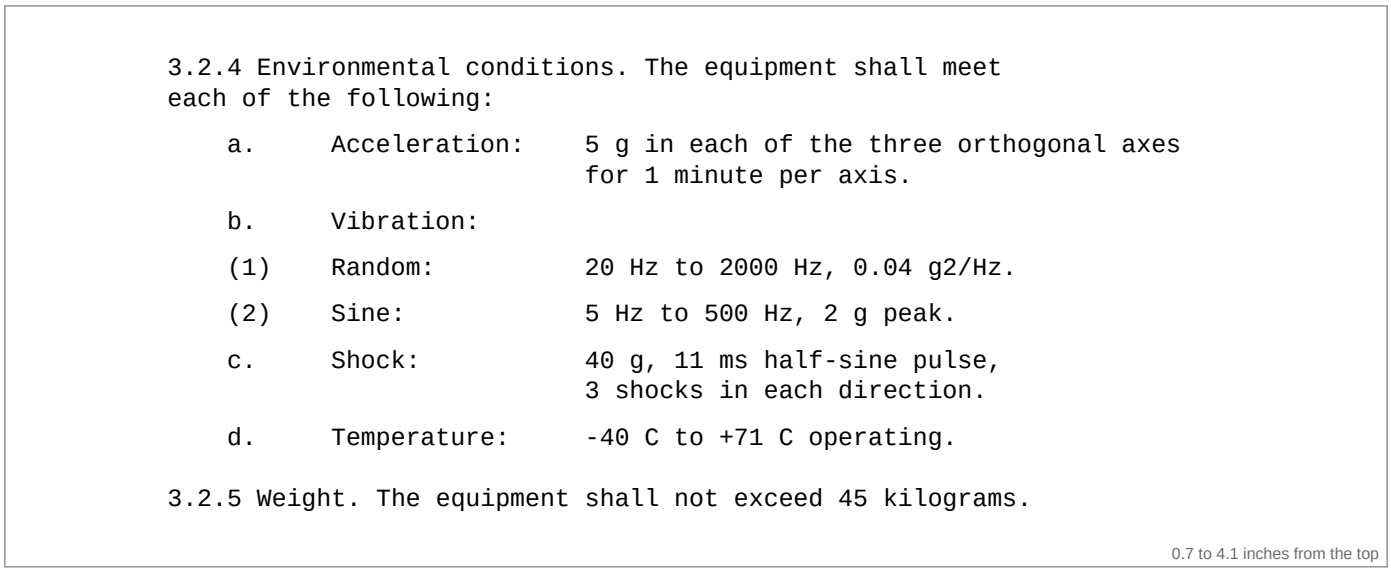
**What the tool must do**

The change of marker style after a line ending ':' (and back again at c.) shows that (1) and (2) belong to b: rows 3.2.4 b.1 and b.2.

**Source**

Review round 3 (found while checking the round-2 fixes).

**The page (as the scan document has it)**



```
3.2.4 Environmental conditions. The equipment shall meet
each of the following:

    a.     Acceleration:    5 g in each of the three orthogonal axes
                            for 1 minute per axis.

    b.     Vibration:

    (1)    Random:          20 Hz to 2000 Hz, 0.04 g2/Hz.

    (2)    Sine:            5 Hz to 500 Hz, 2 g peak.

    c.     Shock:           40 g, 11 ms half-sine pulse,
                            3 shocks in each direction.

    d.     Temperature:     -40 C to +71 C operating.

3.2.5 Weight. The equipment shall not exceed 45 kilograms.
```

0.7 to 4.1 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The equipment shall meet each of the following: a. Acceleration: 5 g in each of the three orthogonal axes for 1 minute per axis. | |
| 3.2.4 b.1 | The equipment shall meet each of the following: b. Vibration: (1) Random: 20 Hz to 2000 Hz, 0.04 g2/Hz. | |
| 3.2.4 b.2 | The equipment shall meet each of the following: b. Vibration: (2) Sine: 5 Hz to 500 Hz, 2 g peak. | |
| 3.2.4 c | The equipment shall meet each of the following: c. Shock: 40 g, 11 ms half-sine pulse, 3 shocks in each direction. | |
| 3.2.4 d | The equipment shall meet each of the following: d. Temperature: -40 C to +71 C operating. | |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

# TAB-31  A label that wraps and is longer than its value

Lists on tab stops | document: documents/TAB-31_a_label_that_wraps_and_is_longer_than_its_value.pdf | typewriter 11 pt, 300 dpi

**Situation**

'b. Operating' / 'temperature:' - a two-line label beside a ONE-line value, so the label's second line has nothing beside it.

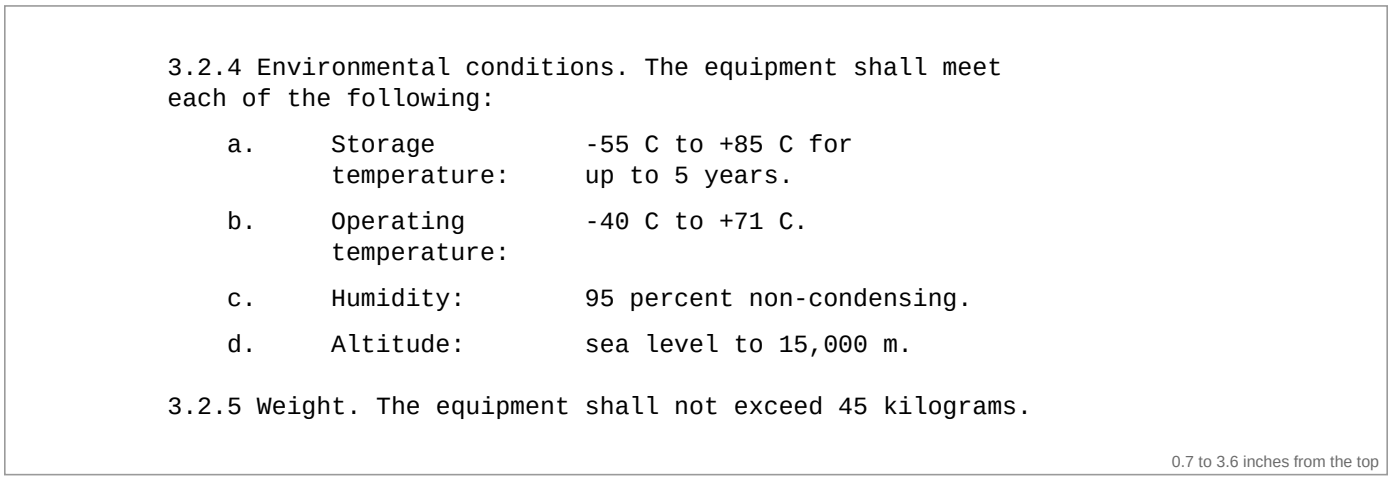
**What the tool must do**

The label is read first, then the value: 'b. Operating temperature: -40 C to +71 C.' (round 2: 'b. Operating -40 C to +71 C. temperature:').

**Source**

Review round 2, finding layout/F5.

**The page (as the scan document has it)**



```
3.2.4 Environmental conditions. The equipment shall meet
each of the following:

    a.     Storage          -55 C to +85 C for
           temperature:     up to 5 years.

    b.     Operating        -40 C to +71 C.
           temperature:

    c.     Humidity:        95 percent non-condensing.

    d.     Altitude:        sea level to 15,000 m.

3.2.5 Weight. The equipment shall not exceed 45 kilograms.
```

0.7 to 3.6 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The equipment shall meet each of the following: a. Storage temperature: -55 C to +85 C for up to 5 years. | |
| 3.2.4 b | The equipment shall meet each of the following: b. Operating temperature: -40 C to +71 C. | |
| 3.2.4 c | The equipment shall meet each of the following: c. Humidity: 95 percent non-condensing. | |
| 3.2.4 d | The equipment shall meet each of the following: d. Altitude: sea level to 15,000 m. | |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

# TAB-32 A label that wraps and is longer than its value, proportional font

Lists on tab stops | document: documents/TAB-32_a_label_that_wraps_and_is_longer_than_its_value_proportional.pdf | proportional 11 pt, 300 dpi

**Situation**

TAB-31 in a proportional (serif) font.

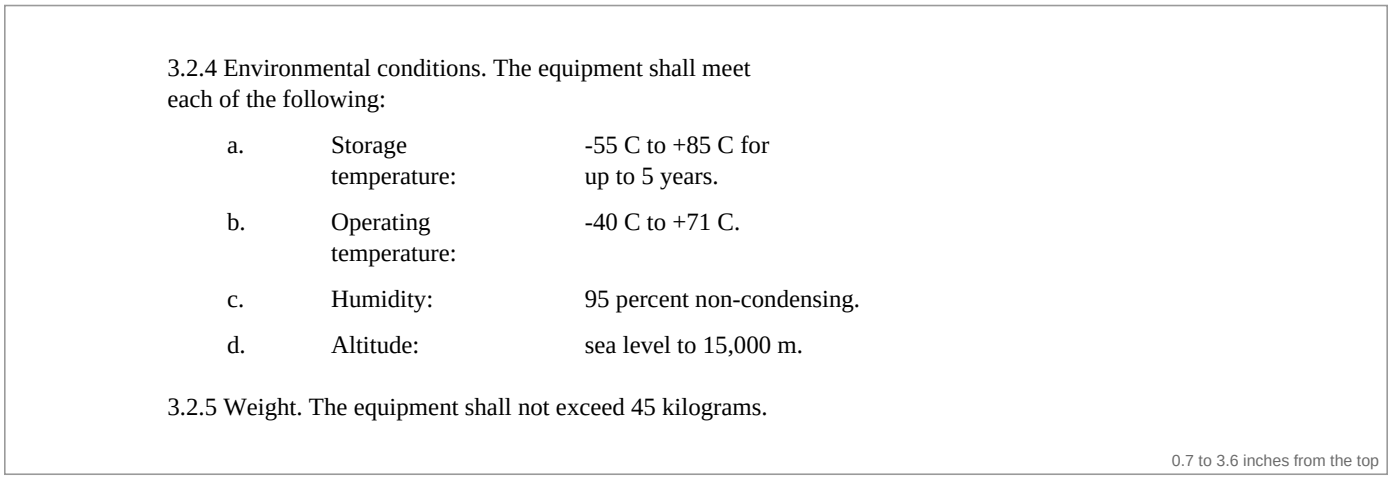
**What the tool must do**

As TAB-31. On this scan the first cells of the would-be table are the halves of wrapped labels ('Storage' / 'temperature:'), which must be read as one label each, not as a table (round 3).

**Source**

Review round 2, finding layout/F5; real-scan failure found in round 3.

**The page (as the scan document has it)**



3.2.4 Environmental conditions. The equipment shall meet each of the following:

| | | |
|---|---|---|
| a. | Storage temperature: | -55 C to +85 C for up to 5 years. |
| b. | Operating temperature: | -40 C to +71 C. |
| c. | Humidity: | 95 percent non-condensing. |
| d. | Altitude: | sea level to 15,000 m. |

3.2.5 Weight. The equipment shall not exceed 45 kilograms.

0.7 to 3.6 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The equipment shall meet each of the following: a. Storage temperature: -55 C to +85 C for up to 5 years. | |
| 3.2.4 b | The equipment shall meet each of the following: b. Operating temperature: -40 C to +71 C. | |
| 3.2.4 c | The equipment shall meet each of the following: c. Humidity: 95 percent non-condensing. | |
| 3.2.4 d | The equipment shall meet each of the following: d. Altitude: sea level to 15,000 m. | |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

# TAB-33  A wrapped label whose second line has four words

Lists on tab stops | document: documents/TAB-33_a_wrapped_label_whose_second_line_has_four_words.pdf | proportional 11 pt, 300 dpi

**Situation**

'a. Peak power' / 'consumption in any mode:' beside '150 W for not more than' / '10 seconds.'; serif font.

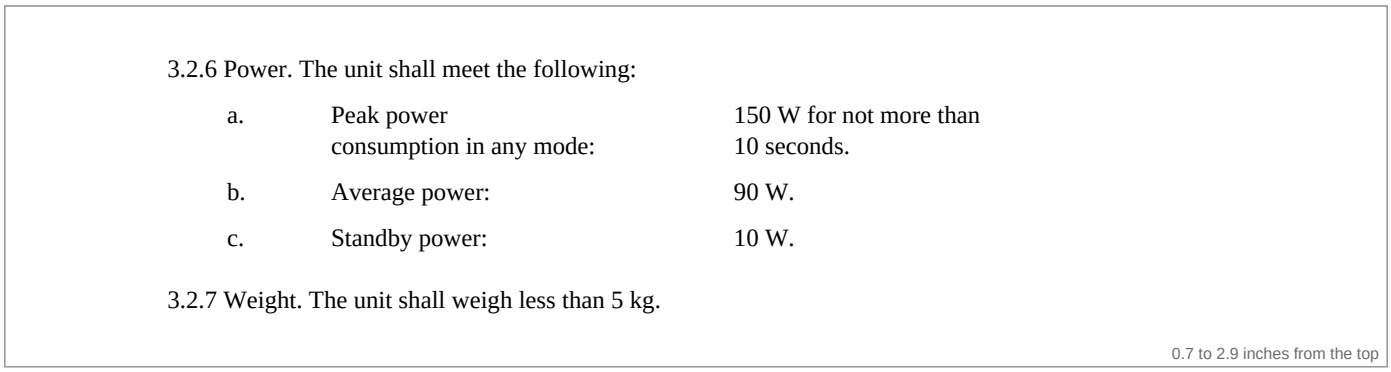
**What the tool must do**

'a. Peak power consumption in any mode: 150 W for not more than 10 seconds.' - a long second label line does not stop the label from being read first.

**Source**

Review round 2, finding layout/F5.

**The page (as the scan document has it)**



3.2.6 Power. The unit shall meet the following:

| | | |
|---|---|---|
| a. | Peak power consumption in any mode: | 150 W for not more than 10 seconds. |
| b. | Average power: | 90 W. |
| c. | Standby power: | 10 W. |

3.2.7 Weight. The unit shall weigh less than 5 kg.

0.7 to 2.9 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.6 a | The unit shall meet the following: a. Peak power consumption in any mode: 150 W for not more than 10 seconds. | |
| 3.2.6 b | The unit shall meet the following: b. Average power: 90 W. | |
| 3.2.6 c | The unit shall meet the following: c. Standby power: 10 W. | |
| 3.2.7 | The unit shall weigh less than 5 kg. | |

# TAB-34  A wrapped label whose first line ends with an abbreviation

Lists on tab stops | document: documents/TAB-34_a_wrapped_label_whose_first_line_ends_with_an_abbreviation.pdf | typewriter 11 pt, 300 dpi

### Situation

'a. Max. oper.' / 'temperature:' beside '+71 C with the unit powered' / 'and cooled.'

### What the tool must do

The full stop of 'oper.' does not end the label: 'a. Max. oper. temperature: +71 C with the unit powered and cooled.' (a review note about the full stops is fine).

### Source

Review round 2, finding layout/F5.

### The page (as the scan document has it)

```
3.2.4 Environmental conditions. The equipment shall meet
each of the following:

    a.     Max. oper.       +71 C with the unit powered
           temperature:     and cooled.

    b.     Min. oper.       -40 C after a 2 hour
           temperature:     cold soak.

    c.     Humidity:        95 percent non-condensing.

3.2.5 Weight. The equipment shall not exceed 45 kilograms.
```

0.7 to 3.3 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The equipment shall meet each of the following: a. Max. oper. temperature: +71 C with the unit powered and cooled. | |
| 3.2.4 b | The equipment shall meet each of the following: b. Min. oper. temperature: -40 C after a 2 hour cold soak. | |
| 3.2.4 c | The equipment shall meet each of the following: c. Humidity: 95 percent non-condensing. | |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

# TAB-35  Values that run on in lower case, each ending 'and'

Lists on tab stops | document: documents/TAB-35_values_that_run_on_in_lower_case_each_ending_and.pdf | typewriter 11 pt, 300 dpi

### Situation

'a. Search: the radar shall detect targets of 1 m2' / 'at 100 km range and', 'b. Track: maintain track on up to 64 targets with' / 'an update rate of 1 Hz and', 'c. Identify: interrogate ...'.

### What the tool must do

Each item keeps its own value although the values read like one running sentence.

### Source

Review round 2 (tab-list attack).

### The page (as the scan document has it)

```
3.2.6 Modes. In each mode the radar shall operate as follows:

    a.     Search:          the radar shall detect targets of 1 m2
                            at 100 km range and

    b.     Track:           maintain track on up to 64 targets with
                            an update rate of 1 Hz and

    c.     Identify:        interrogate each target in track using
                            Mode 5 IFF.

3.2.7 Weight. The radar shall weigh less than 45 kg.
```

0.7 to 3.3 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.6 a | In each mode the radar shall operate as follows: a. Search: the radar shall detect targets of 1 m2 at 100 km range and | |
| 3.2.6 b | In each mode the radar shall operate as follows: b. Track: maintain track on up to 64 targets with an update rate of 1 Hz and | |
| 3.2.6 c | In each mode the radar shall operate as follows: c. Identify: interrogate each target in track using Mode 5 IFF. | |
| 3.2.7 | The radar shall weigh less than 45 kg. | |

# TAB-36  Long labels ('Mean time between failures:')

Lists on tab stops | document: documents/TAB-36_long_labels_mean_time_between_failures.pdf | typewriter 11 pt, 300 dpi

**Situation**

Four items whose labels have three to four words ('Mean time between failures:', 'Built-in test coverage:').

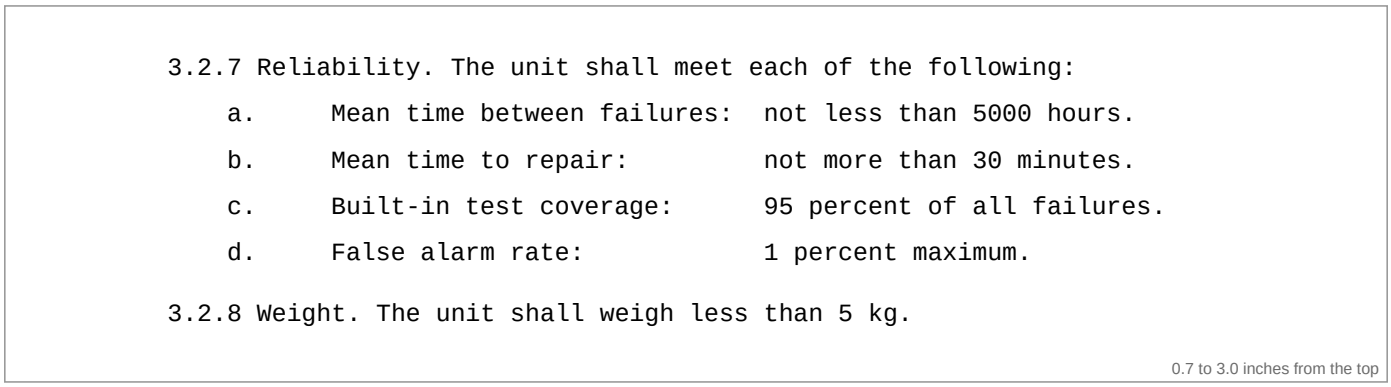
**What the tool must do**

Long labels are labels: one row per item.

**Source**

Review round 2 (tab-list attack).

**The page (as the scan document has it)**



```
3.2.7 Reliability. The unit shall meet each of the following:
    a.     Mean time between failures:  not less than 5000 hours.
    b.     Mean time to repair:         not more than 30 minutes.
    c.     Built-in test coverage:      95 percent of all failures.
    d.     False alarm rate:            1 percent maximum.
3.2.8 Weight. The unit shall weigh less than 5 kg.
```

0.7 to 3.0 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.7 a | The unit shall meet each of the following: a. Mean time between failures: not less than 5000 hours. | |
| 3.2.7 b | The unit shall meet each of the following: b. Mean time to repair: not more than 30 minutes. | |
| 3.2.7 c | The unit shall meet each of the following: c. Built-in test coverage: 95 percent of all failures. | |
| 3.2.7 d | The unit shall meet each of the following: d. False alarm rate: 1 percent maximum. | |
| 3.2.8 | The unit shall weigh less than 5 kg. | |

# TAB-37  A value that wraps back under the label column

Lists on tab stops | document: documents/TAB-37_a_value_that_wraps_back_under_the_label_column.pdf | typewriter 11 pt, 300 dpi

### Situation

'a. Operating temperature: -40 C to +71 C with the unit powered' and the next line starting under the LABEL: 'and cooling air supplied at the rate given in Figure 3.'

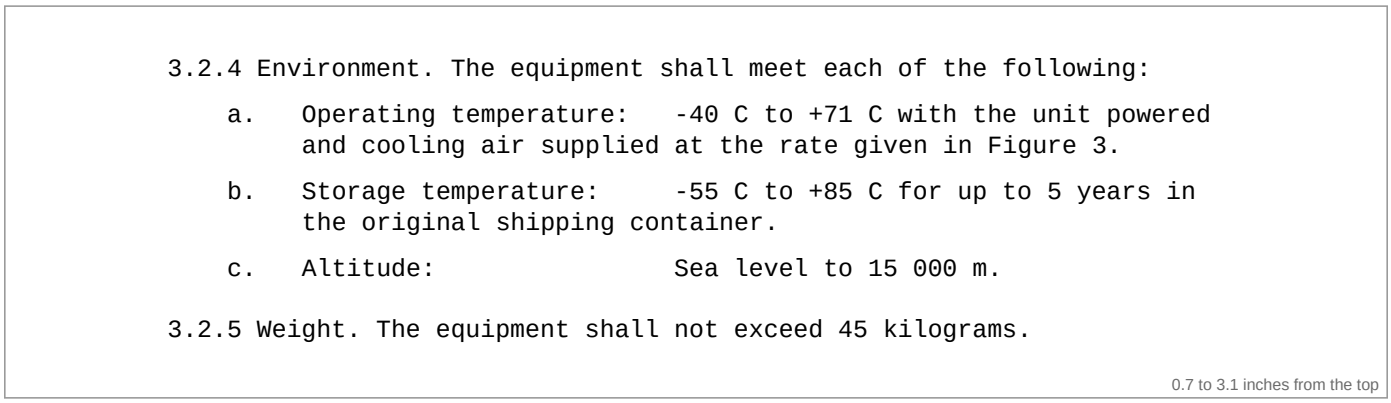
### What the tool must do

The value is read in order: 'a. Operating temperature: -40 C to +71 C with the unit powered and cooling air supplied at the rate given in Figure 3.' (round 1: the second value line was placed first).

### Source

Review round 1, finding layout/F4.

### The page (as the scan document has it)



```
3.2.4 Environment. The equipment shall meet each of the following:

    a.   Operating temperature:   -40 C to +71 C with the unit powered
         and cooling air supplied at the rate given in Figure 3.

    b.   Storage temperature:     -55 C to +85 C for up to 5 years in
         the original shipping container.

    c.   Altitude:                Sea level to 15 000 m.

3.2.5 Weight. The equipment shall not exceed 45 kilograms.
```

0.7 to 3.1 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The equipment shall meet each of the following: a. Operating temperature: -40 C to +71 C with the unit powered and cooling air supplied at the rate given in Figure 3. | |
| 3.2.4 b | The equipment shall meet each of the following: b. Storage temperature: -55 C to +85 C for up to 5 years in the original shipping container. | |
| 3.2.4 c | The equipment shall meet each of the following: c. Altitude: Sea level to 15 000 m. | |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

# TAB-38  A value that wraps back under the label column, proportional font

Lists on tab stops | document: documents/TAB-38_a_value_that_wraps_back_under_the_label_column_proportional_.pdf | proportional 11 pt, 300 dpi

### Situation

TAB-37 in a proportional (serif) font.

### What the tool must do

As TAB-37: three items, each with its whole value.

### Source

Review round 1, finding layout/F4.

**KNOWN LIMITATION** - the tool still gets this case wrong: In the proportional font the labels and values of items b and c (from 'Storage temperature:' down) are taken as a two-column table: item b's row keeps only its marker, without its letter in the Doc Ref ('3.2.4 | ... b.'), and item c has no row. The mono page (TAB-37) is read correctly.

### The page (as the scan document has it)

3.2.4 Environment. The equipment shall meet each of the following:

a. Operating temperature: -40 C to +71 C with the unit powered and cooling air supplied at the rate given in Figure 3.

b. Storage temperature: -55 C to +85 C for up to 5 years in the original shipping container.

c. Altitude: Sea level to 15 000 m.

3.2.5 Weight. The equipment shall not exceed 45 kilograms.

0.7 to 3.1 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The equipment shall meet each of the following: a. Operating temperature: -40 C to +71 C with the unit powered and cooling air supplied at the rate given in Figure 3. | |
| 3.2.4 b | The equipment shall meet each of the following: b. Storage temperature: -55 C to +85 C for up to 5 years in the original shipping container. | |
| 3.2.4 c | The equipment shall meet each of the following: c. Altitude: Sea level to 15 000 m. | |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

# TAB-39 A sentence under the label column after a two-line value

Lists on tab stops | document: documents/TAB-39_a_sentence_under_the_label_column_after_a_two_line_value.pdf | typewriter 11 pt, 300 dpi

### Situation

Item a: 'Acceleration:' with a two-line value, then a sentence starting under the LABEL: 'The unit shall be powered and operating during the test.'

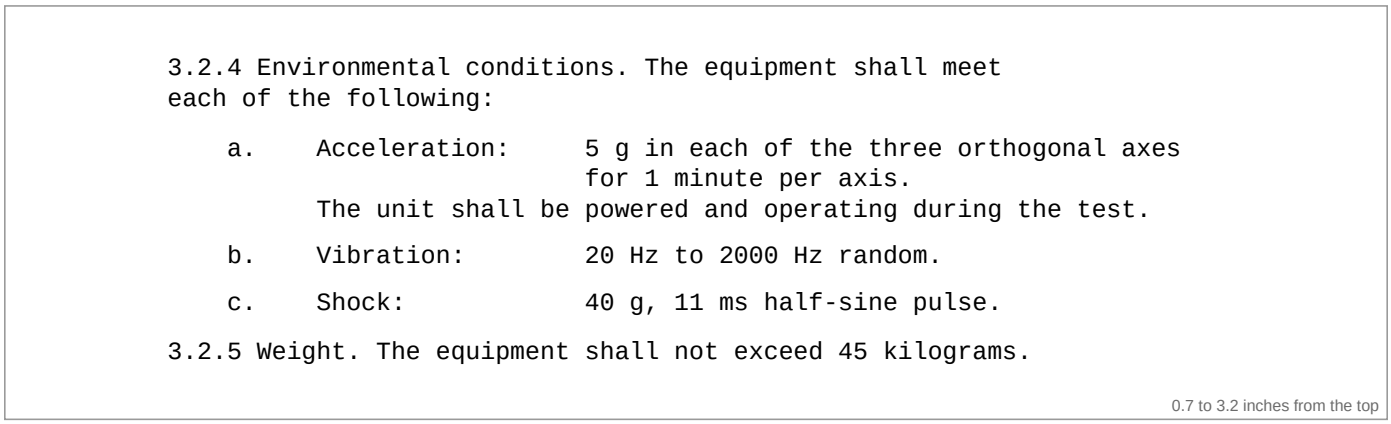
### What the tool must do

The sentence stays in item a, after its value, in reading order (round 1: the rows were reordered).

### Source

Review round 1, finding lists/F3.

### The page (as the scan document has it)



```
3.2.4 Environmental conditions. The equipment shall meet
each of the following:

    a.    Acceleration:     5 g in each of the three orthogonal axes
                            for 1 minute per axis.
          The unit shall be powered and operating during the test.

    b.    Vibration:        20 Hz to 2000 Hz random.

    c.    Shock:            40 g, 11 ms half-sine pulse.

3.2.5 Weight. The equipment shall not exceed 45 kilograms.
```

0.7 to 3.2 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The equipment shall meet each of the following: a. Acceleration: 5 g in each of the three orthogonal axes for 1 minute per axis. The unit shall be powered and operating during the test. | |
| 3.2.4 b | The equipment shall meet each of the following: b. Vibration: 20 Hz to 2000 Hz random. | |
| 3.2.4 c | The equipment shall meet each of the following: c. Shock: 40 g, 11 ms half-sine pulse. | |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

# TAB-40  A 'Note:' line inside a tabbed item

Lists on tab stops | document: documents/TAB-40_a_note_line_inside_a_tabbed_item.pdf | typewriter 11 pt, 300 dpi

### Situation

Item a: 'Acceleration: 5 g in each of the three axes.', then under the label 'Note: The unit shall be powered during this test.', then items b and c.

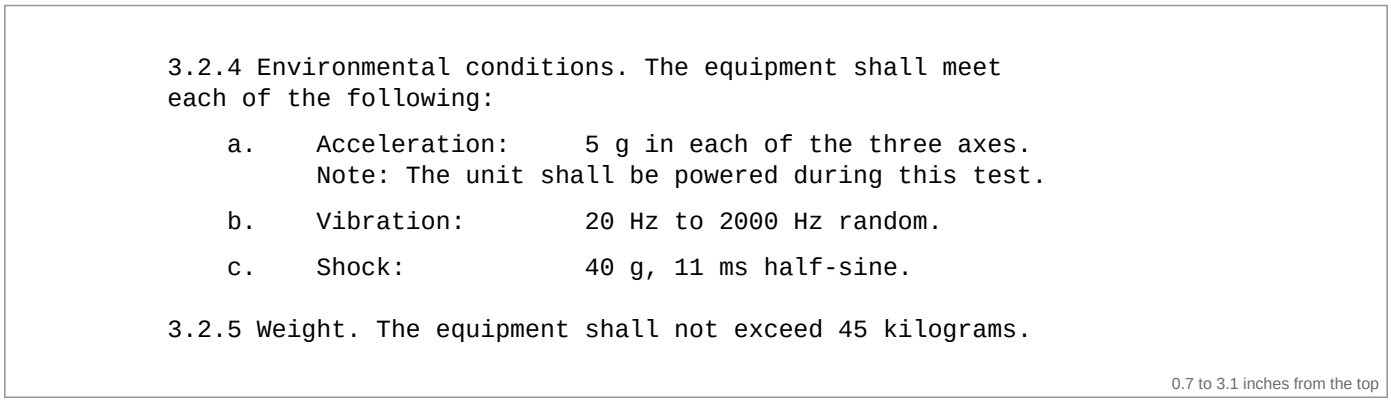
### What the tool must do

The note stays in item a; it is not given a marker of its own, and b and c keep their values (round 1: the note was split off and two rows got Doc Ref '3.2.4 b').

### Source

Review round 1, finding layout/F11.

### The page (as the scan document has it)



```
3.2.4 Environmental conditions. The equipment shall meet
each of the following:

    a.    Acceleration:     5 g in each of the three axes.
          Note: The unit shall be powered during this test.

    b.    Vibration:        20 Hz to 2000 Hz random.

    c.    Shock:            40 g, 11 ms half-sine.

3.2.5 Weight. The equipment shall not exceed 45 kilograms.
```

0.7 to 3.1 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The equipment shall meet each of the following: a. Acceleration: 5 g in each of the three axes. Note: The unit shall be powered during this test. | |
| 3.2.4 b | The equipment shall meet each of the following: b. Vibration: 20 Hz to 2000 Hz random. | |
| 3.2.4 c | The equipment shall meet each of the following: c. Shock: 40 g, 11 ms half-sine. | |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

# TAB-41 A 'Note:' line inside a tabbed item, proportional font

Lists on tab stops | document: documents/TAB-41_a_note_line_inside_a_tabbed_item_proportional_font.pdf | proportional 11 pt, 300 dpi

**Situation**

TAB-40 in a proportional (serif) font.

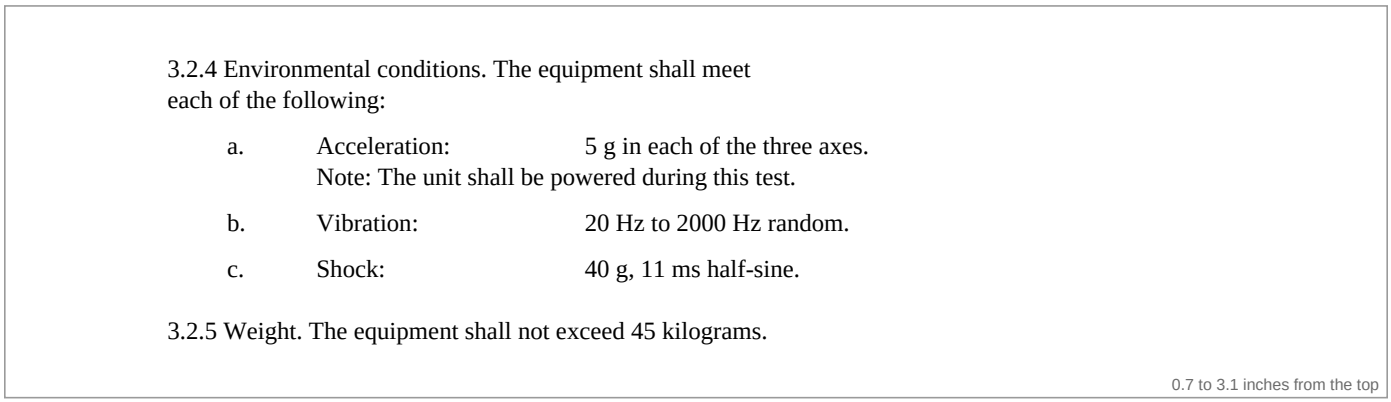
**What the tool must do**

As TAB-40.

**Source**

Review round 1, finding layout/F11.

**The page (as the scan document has it)**



3.2.4 Environmental conditions. The equipment shall meet each of the following:

a. Acceleration: 5 g in each of the three axes.
Note: The unit shall be powered during this test.

b. Vibration: 20 Hz to 2000 Hz random.

c. Shock: 40 g, 11 ms half-sine.

3.2.5 Weight. The equipment shall not exceed 45 kilograms.

0.7 to 3.1 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The equipment shall meet each of the following: a. Acceleration: 5 g in each of the three axes. Note: The unit shall be powered during this test. | |
| 3.2.4 b | The equipment shall meet each of the following: b. Vibration: 20 Hz to 2000 Hz random. | |
| 3.2.4 c | The equipment shall meet each of the following: c. Shock: 40 g, 11 ms half-sine. | |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

# TAB-42 Sub-labels without markers under 'a. Vibration: as follows:'

Lists on tab stops | document: documents/TAB-42_sub_labels_without_markers_under_a_vibration_as_follows.pdf | typewriter 11 pt, 300 dpi

**Situation**

'a. Vibration: as follows:' then, in the label column with no marker, 'Random: 20 Hz ...' and 'Sine: 5 g peak ...', then b. and c.

**What the tool must do**

The sub-labels belong to a (two rows '3.2.4 a'); they are not given made-up letters of their own (round 1: duplicate refs, parent lost).

**Source**

Review round 1, finding lists/F4.

**The page (as the scan document has it)**

```
3.2.4 Environmental conditions. The equipment shall meet
each of the following:

    a.    Vibration:        as follows:
          Random:           20 Hz to 2000 Hz, 0.04 g2/Hz.
          Sine:             5 g peak, 5 Hz to 500 Hz.

    b.    Shock:            40 g, 11 ms half-sine pulse.

    c.    Temperature:      -40 C to +71 C operating.

3.2.5 Weight. The equipment shall not exceed 45 kilograms.
```

0.7 to 3.2 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The equipment shall meet each of the following: a. Vibration: as follows: Random: 20 Hz to 2000 Hz, 0.04 g2/Hz. | |
| 3.2.4 a | The equipment shall meet each of the following: a. Vibration: as follows: Sine: 5 g peak, 5 Hz to 500 Hz. | |
| 3.2.4 b | The equipment shall meet each of the following: b. Shock: 40 g, 11 ms half-sine pulse. | |
| 3.2.4 c | The equipment shall meet each of the following: c. Temperature: -40 C to +71 C operating. | |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

# TAB-43  A value that wraps back under a label without a colon, then a sub-list

Lists on tab stops | document: documents/TAB-43_a_value_that_wraps_back_under_a_label_without_a_colon_then_a.pdf | typewriter 11 pt, 300 dpi

### Situation

'a. Vibration 20 Hz to 2000 Hz random, applied in', then under the label 'each of three axes as follows:', then '(1) Level: ...' and '(2) Duration: ...', then 'b. Shock 40 g ...'.

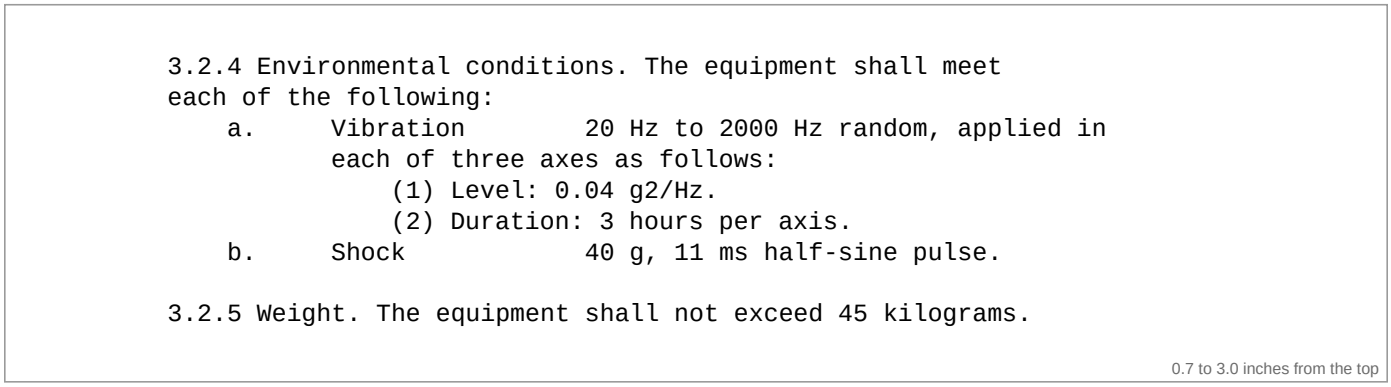
### What the tool must do

The value stays in order ('... applied in each of three axes as follows:') and the sub-items are filed under a: 3.2.4 a.1 and a.2 (round 3: the value line was taken for the end of a wrapped label, the text was reordered and the sub-items lost their parent).

### Source

Review round 3, finding layout/F7.

### The page (as the scan document has it)



```
3.2.4 Environmental conditions. The equipment shall meet
each of the following:
    a.     Vibration        20 Hz to 2000 Hz random, applied in
           each of three axes as follows:
               (1) Level: 0.04 g2/Hz.
               (2) Duration: 3 hours per axis.
    b.     Shock            40 g, 11 ms half-sine pulse.

3.2.5 Weight. The equipment shall not exceed 45 kilograms.
```

0.7 to 3.0 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a.1 | The equipment shall meet each of the following: a. Vibration 20 Hz to 2000 Hz random, applied in each of three axes as follows: (1) Level: 0.04 g2/Hz. | |
| 3.2.4 a.2 | The equipment shall meet each of the following: a. Vibration 20 Hz to 2000 Hz random, applied in each of three axes as follows: (2) Duration: 3 hours per axis. | |
| 3.2.4 b | The equipment shall meet each of the following: b. Shock 40 g, 11 ms half-sine pulse. | |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

# TAB-44  A value '1.5 V peak to peak. Measured at the' ...

Lists on tab stops | document: documents/TAB-44_a_value_1_5_v_peak_to_peak_measured_at_the.pdf | typewriter 11 pt, 300 dpi

### Situation

'b. Ripple: 1.5 V peak to peak. Measured at the' / 'output terminals.' between two one-line items.

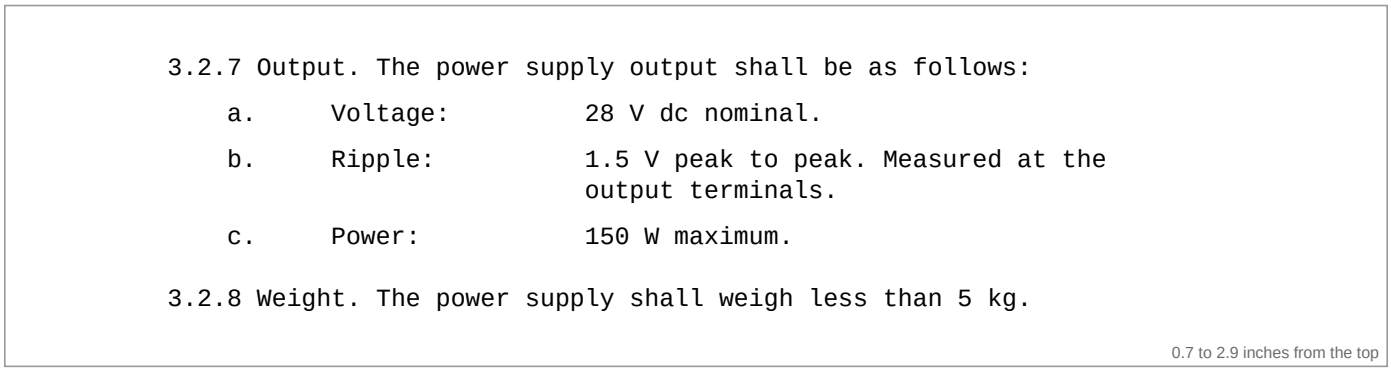
### What the tool must do

'1.5 V ...' is a quantity: it does not start paragraph 1.5, cut the list into columns or end the list (round 2: the values were attached to the wrong items and 3.2.7 c appeared twice).

### Source

Review round 2, finding extraction/F2.

### The page (as the scan document has it)



```
3.2.7 Output. The power supply output shall be as follows:

    a.     Voltage:         28 V dc nominal.

    b.     Ripple:          1.5 V peak to peak. Measured at the
                            output terminals.

    c.     Power:           150 W maximum.

3.2.8 Weight. The power supply shall weigh less than 5 kg.
```

0.7 to 2.9 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.7 a | The power supply output shall be as follows: a. Voltage: 28 V dc nominal. | |
| 3.2.7 b | The power supply output shall be as follows: b. Ripple: 1.5 V peak to peak. Measured at the output terminals. | |
| 3.2.7 c | The power supply output shall be as follows: c. Power: 150 W maximum. | |
| 3.2.8 | The power supply shall weigh less than 5 kg. | |

# TAB-45  A value line '2.5 G rms overall. Duration 3 hours per axis.'

Lists on tab stops | document: documents/TAB-45_a_value_line_2_5_g_rms_overall_duration_3_hours_per_axis.pdf | typewriter 11 pt, 300 dpi

**Situation**

Item b's value has a second line that starts with a decimal quantity and a capital unit: '2.5 G rms overall. Duration 3 hours per axis.'

**What the tool must do**

The line stays in item b; '2.5 G' is not paragraph 2.5.

**Source**

Review round 2, finding extraction/F2.

**The page (as the scan document has it)**

```
3.2.4 Environment. The unit shall meet each of the following:

    a.     Acceleration:    5 g in each of the three axes.

    b.     Vibration:       20 Hz to 2000 Hz random.
                            2.5 G rms overall. Duration 3 hours
                            per axis.

    c.     Shock:           40 g, 11 ms half-sine.

    d.     Humidity:        95 percent, non-condensing.

3.2.5 Weight. The unit shall weigh less than 5 kg.
```

0.7 to 3.4 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The unit shall meet each of the following: a. Acceleration: 5 g in each of the three axes. | |
| 3.2.4 b | The unit shall meet each of the following: b. Vibration: 20 Hz to 2000 Hz random. 2.5 G rms overall. Duration 3 hours per axis. | |
| 3.2.4 c | The unit shall meet each of the following: c. Shock: 40 g, 11 ms half-sine. | |
| 3.2.4 d | The unit shall meet each of the following: d. Humidity: 95 percent, non-condensing. | |
| 3.2.5 | The unit shall weigh less than 5 kg. | |

# TAB-46  A nested (1)/(2) list inside a value that ends with ':'

Lists on tab stops | document: documents/TAB-46_a_nested_1_2_list_inside_a_value_that_ends_with.pdf | typewriter 11 pt, 300 dpi

### Situation

'a. Search: The search mode shall provide:' then '(1) Long range search to 150 km.' and '(2) Short range search to 50 km.' a little right of the value column.

### What the tool must do

The value's own lead-in ('The search mode shall provide:') leads into the sub-items: rows 3.2.4 a.1 and a.2.

### Source

Review round 1 (layout attack, control page).

### The page (as the scan document has it)

```
3.2.4 Modes. The radar shall provide each of the following modes:

    a.   Search:        The search mode shall provide:
                           (1) Long range search to 150 km.
                           (2) Short range search to 50 km.

    b.   Track:         Track while scan of 200 targets.

    c.   Map:           Real beam ground mapping to
                        80 km range.

3.2.5 Weight. The radar shall not exceed 45 kilograms.
```

0.7 to 3.3 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a.1 | The radar shall provide each of the following modes: a. Search: The search mode shall provide: (1) Long range search to 150 km. | |
| 3.2.4 a.2 | The radar shall provide each of the following modes: a. Search: The search mode shall provide: (2) Short range search to 50 km. | |
| 3.2.4 b | The radar shall provide each of the following modes: b. Track: Track while scan of 200 targets. | |
| 3.2.4 c | The radar shall provide each of the following modes: c. Map: Real beam ground mapping to 80 km range. | |
| 3.2.5 | The radar shall not exceed 45 kilograms. | |

# TAB-47  Each label on its own line, its text on the lines below

Lists on tab stops | document: documents/TAB-47_each_label_on_its_own_line_its_text_on_the_lines_below.pdf | typewriter 11 pt, 300 dpi

**Situation**

'a. Operator display:' with nothing beside it, then the item's two-line text below, starting under the label.

**What the tool must do**

The text belongs to its label's item: '3.2.10 a | ... a. Operator display: The display shall present ...'.

**Source**

Review round 2 (tab-list attack).

**The page (as the scan document has it)**

```
3.2.10 Displays. The system shall provide the following displays:

    a.     Operator display:
           The display shall present the current mode, the
           range scale and all tracks in the selected sector.

    b.     Maintainer display:
           The display shall present the current mode, the
           range scale and all tracks in the selected sector.

3.2.11 Weight. The system shall weigh less than 45 kg.
```

0.7 to 3.2 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.10 a | The system shall provide the following displays: a. Operator display: The display shall present the current mode, the range scale and all tracks in the selected sector. | |
| 3.2.10 b | The system shall provide the following displays: b. Maintainer display: The display shall present the current mode, the range scale and all tracks in the selected sector. | |
| 3.2.11 | The system shall weigh less than 45 kg. | |

# TAB-48  A whole page typed double-spaced

Lists on tab stops | document: documents/TAB-48_a_whole_page_typed_double_spaced.pdf | typewriter 11 pt, 300 dpi

### Situation

A double-spaced typewritten page: the lead-in, a list whose values wrap (a blank line between every line), a paragraph, then 3.2.5.

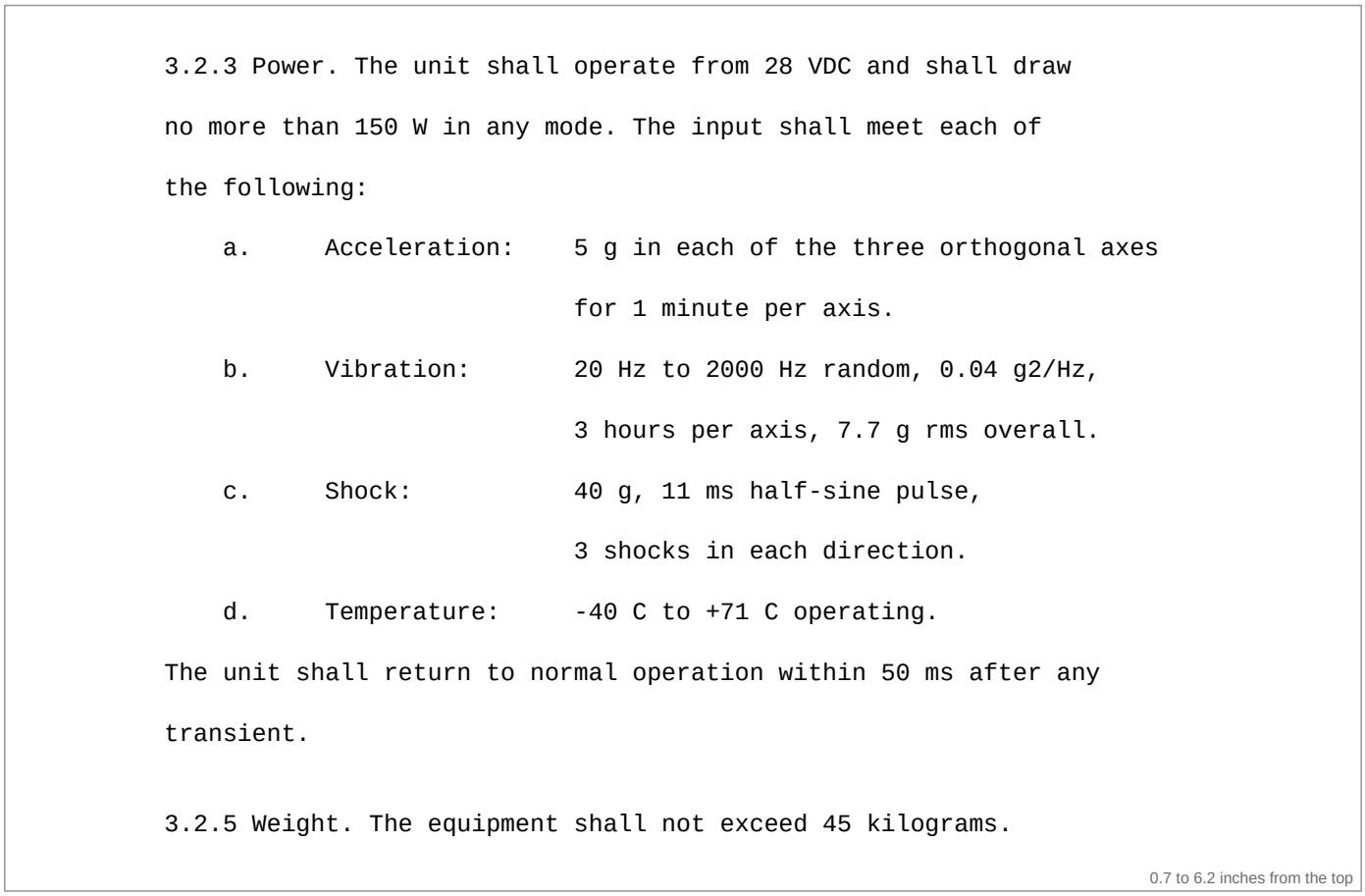
### What the tool must do

Double spacing is the page's normal spacing, not a blank line after every item: the values stay whole, and the paragraph after the list is a row of its own.

### Source

Review round 2 (tab-list attack).

### The page (as the scan document has it)



```
3.2.3 Power. The unit shall operate from 28 VDC and shall draw

no more than 150 W in any mode. The input shall meet each of

the following:

    a.     Acceleration:    5 g in each of the three orthogonal axes

                            for 1 minute per axis.

    b.     Vibration:       20 Hz to 2000 Hz random, 0.04 g2/Hz,

                            3 hours per axis, 7.7 g rms overall.

    c.     Shock:           40 g, 11 ms half-sine pulse,

                            3 shocks in each direction.

    d.     Temperature:     -40 C to +71 C operating.

The unit shall return to normal operation within 50 ms after any

transient.


3.2.5 Weight. The equipment shall not exceed 45 kilograms.
```

0.7 to 6.2 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.3 | The unit shall operate from 28 VDC and shall draw no more than 150 W in any mode. | |
| 3.2.3 a | The input shall meet each of the following: a. Acceleration: 5 g in each of the three orthogonal axes for 1 minute per axis. | |
| 3.2.3 b | The input shall meet each of the following: b. Vibration: 20 Hz to 2000 Hz random, 0.04 g2/Hz, 3 hours per axis, 7.7 g rms overall. | |
| 3.2.3 c | The input shall meet each of the following: c. Shock: 40 g, 11 ms half-sine pulse, 3 shocks in each direction. | |
| 3.2.3 d | The input shall meet each of the following: d. Temperature: -40 C to +71 C operating. | |
| (empty) | The unit shall return to normal operation within 50 ms after any transient. | |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

# TAB-49  A second paragraph of a value, after a blank line, in the value column

Lists on tab stops | document: documents/TAB-49_a_second_paragraph_of_a_value_after_a_blank_line_in_the_valu.pdf | typewriter 11 pt, 300 dpi

### Situation

Each item: its value, a blank line, then a second sentence in the VALUE column ('The unit shall operate during the test.').

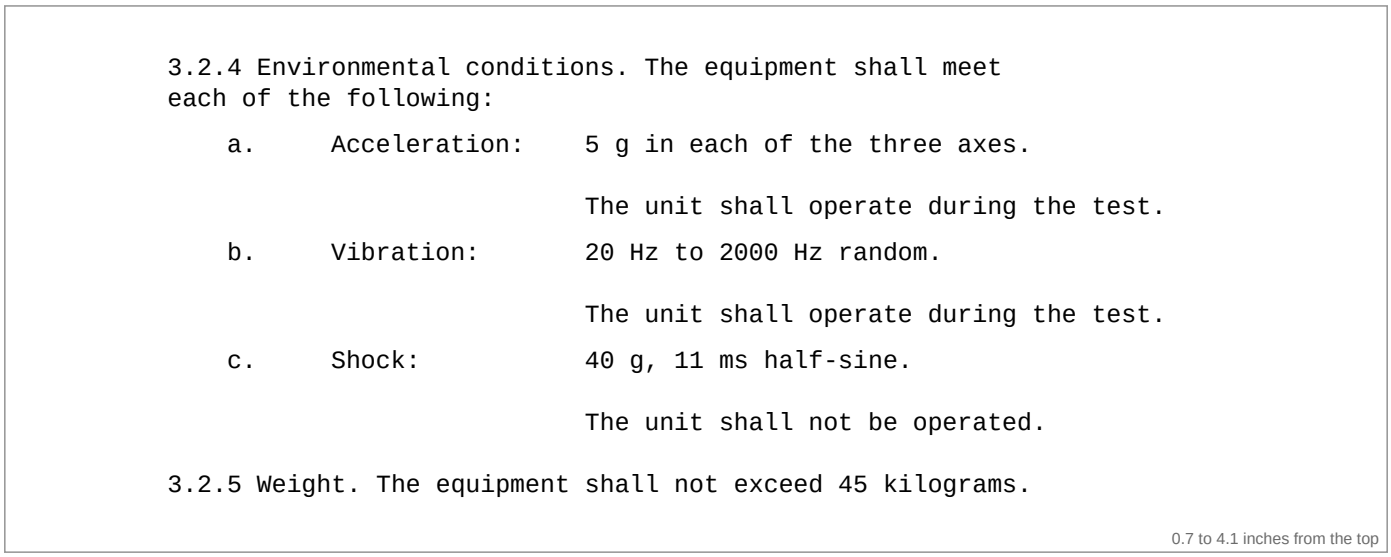
### What the tool must do

A paragraph in the value column after a blank line is still part of that item's value.

### Source

Review round 2 (tab-list attack).

### The page (as the scan document has it)



```
3.2.4 Environmental conditions. The equipment shall meet
each of the following:

    a.     Acceleration:    5 g in each of the three axes.

                            The unit shall operate during the test.

    b.     Vibration:       20 Hz to 2000 Hz random.

                            The unit shall operate during the test.

    c.     Shock:           40 g, 11 ms half-sine.

                            The unit shall not be operated.

3.2.5 Weight. The equipment shall not exceed 45 kilograms.
```

0.7 to 4.1 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The equipment shall meet each of the following: a. Acceleration: 5 g in each of the three axes. The unit shall operate during the test. | |
| 3.2.4 b | The equipment shall meet each of the following: b. Vibration: 20 Hz to 2000 Hz random. The unit shall operate during the test. | |
| 3.2.4 c | The equipment shall meet each of the following: c. Shock: 40 g, 11 ms half-sine. The unit shall not be operated. | |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

# TAB-50  A value that ends with a colon of its own

Lists on tab stops | document: documents/TAB-50_a_value_that_ends_with_a_colon_of_its_own.pdf | typewriter 11 pt, 300 dpi

### Situation

'a. Output voltage: The output voltage shall be:' / '28 VDC +/- 2 V at the connector.'

### What the tool must do

A colon at the end of a value line does not turn the item into the lead-in of a sub-list; the next value line is more of the same value.

### Source

Review round 2 (tab-list attack).

### The page (as the scan document has it)

```
3.2.9 Power supply. The power supply shall provide the following:

    a.     Output voltage:  The output voltage shall be:
                            28 VDC +/- 2 V at the connector.

    b.     Output current:  5 A maximum.

    c.     Efficiency:      85 percent minimum.

3.2.10 Weight. The power supply shall weigh less than 5 kg.
```

0.7 to 2.9 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.9 a | The power supply shall provide the following: a. Output voltage: The output voltage shall be: 28 VDC +/- 2 V at the connector. | |
| 3.2.9 b | The power supply shall provide the following: b. Output current: 5 A maximum. | |
| 3.2.9 c | The power supply shall provide the following: c. Efficiency: 85 percent minimum. | |
| 3.2.10 | The power supply shall weigh less than 5 kg. | |

# TAB-51  Contents of a report: 'Title:', 'Procedure:', 'Remarks:', 'Signature:'

Lists on tab stops | document: documents/TAB-51_contents_of_a_report_title_procedure_remarks_signature.pdf | typewriter 11 pt, 300 dpi

**Situation**

'4.3 Test report. Each test report shall contain the following:' then a. Title:, b. Procedure:, c. Remarks:, d. Signature: with lower-case values.

**What the tool must do**

Four items. Words such as 'Remarks:' are labels of this list, not notes.

**Source**

Review round 2 (tab-list attack).

**The page (as the scan document has it)**

```
4.3 Test report. Each test report shall contain the following:
    a.    Title:          the title of the test.
    b.    Procedure:      the procedure used.
    c.    Remarks:        any deviation from the procedure.
    d.    Signature:      the signature of the inspector.

4.4 Records. The contractor shall keep test reports for five years.
```

0.7 to 3.0 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 4.3 a | Each test report shall contain the following: a. Title: the title of the test. | |
| 4.3 b | Each test report shall contain the following: b. Procedure: the procedure used. | |
| 4.3 c | Each test report shall contain the following: c. Remarks: any deviation from the procedure. | |
| 4.3 d | Each test report shall contain the following: d. Signature: the signature of the inspector. | |
| 4.4 | The contractor shall keep test reports for five years. | |

# TAB-52  An item whose value is empty

Lists on tab stops | document: documents/TAB-52_an_item_whose_value_is_empty.pdf | typewriter 11 pt, 300 dpi

### Situation

'b. Vibration:' with nothing beside it and nothing under it, between complete items.

### What the tool must do

b is still an item of its own ('b. Vibration:'); c and d keep their values.

### Source

Review round 2 (tab-list attack).

### The page (as the scan document has it)

```
3.2.4 Environmental conditions. The equipment shall meet
each of the following:

    a.     Acceleration:    5 g in each of the three orthogonal axes
                            for 1 minute per axis.

    b.     Vibration:

    c.     Shock:           40 g, 11 ms half-sine pulse,
                            3 shocks in each direction.

    d.     Temperature:     -40 C to +71 C operating.

3.2.5 Weight. The equipment shall not exceed 45 kilograms.
```

0.7 to 3.6 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The equipment shall meet each of the following: a. Acceleration: 5 g in each of the three orthogonal axes for 1 minute per axis. | |
| 3.2.4 b | The equipment shall meet each of the following: b. Vibration: | |
| 3.2.4 c | The equipment shall meet each of the following: c. Shock: 40 g, 11 ms half-sine pulse, 3 shocks in each direction. | |
| 3.2.4 d | The equipment shall meet each of the following: d. Temperature: -40 C to +71 C operating. | |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

# TAB-53  Labels without colons

Lists on tab stops | document: documents/TAB-53_labels_without_colons.pdf | typewriter 11 pt, 300 dpi

### Situation

The list of TAB-01's shape with every label's colon left out ('Acceleration', 'Vibration').

### What the tool must do

Still one row per item with its label and value.

### Source

Review round 2 (tab-list attack).

### The page (as the scan document has it)

```
3.2.4 Environmental conditions. The equipment shall meet
each of the following:

    a.     Acceleration     5 g in each of the three orthogonal axes
                            for 1 minute per axis.

    b.     Vibration        20 Hz to 2000 Hz random, 0.04 g2/Hz,
                            3 hours per axis, 7.7 g rms overall.

    c.     Shock            40 g, 11 ms half-sine pulse,
                            3 shocks in each direction.

    d.     Temperature      -40 C to +71 C operating.

3.2.5 Weight. The equipment shall not exceed 45 kilograms.
```

0.7 to 3.8 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The equipment shall meet each of the following: a. Acceleration 5 g in each of the three orthogonal axes for 1 minute per axis. | |
| 3.2.4 b | The equipment shall meet each of the following: b. Vibration 20 Hz to 2000 Hz random, 0.04 g2/Hz, 3 hours per axis, 7.7 g rms overall. | |
| 3.2.4 c | The equipment shall meet each of the following: c. Shock 40 g, 11 ms half-sine pulse, 3 shocks in each direction. | |
| 3.2.4 d | The equipment shall meet each of the following: d. Temperature -40 C to +71 C operating. | |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

# TAB-54  A value line '12.5 V. The auxiliary output shall be' after a finished value line

Lists on tab stops | document: documents/TAB-54_a_value_line_12_5_v_the_auxiliary_output_shall_be_after_a_fi.pdf | typewriter 11 pt, 300 dpi

### Situation

'a. Main output: The output shall be 28 VDC nominal.' then '12.5 V. The auxiliary output shall be' / 'regulated to 1 percent.' in the value column; item b's value wraps back under its label.

### What the tool must do

'12.5 V.' is a quantity, not paragraph 12.5: it stays in item a.

### Source

Review round 2 (tab-list attack).

### The page (as the scan document has it)

```
        3.2.9 Power supply. The power supply shall provide the following:

            a.     Main output:     The output shall be 28 VDC nominal.
                                    12.5 V. The auxiliary output shall be
                                    regulated to 1 percent.

            b.     Efficiency:      85 percent minimum at full load with
                   the cooling air at the temperature given in Table II.

        3.2.10 Weight. The power supply shall weigh less than 5 kg.
```

0.7 to 3.0 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.9 a | The power supply shall provide the following: a. Main output: The output shall be 28 VDC nominal. 12.5 V. The auxiliary output shall be regulated to 1 percent. | |
| 3.2.9 b | The power supply shall provide the following: b. Efficiency: 85 percent minimum at full load with the cooling air at the temperature given in Table II. | |
| 3.2.10 | The power supply shall weigh less than 5 kg. | |

# List markers and list structure

List markers and list structure: markers that the scan lost or misread, lists that skip on purpose, nested lists, items that wrap back to the margin, sentences and notes between items, and a second lead-in inside a paragraph. LST-01 and LST-02 are the ordinary tight lists of the earlier 'list suite' (controls that must not change). The later cases are the real-OCR pages of the review rounds, one situation per document.

## LST-01  An ordinary tight lettered list (control)

List markers and list structure | document: documents/LST-01_an_ordinary_tight_lettered_list_control.pdf | typewriter 11 pt, 300 dpi

**Situation**

'a. Search mode.' / 'b. Track mode.' - a marker and one space, no tabs, no gap between items.

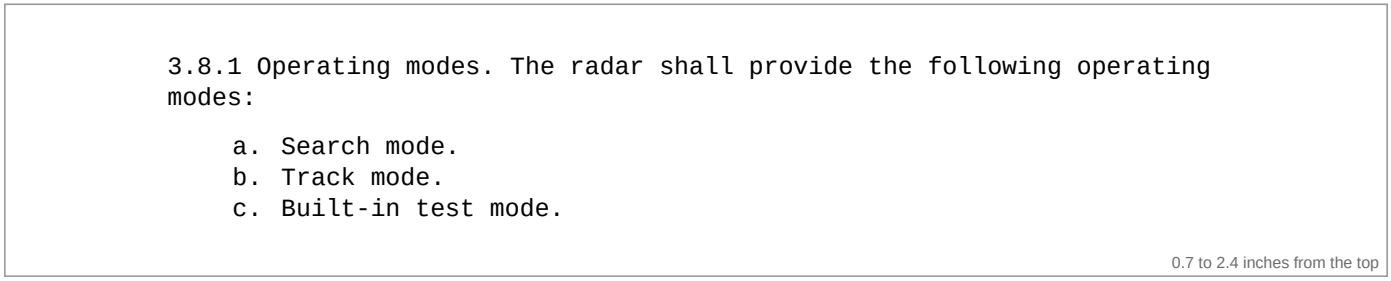
**What the tool must do**

The plainest list there is: one row per item under its lead-in. The tab-list rules must not change it.

**Source**

List suite p08 (control).

**The page (as the scan document has it)**



3.8.1 Operating modes. The radar shall provide the following operating modes:

a. Search mode.
b. Track mode.
c. Built-in test mode.

0.7 to 2.4 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.8.1 a | The radar shall provide the following operating modes: a. Search mode. | |
| 3.8.1 b | The radar shall provide the following operating modes: b. Track mode. | |
| 3.8.1 c | The radar shall provide the following operating modes: c. Built-in test mode. | |

# LST-02  An ordinary tight numbered list whose last item wraps (control)

List markers and list structure | document: documents/LST-02_an_ordinary_tight_numbered_list_whose_last_item_wraps_contro.pdf | typewriter 11 pt, 300 dpi

### Situation

'1.' to '4.' with one space after the marker, no gap between items; item 4 wraps onto a second line aligned with its text.

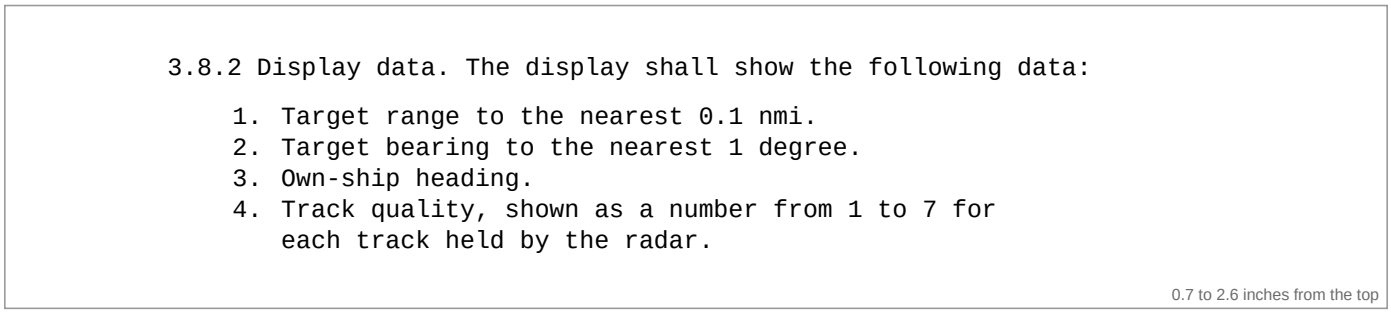
### What the tool must do

One row per item; the wrapped line stays in item 4. The tab-list rules must not change it.

### Source

List suite p08 (control).

### The page (as the scan document has it)



3.8.2 Display data. The display shall show the following data:

1. Target range to the nearest 0.1 nmi.
2. Target bearing to the nearest 1 degree.
3. Own-ship heading.
4. Track quality, shown as a number from 1 to 7 for each track held by the radar.

0.7 to 2.6 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.8.2 1 | The display shall show the following data: 1. Target range to the nearest 0.1 nmi. | |
| 3.8.2 2 | The display shall show the following data: 2. Target bearing to the nearest 1 degree. | |
| 3.8.2 3 | The display shall show the following data: 3. Own-ship heading. | |
| 3.8.2 4 | The display shall show the following data: 4. Track quality, shown as a number from 1 to 7 for each track held by the radar. | |

# LST-03  A lost marker: the item starts in the marker column

List markers and list structure | document: documents/LST-03_a_lost_marker_the_item_starts_in_the_marker_column.pdf | typewriter 11 pt, 300 dpi

### Situation

Items a., b., d. have their markers; the third item 'Map. Ground mapping at 10 m resolution.' has lost its 'c.' and starts where the markers are. A blank line between items.

### What the tool must do

The neighbours b. and d. confirm the missing letter: the row is '3.2.4 c' with 'c.' put back, and Review Reasons says the marker was taken from the sequence (the letter came from the tool, not the page).

### Source

Review round 2 (lists), real OCR page f8_marker_col.

### The page (as the scan document has it)

```
3.2.4 Modes. The radar shall provide the following modes:

     a. Search. Detection of targets out to 100 nmi.

     b. Track. Tracking of up to 20 targets.

     Map. Ground mapping at 10 m resolution.

     d. Test. Built-in test of all modules.

3.2.5 Weight. The radar shall weigh 45 kg.
```

0.7 to 3.4 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The radar shall provide the following modes: a. Search. Detection of targets out to 100 nmi. | |
| 3.2.4 b | The radar shall provide the following modes: b. Track. Tracking of up to 20 targets. | |
| 3.2.4 c | The radar shall provide the following modes: c. Map. Ground mapping at 10 m resolution. | review note: List marker missing |
| 3.2.4 d | The radar shall provide the following modes: d. Test. Built-in test of all modules. | |
| 3.2.5 | The radar shall weigh 45 kg. | |

# LST-04  A lost marker: the item starts in the text column

List markers and list structure | document: documents/LST-04_a_lost_marker_the_item_starts_in_the_text_column.pdf | typewriter 11 pt, 300 dpi

**Situation**

As LST-03, but 'Map. Ground mapping ...' starts where the TEXT of the items starts (under 'Search.').

**What the tool must do**

As LST-03: '3.2.4 c' with a review note. The item is not joined to b. as a second line.

**Source**

Review round 2 (lists), real OCR page f11_text_col.

**The page (as the scan document has it)**

```
        3.2.4 Modes. The radar shall provide the following modes:

             a. Search. Detection of targets out to 100 nmi.

             b. Track. Tracking of up to 20 targets.

                Map. Ground mapping at 10 m resolution.

             d. Test. Built-in test of all modules.

        3.2.5 Weight. The radar shall weigh 45 kg.
```

0.7 to 3.4 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The radar shall provide the following modes: a. Search. Detection of targets out to 100 nmi. | |
| 3.2.4 b | The radar shall provide the following modes: b. Track. Tracking of up to 20 targets. | |
| 3.2.4 c | The radar shall provide the following modes: c. Map. Ground mapping at 10 m resolution. | review note: List marker missing |
| 3.2.4 d | The radar shall provide the following modes: d. Test. Built-in test of all modules. | |
| 3.2.5 | The radar shall weigh 45 kg. | |

# LST-05  A lost marker in the marker column, between long paragraphs

List markers and list structure | document: documents/LST-05_a_lost_marker_in_the_marker_column_between_long_paragraphs.pdf | typewriter 11 pt, 300 dpi

### Situation

LST-03 with a four-line paragraph 3.2.3 above the list and a three-line paragraph 3.2.5 below it.

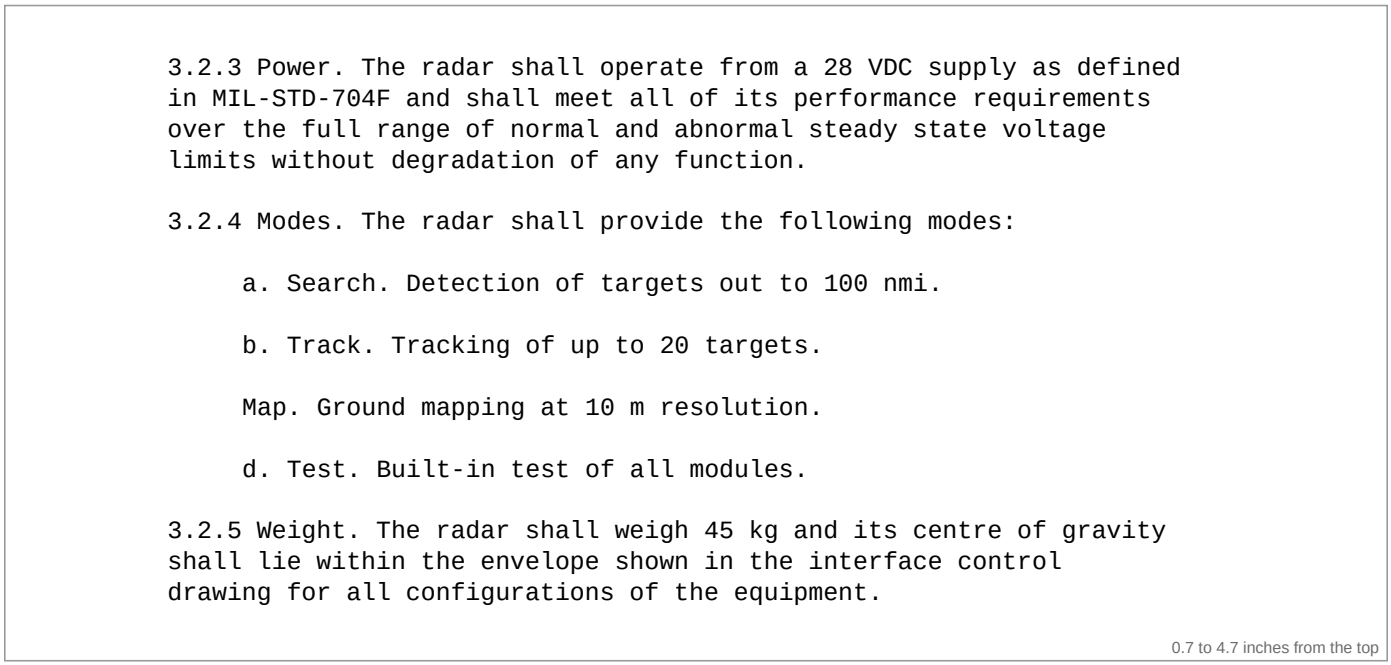
### What the tool must do

As LST-03. The long lines around the list must not change how the list is read.

### Source

Review round 2 (lists), real OCR page f8b_marker_col.

### The page (as the scan document has it)



3.2.3 Power. The radar shall operate from a 28 VDC supply as defined in MIL-STD-704F and shall meet all of its performance requirements over the full range of normal and abnormal steady state voltage limits without degradation of any function.

3.2.4 Modes. The radar shall provide the following modes:

a. Search. Detection of targets out to 100 nmi.

b. Track. Tracking of up to 20 targets.

Map. Ground mapping at 10 m resolution.

d. Test. Built-in test of all modules.

3.2.5 Weight. The radar shall weigh 45 kg and its centre of gravity shall lie within the envelope shown in the interface control drawing for all configurations of the equipment.

0.7 to 4.7 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.3 | The radar shall operate from a 28 VDC supply as defined in MIL-STD-704F and shall meet all of its performance requirements over the full range of normal and abnormal steady state voltage limits without degradation of any function. | |
| 3.2.4 a | The radar shall provide the following modes: a. Search. Detection of targets out to 100 nmi. | |
| 3.2.4 b | The radar shall provide the following modes: b. Track. Tracking of up to 20 targets. | |
| 3.2.4 c | The radar shall provide the following modes: c. Map. Ground mapping at 10 m resolution. | review note: List marker missing |
| 3.2.4 d | The radar shall provide the following modes: d. Test. Built-in test of all modules. | |
| 3.2.5 | The radar shall weigh 45 kg and its centre of gravity shall lie within the envelope shown in the interface control drawing for all configurations of the equipment. | |

# LST-06  A lost marker in the text column, between long paragraphs

List markers and list structure | document: documents/LST-06_a_lost_marker_in_the_text_column_between_long_paragraphs.pdf | typewriter 11 pt, 300 dpi

### Situation

LST-04 with a four-line paragraph 3.2.3 above the list and a three-line paragraph 3.2.5 below it.

### What the tool must do

As LST-04.

### Source

Review round 2 (lists), real OCR page f11b_text_col.

### The page (as the scan document has it)

```
3.2.3 Power. The radar shall operate from a 28 VDC supply as defined
in MIL-STD-704F and shall meet all of its performance requirements
over the full range of normal and abnormal steady state voltage
limits without degradation of any function.

3.2.4 Modes. The radar shall provide the following modes:

     a. Search. Detection of targets out to 100 nmi.

     b. Track. Tracking of up to 20 targets.

        Map. Ground mapping at 10 m resolution.

     d. Test. Built-in test of all modules.

3.2.5 Weight. The radar shall weigh 45 kg and its centre of gravity
shall lie within the envelope shown in the interface control
drawing for all configurations of the equipment.
```

0.7 to 4.7 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.3 | The radar shall operate from a 28 VDC supply as defined in MIL-STD-704F and shall meet all of its performance requirements over the full range of normal and abnormal steady state voltage limits without degradation of any function. | |
| 3.2.4 a | The radar shall provide the following modes: a. Search. Detection of targets out to 100 nmi. | |
| 3.2.4 b | The radar shall provide the following modes: b. Track. Tracking of up to 20 targets. | |
| 3.2.4 c | The radar shall provide the following modes: c. Map. Ground mapping at 10 m resolution. | review note: List marker missing |
| 3.2.4 d | The radar shall provide the following modes: d. Test. Built-in test of all modules. | |
| 3.2.5 | The radar shall weigh 45 kg and its centre of gravity shall lie within the envelope shown in the interface control drawing for all configurations of the equipment. | |

# LST-07  The first two markers lost in a tab list

List markers and list structure | document: documents/LST-07_the_first_two_markers_lost_in_a_tab_list.pdf | typewriter 11 pt, 300 dpi

### Situation

A list on tab stops whose first two items have lost their markers ('Acceleration: ...', 'Vibration: ...'); the third keeps 'c.'.

### What the tool must do

The markers are counted back from 'c.': rows a and b, each with a review note, then c.

### Source

Review round 2 (lists), real OCR page f10_twolost.

### The page (as the scan document has it)

```
3.2.4 Environmental conditions. The equipment shall meet
each of the following:

          Acceleration:     5 g in each of the three orthogonal axes
                            for 1 minute per axis.

          Vibration:        20 Hz to 2000 Hz random, 0.04 g2/Hz,
                            3 hours per axis, 7.7 g rms overall.

    c.    Shock:            40 g, 11 ms half-sine pulse,
                            3 shocks in each direction.

3.2.5 Weight. The equipment shall not exceed 45 kilograms.
```

0.7 to 3.5 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The equipment shall meet each of the following: a. Acceleration: 5 g in each of the three orthogonal axes for 1 minute per axis. | review note: List marker missing |
| 3.2.4 b | The equipment shall meet each of the following: b. Vibration: 20 Hz to 2000 Hz random, 0.04 g2/Hz, 3 hours per axis, 7.7 g rms overall. | review note: List marker missing |
| 3.2.4 c | The equipment shall meet each of the following: c. Shock: 40 g, 11 ms half-sine pulse, 3 shocks in each direction. | |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

# LST-08  The first two markers lost in a tab list, proportional font

List markers and list structure | document: documents/LST-08_the_first_two_markers_lost_in_a_tab_list_proportional_font.pdf | proportional 11 pt, 300 dpi

**Situation**

LST-07 in a proportional (serif) font.

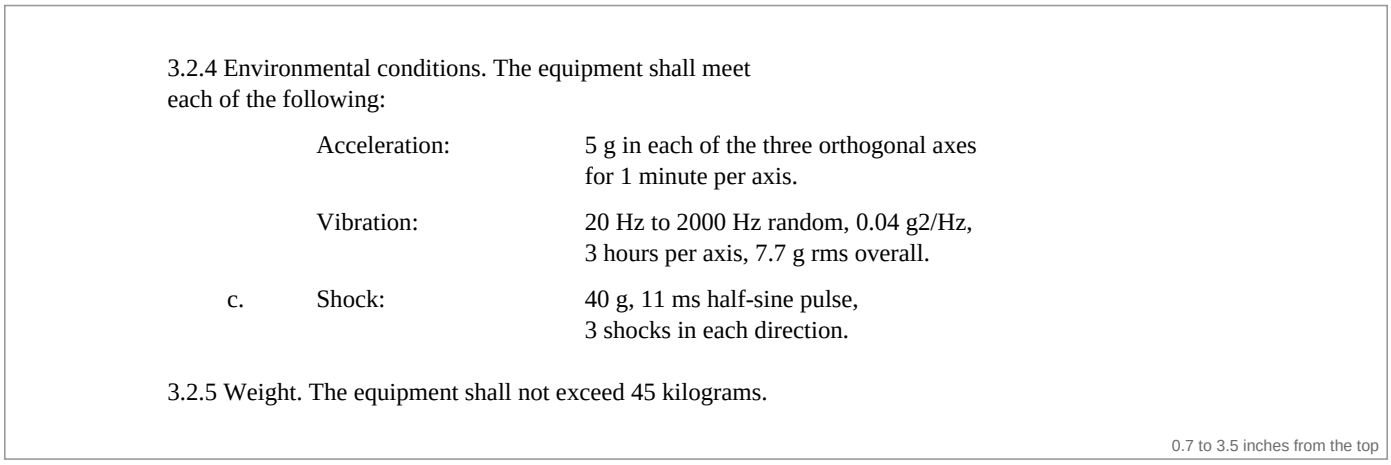
**What the tool must do**

As LST-07.

**Source**

Review round 2 (lists), real OCR page f10_twolost_serif.

**The page (as the scan document has it)**



3.2.4 Environmental conditions. The equipment shall meet each of the following:

| | | |
|---|---|---|
| | Acceleration: | 5 g in each of the three orthogonal axes for 1 minute per axis. |
| | Vibration: | 20 Hz to 2000 Hz random, 0.04 g2/Hz, 3 hours per axis, 7.7 g rms overall. |
| c. | Shock: | 40 g, 11 ms half-sine pulse, 3 shocks in each direction. |

3.2.5 Weight. The equipment shall not exceed 45 kilograms.

0.7 to 3.5 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The equipment shall meet each of the following: a. Acceleration: 5 g in each of the three orthogonal axes for 1 minute per axis. | review note: List marker missing |
| 3.2.4 b | The equipment shall meet each of the following: b. Vibration: 20 Hz to 2000 Hz random, 0.04 g2/Hz, 3 hours per axis, 7.7 g rms overall. | review note: List marker missing |
| 3.2.4 c | The equipment shall meet each of the following: c. Shock: 40 g, 11 ms half-sine pulse, 3 shocks in each direction. | |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

# LST-09  A figure caption between two items, and the next item's marker lost

List markers and list structure | document: documents/LST-09_a_figure_caption_between_two_items_and_the_next_item_s_marke.pdf | typewriter 11 pt, 300 dpi

### Situation

Items a. and b. on tab stops, then a figure caption 'Figure 3. Shock pulse shape.' in the middle of the list, then 'Shock: ...' without its marker, then d.

### What the tool must do

The caption is a Figure Caption item of its own and does not end the list; the item after it is '3.2.4 c' (marker taken from the sequence, with a review note), and d. follows.

### Source

Review round 2 (lists), real OCR page f9_caption.

### The page (as the scan document has it)

```
3.2.4 Environmental conditions. The equipment shall meet
each of the following:

    a.    Acceleration:     5 g in each of the three orthogonal axes
                            for 1 minute per axis.

    b.    Vibration:        20 Hz to 2000 Hz random, 0.04 g2/Hz,
                            3 hours per axis, 7.7 g rms overall.

                      Figure 3. Shock pulse shape.

          Shock:            40 g, 11 ms half-sine pulse,
                            3 shocks in each direction.

    d.    Temperature:      -40 C to +71 C operating.

3.2.5 Weight. The equipment shall not exceed 45 kilograms.
```

0.7 to 4.3 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The equipment shall meet each of the following: a. Acceleration: 5 g in each of the three orthogonal axes for 1 minute per axis. | |
| 3.2.4 b | The equipment shall meet each of the following: b. Vibration: 20 Hz to 2000 Hz random, 0.04 g2/Hz, 3 hours per axis, 7.7 g rms overall. | |
| 3.2.4 c | The equipment shall meet each of the following: c. Shock: 40 g, 11 ms half-sine pulse, 3 shocks in each direction. | review note: List marker missing |
| 3.2.4 d | The equipment shall meet each of the following: d. Temperature: -40 C to +71 C operating. | |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

Also expected: caption(s): FIGURE 3.

# LST-10  A lost marker in a single-spaced list

List markers and list structure | document: documents/LST-10_a_lost_marker_in_a_single_spaced_list.pdf | typewriter 11 pt, 300 dpi

### Situation

'a. Search mode.' / ' Track mode.' / 'c. Ground mapping mode.' with no gap between items: the second item has lost its 'b.' and starts where a wrapped line of 'a.' would.

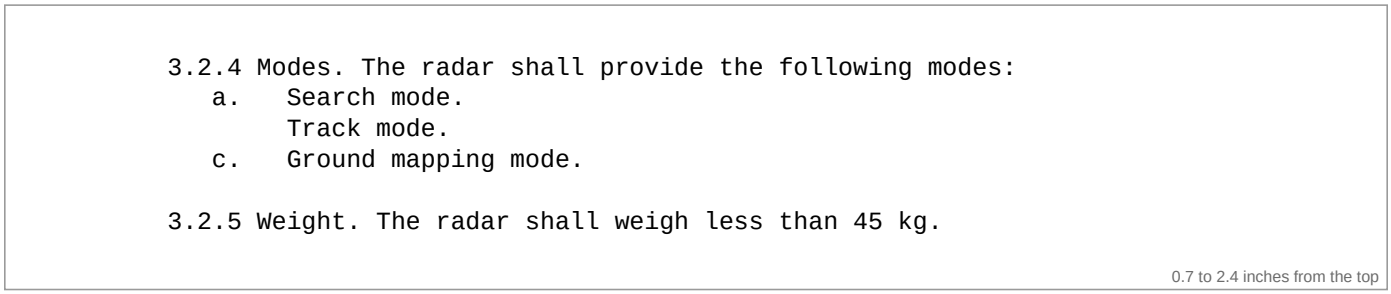
### What the tool must do

The markers skip exactly one letter (a. then c.) and one line in a's text column starts a new sentence after a finished one: it is split off as item b, with a review note.

### Source

Review round 3 (lists).

### The page (as the scan document has it)



```
3.2.4 Modes. The radar shall provide the following modes:
   a.   Search mode.
        Track mode.
   c.   Ground mapping mode.

3.2.5 Weight. The radar shall weigh less than 45 kg.
```

0.7 to 2.4 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The radar shall provide the following modes: a. Search mode. | |
| 3.2.4 b | The radar shall provide the following modes: b. Track mode. | review note: List marker missing |
| 3.2.4 c | The radar shall provide the following modes: c. Ground mapping mode. | |
| 3.2.5 | The radar shall weigh less than 45 kg. | |

# LST-11  A lost marker after an item of three lines

List markers and list structure | document: documents/LST-11_a_lost_marker_after_an_item_of_three_lines.pdf | typewriter 11 pt, 300 dpi

### Situation

Item a. wraps onto three lines; after a blank line 'Target range and bearing.' starts in a's text column (its 'b.' lost); then c.

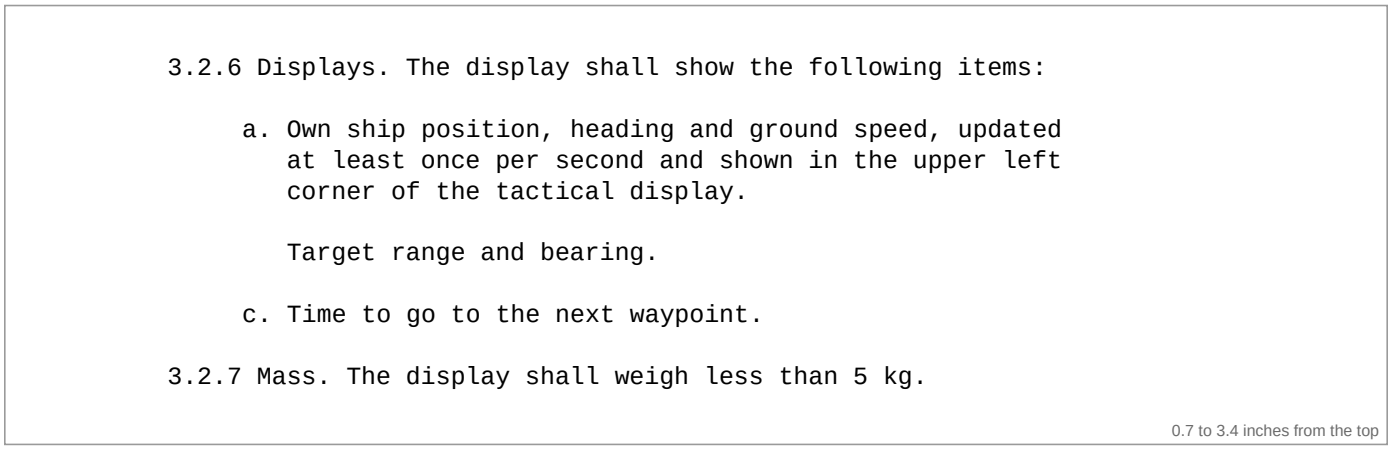
### What the tool must do

The blank line ends item a (its sentence is finished): the next paragraph is item b with its marker taken from the sequence - not a fourth line of a (round 2: a's line spacing made the blank line look like an ordinary line gap).

### Source

Review round 2, finding contstem/a4b; real OCR page para_after_long_item.

### The page (as the scan document has it)



```
3.2.6 Displays. The display shall show the following items:

     a. Own ship position, heading and ground speed, updated
        at least once per second and shown in the upper left
        corner of the tactical display.

        Target range and bearing.

     c. Time to go to the next waypoint.

3.2.7 Mass. The display shall weigh less than 5 kg.
```

0.7 to 3.4 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.6 a | The display shall show the following items: a. Own ship position, heading and ground speed, updated at least once per second and shown in the upper left corner of the tactical display. | |
| 3.2.6 b | The display shall show the following items: b. Target range and bearing. | review note: List marker missing |
| 3.2.6 c | The display shall show the following items: c. Time to go to the next waypoint. | |
| 3.2.7 | The display shall weigh less than 5 kg. | |

# LST-12  A marker misread as the next one ('c.', 'c.')

List markers and list structure | document: documents/LST-12_a_marker_misread_as_the_next_one_c_c.pdf | typewriter 11 pt, 300 dpi

### Situation

'a. Search mode, ...' with a second sentence on its own line, then 'c. Track mode.', 'c. Ground mapping mode.', 'd. Test mode.' - the scan read 'b.' as 'c.'.

### What the tool must do

Two 'c.' in a row, with d. after them: the first is a misread 'b.' (not a lost item): rows a, b (with a review note), c, d; item a keeps its second sentence.

### Source

Review round 3, finding V2 (misread split); real OCR page r5_v2_misread_split.

### The page (as the scan document has it)

```
3.2.4 Modes. The radar shall provide the following modes:

     a. Search mode, with a scan of 120 degrees in azimuth.
        The scan rate shall be 60 degrees per second.
     c. Track mode.
     c. Ground mapping mode.
     d. Test mode.

3.2.5 Weight. The radar shall weigh less than 45 kg.
```

0.7 to 3.0 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The radar shall provide the following modes: a. Search mode, with a scan of 120 degrees in azimuth. The scan rate shall be 60 degrees per second. | |
| 3.2.4 b | The radar shall provide the following modes: b. Track mode. | review note: List marker read as 'c.' |
| 3.2.4 c | The radar shall provide the following modes: c. Ground mapping mode. | |
| 3.2.4 d | The radar shall provide the following modes: d. Test mode. | |
| 3.2.5 | The radar shall weigh less than 45 kg. | |

# LST-13  A numbered item whose second sentence starts on its own line

List markers and list structure | document: documents/LST-13_a_numbered_item_whose_second_sentence_starts_on_its_own_line.pdf | typewriter 11 pt, 300 dpi

### Situation

'1. Search mode.' / '2. Track mode, holding up to twenty targets in track.' / 'Tracks shall be updated once per second.' (on its own line, in the text column) / '3. Ground mapping mode.'

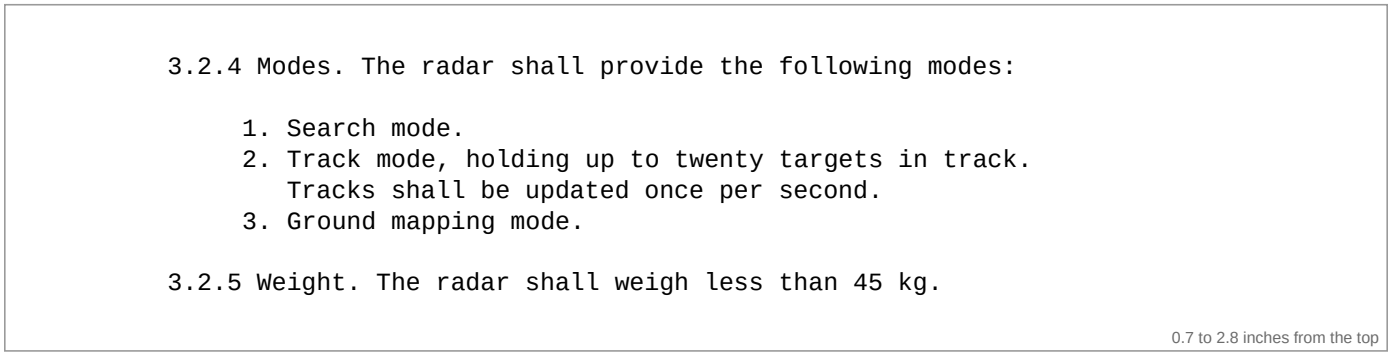
### What the tool must do

Item 2 keeps its second sentence and item 3 stays an item (round 3: item 3 became a section heading, was dropped from the list and became the Section of every later row). 3.2.5 has no Section.

### Source

Review round 3, finding B1; real OCR page r3_numbered_second_sentence.

### The page (as the scan document has it)



```
3.2.4 Modes. The radar shall provide the following modes:

     1. Search mode.
     2. Track mode, holding up to twenty targets in track.
        Tracks shall be updated once per second.
     3. Ground mapping mode.

3.2.5 Weight. The radar shall weigh less than 45 kg.
```

0.7 to 2.8 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 1 | The radar shall provide the following modes: 1. Search mode. | |
| 3.2.4 2 | The radar shall provide the following modes: 2. Track mode, holding up to twenty targets in track. Tracks shall be updated once per second. | |
| 3.2.4 3 | The radar shall provide the following modes: 3. Ground mapping mode. | |
| 3.2.5 | The radar shall weigh less than 45 kg. | Section: (empty) |

# LST-14  A numbered list that skips on purpose (1, 3, 5)

List markers and list structure | document: documents/LST-14_a_numbered_list_that_skips_on_purpose_1_3_5.pdf | typewriter 11 pt, 300 dpi

### Situation

'The following tests of TABLE I shall be repeated after vibration:' then '1. Visual examination.', '3. Insulation resistance, ...' (three lines, the last a sentence of its own), '5. Dielectric withstanding voltage.' - the test numbers of a table.

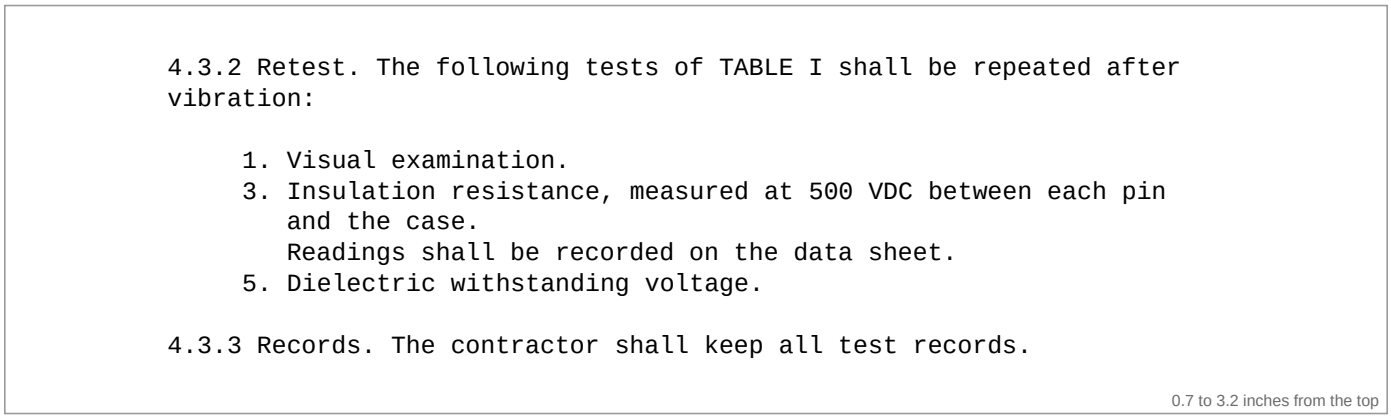
### What the tool must do

A list that counts in twos has no lost items: rows 1, 3 and 5 with their own numbers and no review note about markers; item 3 keeps all its lines.

### Source

Review round 3, finding V2 (skip split); real OCR page r4_v2_skip_split.

### The page (as the scan document has it)



```
4.3.2 Retest. The following tests of TABLE I shall be repeated after
vibration:

     1. Visual examination.
     3. Insulation resistance, measured at 500 VDC between each pin
        and the case.
        Readings shall be recorded on the data sheet.
     5. Dielectric withstanding voltage.

4.3.3 Records. The contractor shall keep all test records.
```

0.7 to 3.2 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 4.3.2 1 | The following tests of TABLE I shall be repeated after vibration: 1. Visual examination. | no list-marker note |
| 4.3.2 3 | The following tests of TABLE I shall be repeated after vibration: 3. Insulation resistance, measured at 500 VDC between each pin and the case. Readings shall be recorded on the data sheet. | no list-marker note |
| 4.3.2 5 | The following tests of TABLE I shall be repeated after vibration: 5. Dielectric withstanding voltage. | no list-marker note |
| 4.3.3 | The contractor shall keep all test records. | |

# LST-15  Capital letters with a small-letter list under each

List markers and list structure | document: documents/LST-15_capital_letters_with_a_small_letter_list_under_each.pdf | typewriter 11 pt, 300 dpi

### Situation

'A. Operating conditions:' with 'a. Temperature: ...' and 'b. Humidity: ...' indented under it, then 'B. Storage conditions:' with its own a. and b.

### What the tool must do

Each level is judged on its own: A. and B. keep their capitals (they are not misread 'a.'), and the rows are 3.2.4 A.a, A.b, B.a, B.b (round 3: the capitals were lower-cased).

### Source

Review round 3, finding F1; real OCR page r6_caps_with_sublists.

### The page (as the scan document has it)

```
3.2.4 Environment. The unit shall meet the following conditions:

     A. Operating conditions:
         a. Temperature: -40 C to +71 C.
         b. Humidity: 95 percent, non-condensing.
     B. Storage conditions:
         a. Temperature: -55 C to +85 C.
         b. Humidity: 100 percent, condensing.

3.2.5 Weight. The unit shall weigh less than 5 kg.
```

0.7 to 3.2 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 A.a | The unit shall meet the following conditions: A. Operating conditions: a. Temperature: -40 C to +71 C. | |
| 3.2.4 A.b | The unit shall meet the following conditions: A. Operating conditions: b. Humidity: 95 percent, non-condensing. | |
| 3.2.4 B.a | The unit shall meet the following conditions: B. Storage conditions: a. Temperature: -55 C to +85 C. | |
| 3.2.4 B.b | The unit shall meet the following conditions: B. Storage conditions: b. Humidity: 100 percent, condensing. | |
| 3.2.5 | The unit shall weigh less than 5 kg. | |

# LST-16  Lettered items that wrap back to the left margin

List markers and list structure | document: documents/LST-16_lettered_items_that_wrap_back_to_the_left_margin.pdf | typewriter 11 pt, 300 dpi

### Situation

A paragraph, then a list whose items are indented but wrap back to the LEFT MARGIN ('a. Search mode, with a scan ...' / '60 degrees in elevation at all ranges.').

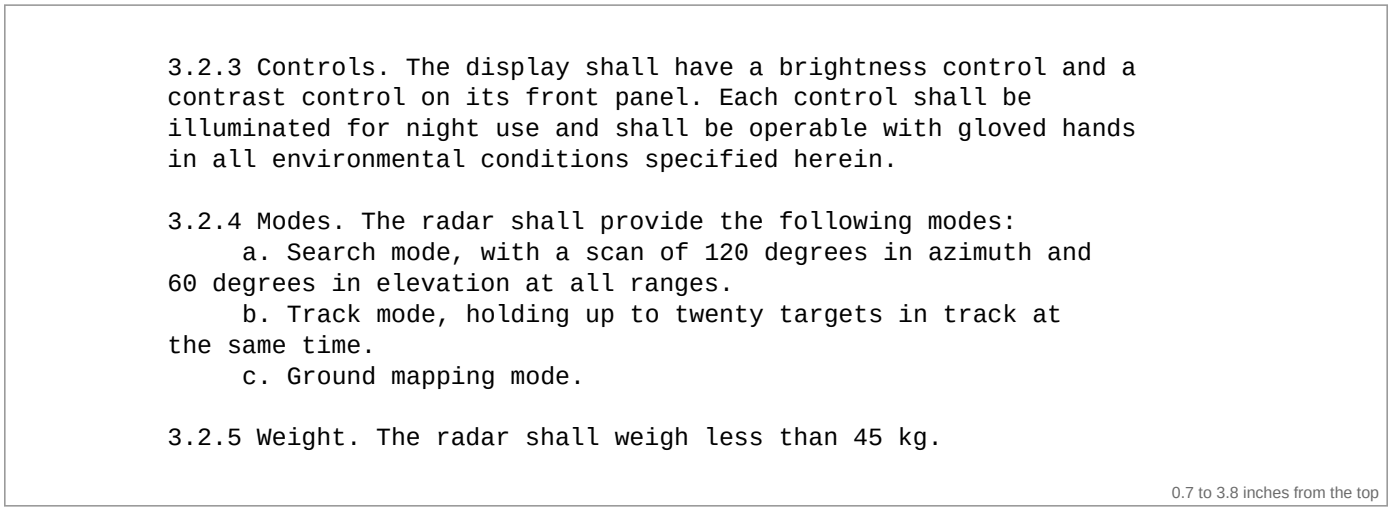
### What the tool must do

The lines at the margin are more of their item, not new paragraphs: three whole items (round 3: the list stopped at the first line that came back to the margin and the rest was lost).

### Source

Review round 3, finding F4; real OCR page r9_flushleft_lettered.

### The page (as the scan document has it)



3.2.3 Controls. The display shall have a brightness control and a contrast control on its front panel. Each control shall be illuminated for night use and shall be operable with gloved hands in all environmental conditions specified herein.

3.2.4 Modes. The radar shall provide the following modes:
a. Search mode, with a scan of 120 degrees in azimuth and 60 degrees in elevation at all ranges.
b. Track mode, holding up to twenty targets in track at the same time.
c. Ground mapping mode.

3.2.5 Weight. The radar shall weigh less than 45 kg.

0.7 to 3.8 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.3 | The display shall have a brightness control and a contrast control on its front panel. | |
| 3.2.3 | Each control shall be illuminated for night use and shall be operable with gloved hands in all environmental conditions specified herein. | |
| 3.2.4 a | The radar shall provide the following modes: a. Search mode, with a scan of 120 degrees in azimuth and 60 degrees in elevation at all ranges. | |
| 3.2.4 b | The radar shall provide the following modes: b. Track mode, holding up to twenty targets in track at the same time. | |
| 3.2.4 c | The radar shall provide the following modes: c. Ground mapping mode. | |
| 3.2.5 | The radar shall weigh less than 45 kg. | |

# LST-17  Capital-lettered items on tab stops whose value wraps back to the margin

List markers and list structure | document: documents/LST-17_capital_lettered_items_on_tab_stops_whose_value_wraps_back_t.pdf | typewriter 11 pt, 300 dpi

### Situation

A paragraph, then 'A. Temperature: -40 C to +71 C when mounted in the' / 'aircraft equipment bay with forced air cooling.' (back at the margin), then B. and C.

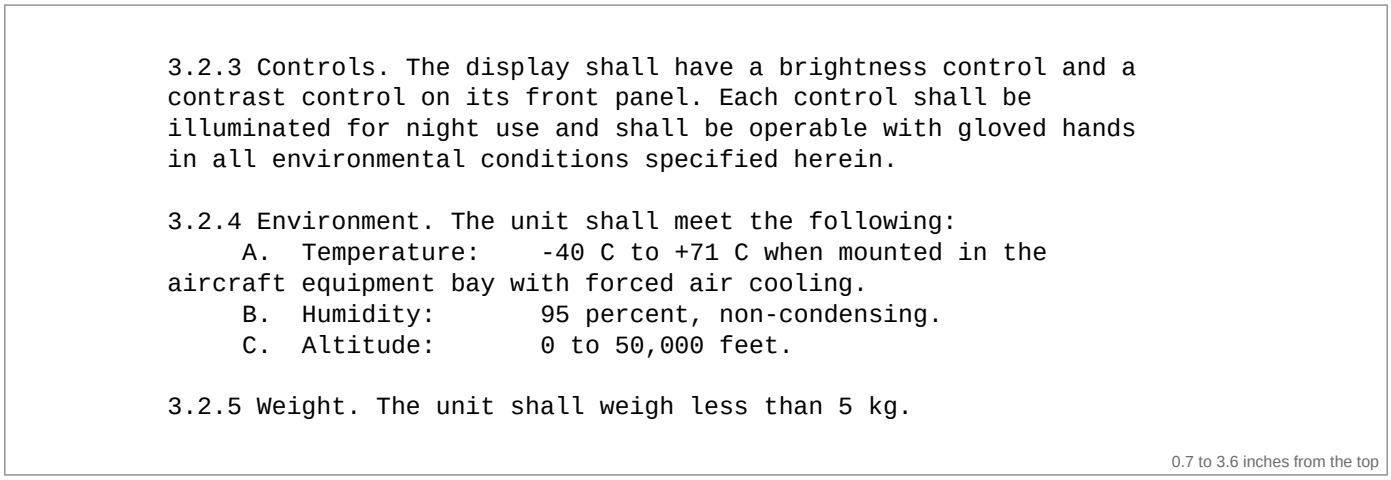
### What the tool must do

Item A keeps its whole value; B. and C. follow; the capitals are kept.

### Source

Review round 3, finding F4; real OCR page r10_flushleft_caps_label.

### The page (as the scan document has it)



```
3.2.3 Controls. The display shall have a brightness control and a
contrast control on its front panel. Each control shall be
illuminated for night use and shall be operable with gloved hands
in all environmental conditions specified herein.

3.2.4 Environment. The unit shall meet the following:
     A.  Temperature:    -40 C to +71 C when mounted in the
aircraft equipment bay with forced air cooling.
     B.  Humidity:       95 percent, non-condensing.
     C.  Altitude:       0 to 50,000 feet.

3.2.5 Weight. The unit shall weigh less than 5 kg.
```

0.7 to 3.6 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.3 | The display shall have a brightness control and a contrast control on its front panel. | |
| 3.2.3 | Each control shall be illuminated for night use and shall be operable with gloved hands in all environmental conditions specified herein. | |
| 3.2.4 A | The unit shall meet the following: A. Temperature: -40 C to +71 C when mounted in the aircraft equipment bay with forced air cooling. | |
| 3.2.4 B | The unit shall meet the following: B. Humidity: 95 percent, non-condensing. | |
| 3.2.4 C | The unit shall meet the following: C. Altitude: 0 to 50,000 feet. | |
| 3.2.5 | The unit shall weigh less than 5 kg. | |

# LST-18  Sub-items without markers under 'a. ... which shall include:'

List markers and list structure | document: documents/LST-18_sub_items_without_markers_under_a_which_shall_include.pdf | typewriter 11 pt, 300 dpi

### Situation

'a. Search mode, which shall include:' then three short lines without markers, indented a little further ('Long range scan', 'Short range scan', 'Sector scan'), then b. and c.

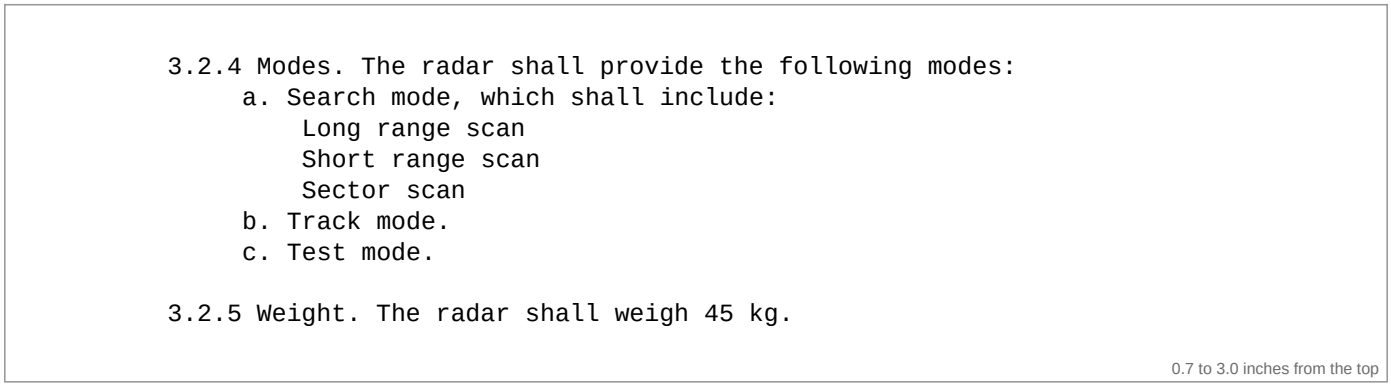
### What the tool must do

The unmarked lines are sub-items of a: three rows '3.2.4 a', each with 'a. Search mode, which shall include:' in front - not joined into one line, and not items of their own (round 1: they were run into a's text).

### Source

Review round 1, finding lists/F9; review round 2, finding contstem/a18; real OCR page r09_subitems.

### The page (as the scan document has it)



```
3.2.4 Modes. The radar shall provide the following modes:
     a. Search mode, which shall include:
         Long range scan
         Short range scan
         Sector scan
     b. Track mode.
     c. Test mode.

3.2.5 Weight. The radar shall weigh 45 kg.
```

0.7 to 3.0 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The radar shall provide the following modes: a. Search mode, which shall include: Long range scan | |
| 3.2.4 a | The radar shall provide the following modes: a. Search mode, which shall include: Short range scan | |
| 3.2.4 a | The radar shall provide the following modes: a. Search mode, which shall include: Sector scan | |
| 3.2.4 b | The radar shall provide the following modes: b. Track mode. | |
| 3.2.4 c | The radar shall provide the following modes: c. Test mode. | |
| 3.2.5 | The radar shall weigh 45 kg. | |

# LST-19  A requirement sentence between items, after a sub-list

List markers and list structure | document: documents/LST-19_a_requirement_sentence_between_items_after_a_sub_list.pdf | typewriter 11 pt, 300 dpi

### Situation

'a. Search mode, which shall include:' with (1) and (2) under it, a blank line, the sentence 'The operator shall be able to select either scan at any time.' in the markers' column, a blank line, then b. and c.

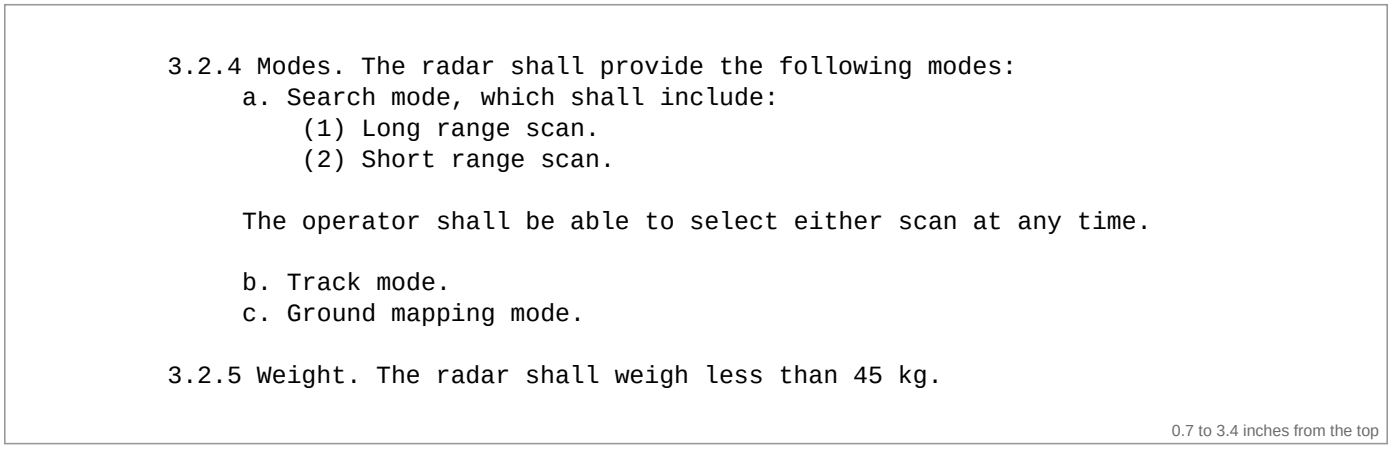
### What the tool must do

The sentence states a requirement of its own: it keeps its own row (with the lead-in, no letter), it is not glued to item (2), and b. and c. are still items (round 2: the sequence rule merged it into the wrong item).

### Source

Review round 2, finding contstem/a6 (merge rule); real OCR page merge_rule.

### The page (as the scan document has it)



```
3.2.4 Modes. The radar shall provide the following modes:
     a. Search mode, which shall include:
         (1) Long range scan.
         (2) Short range scan.

     The operator shall be able to select either scan at any time.

     b. Track mode.
     c. Ground mapping mode.

3.2.5 Weight. The radar shall weigh less than 45 kg.
```

0.7 to 3.4 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a.1 | The radar shall provide the following modes: a. Search mode, which shall include: (1) Long range scan. | |
| 3.2.4 a.2 | The radar shall provide the following modes: a. Search mode, which shall include: (2) Short range scan. | |
| 3.2.4 | The radar shall provide the following modes: The operator shall be able to select either scan at any time. | |
| 3.2.4 b | The radar shall provide the following modes: b. Track mode. | |
| 3.2.4 c | The radar shall provide the following modes: c. Ground mapping mode. | |
| 3.2.5 | The radar shall weigh less than 45 kg. | |

# LST-20  A second lead-in inside the same paragraph

List markers and list structure | document: documents/LST-20_a_second_lead_in_inside_the_same_paragraph.pdf | typewriter 11 pt, 300 dpi

### Situation

'3.2.5 ... shall provide the following interfaces:' with a. and b., a blank line, then 'The unit shall also provide the following discrete signals:' with a new a., b., c.

### What the tool must do

The second list belongs to its own lead-in: its rows read 'The unit shall also provide the following discrete signals: a. Weight on wheels.' and so on.

### Source

Review round 2, finding contstem/a6 (merge rule); real OCR page merge_rule.

**KNOWN LIMITATION** - the tool still gets this case wrong: The second lead-in is not used for the items below it: it becomes a row of its own under the first lead-in ('... the following interfaces: The unit shall also provide the following discrete signals:'), and 'Weight on wheels', 'Master arm' and 'Emergency jettison' get the first lead-in ('The unit shall provide the following interfaces:') in front of them.

### The page (as the scan document has it)

```
3.2.5 Interfaces. The unit shall provide the following interfaces:
     a. One MIL-STD-1553B bus.
     b. Two RS-422 serial ports.

     The unit shall also provide the following discrete signals:
     a. Weight on wheels.
     b. Master arm.
     c. Emergency jettison.

3.2.6 Weight. The unit shall weigh less than 5 kg.
```

0.7 to 3.2 inches from the top

### The rows the tool must produce

| **Doc Ref** | **Content** (the lead-in in grey) | **Also checked** |
|---|---|---|
| 3.2.5 a | The unit shall provide the following interfaces: a. One MIL-STD-1553B bus. | |
| 3.2.5 b | The unit shall provide the following interfaces: b. Two RS-422 serial ports. | |
| 3.2.5 a | The unit shall also provide the following discrete signals: a. Weight on wheels. | |
| 3.2.5 b | The unit shall also provide the following discrete signals: b. Master arm. | |
| 3.2.5 c | The unit shall also provide the following discrete signals: c. Emergency jettison. | |
| 3.2.6 | The unit shall weigh less than 5 kg. | |

# LST-21  Two requirement sentences between a. and b.

List markers and list structure | document: documents/LST-21_two_requirement_sentences_between_a_and_b.pdf | typewriter 11 pt, 300 dpi

### Situation

'a. Search mode.', then two sentences each after a blank line, in the markers' column ('Search mode shall be selected at power-up.', 'Search mode shall cover 360 degrees in azimuth.'), then b. and c.

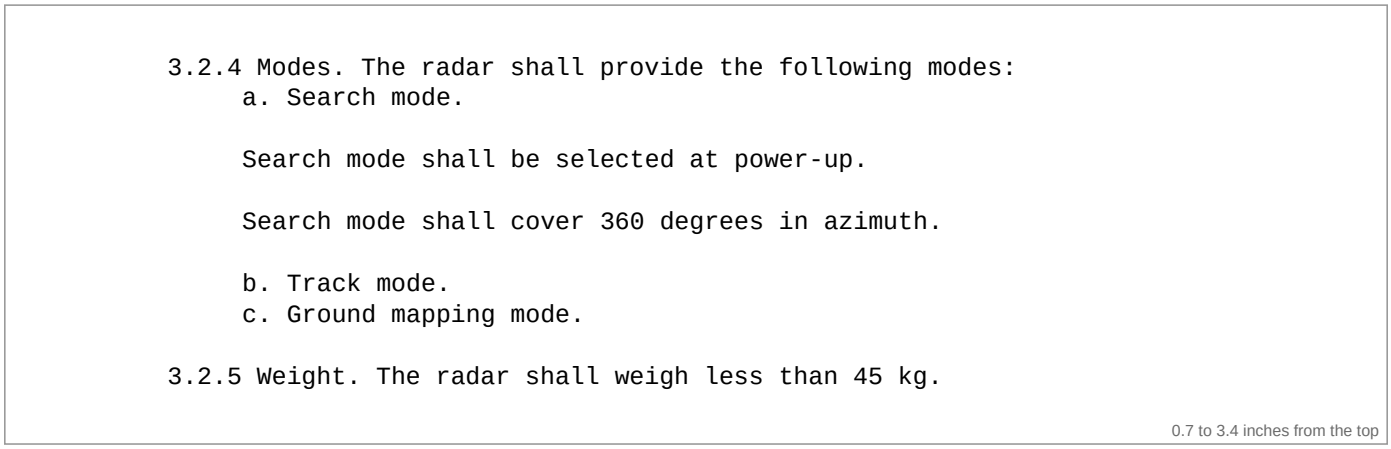
### What the tool must do

Both sentences keep rows of their own (with the lead-in, no letter) and neither is given the letter 'b.' that the next item really carries (round 2: the first was merged into a., the second became an orphan row).

### Source

Review round 2, finding contstem/a16 (merge chain); real OCR page misc1.

### The page (as the scan document has it)



```
3.2.4 Modes. The radar shall provide the following modes:
     a. Search mode.

     Search mode shall be selected at power-up.

     Search mode shall cover 360 degrees in azimuth.

     b. Track mode.
     c. Ground mapping mode.

3.2.5 Weight. The radar shall weigh less than 45 kg.
```

0.7 to 3.4 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The radar shall provide the following modes: a. Search mode. | |
| 3.2.4 | The radar shall provide the following modes: Search mode shall be selected at power-up. | |
| 3.2.4 | The radar shall provide the following modes: Search mode shall cover 360 degrees in azimuth. | |
| 3.2.4 b | The radar shall provide the following modes: b. Track mode. | |
| 3.2.4 c | The radar shall provide the following modes: c. Ground mapping mode. | |
| 3.2.5 | The radar shall weigh less than 45 kg. | |

# LST-22  A NOTE between two items of a tab list

List markers and list structure | document: documents/LST-22_a_note_between_two_items_of_a_tab_list.pdf | typewriter 11 pt, 300 dpi

### Situation

'a. Voltage: ...', 'b. Ripple: ...', then 'NOTE: The limits apply at the equipment connector.' in the markers' column, then 'c. Power: ...'.

### What the tool must do

The NOTE is never given a letter: it keeps a row of its own (with the lead-in), and the next item is still '3.2.10 c' (round 2: the NOTE was made item c and the real c. got a duplicate letter).

### Source

Review round 2, finding numbering/F3; real OCR page r3_swallow.

### The page (as the scan document has it)

```
3.2.10 Power. The equipment shall operate within the
following limits:

    a.    Voltage:          24 to 32 VDC, 28 VDC nominal.

    b.    Ripple:           1.5 V peak-to-peak maximum.

    NOTE: The limits apply at the equipment connector.

    c.    Power:            150 W maximum at 28 VDC.

3.3 Design and construction. The equipment shall be modular.
```

0.7 to 3.2 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.10 a | The equipment shall operate within the following limits: a. Voltage: 24 to 32 VDC, 28 VDC nominal. | |
| 3.2.10 b | The equipment shall operate within the following limits: b. Ripple: 1.5 V peak-to-peak maximum. | |
| 3.2.10 | The equipment shall operate within the following limits: NOTE: The limits apply at the equipment connector. | |
| 3.2.10 c | The equipment shall operate within the following limits: c. Power: 150 W maximum at 28 VDC. | |
| 3.3 | The equipment shall be modular. | |

# LST-23  A capital 'C.' printed in a small-letter list

List markers and list structure | document: documents/LST-23_a_capital_c_printed_in_a_small_letter_list.pdf | typewriter 11 pt, 300 dpi

### Situation

'a. MIL-STD-461G', 'b. MIL-STD-704F', 'C. MIL-STD-810H', 'd. MIL-STD-1275E' - the page itself has 'C.' (the same thing the scan often does to a lone 'c.').

### What the tool must do

The a, b, ?, d sequence says it is c: the row is '3.2.7 c' and Review Reasons says the marker was read as 'C.'.

### Source

Review round 2 (layout, f2_upperC).

### The page (as the scan document has it)

```
3.2.7 Documents. The unit shall comply with each of the following:
        a. MIL-STD-461G
        b. MIL-STD-704F
        C. MIL-STD-810H
        d. MIL-STD-1275E

3.2.8 Weight. The unit shall not exceed 45 kilograms.
```

0.7 to 2.6 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.7 a | The unit shall comply with each of the following: a. MIL-STD-461G | |
| 3.2.7 b | The unit shall comply with each of the following: b. MIL-STD-704F | |
| 3.2.7 c | The unit shall comply with each of the following: c. MIL-STD-810H | review note: List marker read as 'C.' |
| 3.2.7 d | The unit shall comply with each of the following: d. MIL-STD-1275E | |
| 3.2.8 | The unit shall not exceed 45 kilograms. | |

# LST-24  The last item's marker lost in a single-spaced list

List markers and list structure | document: documents/LST-24_the_last_item_s_marker_lost_in_a_single_spaced_list.pdf | typewriter 11 pt, 300 dpi

### Situation

'a. Search mode.' / 'b. Track mode.' / ' Ground mapping mode.' - the last item has lost its 'c.' and starts where a wrapped line of b. would; nothing follows it in the list.

### What the tool must do

Ideally item c, with a review note - or at least a review note on b.

### Source

Review round 3 (lists), known limitation.

**KNOWN LIMITATION** - the tool still gets this case wrong: No marker after the gap confirms the sequence, so the line is read as the second line of item b ('b. Track mode. Ground mapping mode.'), and nothing is flagged. A lost marker in the middle of a list is found (LST-10).

### The page (as the scan document has it)

```
3.2.4 Modes. The radar shall provide the following modes:
   a.   Search mode.
   b.   Track mode.
        Ground mapping mode.

3.2.5 Weight. The radar shall weigh less than 45 kg.
```

0.7 to 2.4 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The radar shall provide the following modes: a. Search mode. | |
| 3.2.4 b | The radar shall provide the following modes: b. Track mode. | |
| 3.2.4 c | The radar shall provide the following modes: c. Ground mapping mode. | review note: List marker missing |
| 3.2.5 | The radar shall weigh less than 45 kg. | |

# LST-25  Lettered items without full stops, then a paragraph

List markers and list structure | document: documents/LST-25_lettered_items_without_full_stops_then_a_paragraph.pdf | typewriter 11 pt, 300 dpi

**Situation**

'a. Search mode' / 'b. Track mode' (no full stops), a blank line, then 'The operator shall select the mode from the front panel.' in the items' text column.

**What the tool must do**

The paragraph is a requirement of its own (with the lead-in, no letter), not more of item b.

**Source**

Review round 3, finding B3; still wrong when the case library was written, fixed with it.

**The page (as the scan document has it)**

```
3.2.4 Modes. The radar shall provide the following modes:
     a. Search mode
     b. Track mode

        The operator shall select the mode from the front panel.

3.2.5 Weight. The radar shall weigh less than 45 kg.
```

0.7 to 2.6 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The radar shall provide the following modes: a. Search mode | |
| 3.2.4 b | The radar shall provide the following modes: b. Track mode | |
| 3.2.4 | The radar shall provide the following modes: The operator shall select the mode from the front panel. | |
| 3.2.5 | The radar shall weigh less than 45 kg. | |

# Lead-ins

Lead-ins: the sentence that introduces a list. It normally ends with ':', but the colon is often left out ('... shall show the following information.', '... listed below.'), and the same words can appear in running text without any list ('... retested on the following day.'). These cases check which lines are taken as the items of a lead-in, and that a sentence after the list is not swallowed as one more item.

## LEAD-01  No colon; items separated by blank lines; then a requirement at the same indent

Lead-ins | document: documents/LEAD-01_no_colon_items_separated_by_blank_lines_then_a_requirement_a.pdf | typewriter 11 pt, 300 dpi

**Situation**

A paragraph, then '3.2.6 ... the following information.' (no colon), three short indented lines each after a blank line, then - at the same indent - 'The display shall be readable in direct sunlight.'

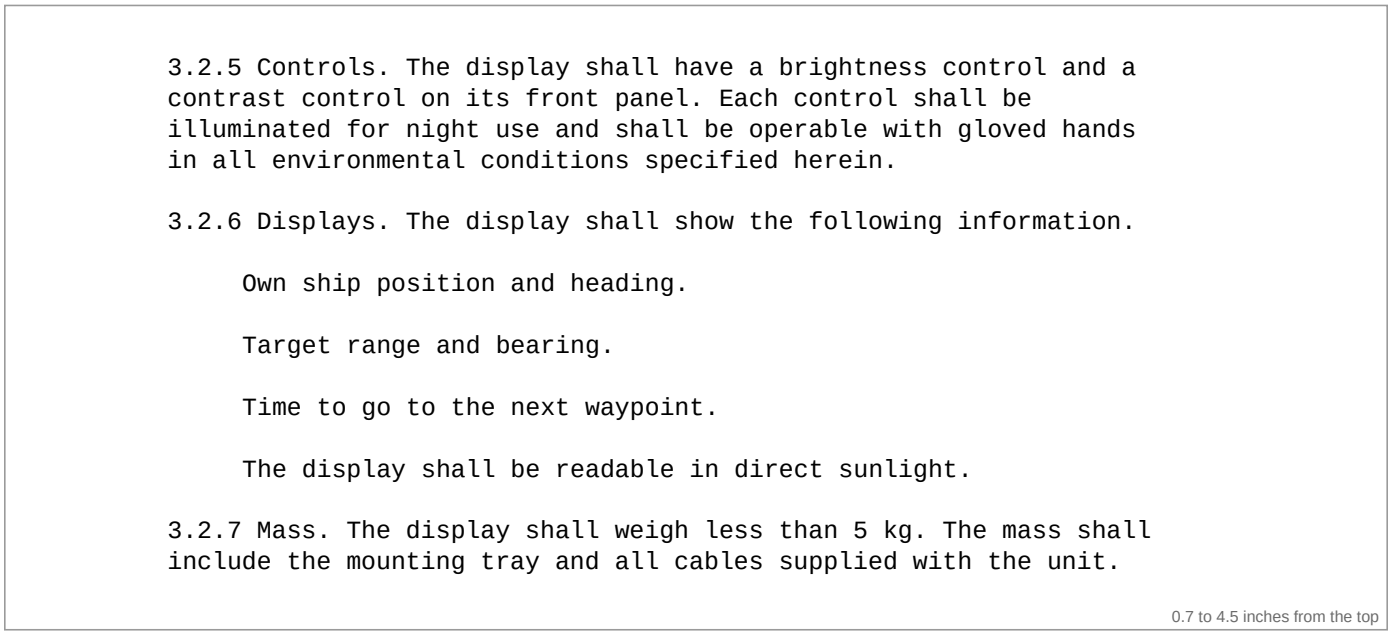
**What the tool must do**

The three short lines are the items of the lead-in (each row: the lead-in + the item). The sentence after them states a requirement of its own and keeps its own row (round 3: it was taken as a fourth item).

**Source**

Review round 3, finding F3; real OCR page r2d_colonless_swallow_ctx_blanks.

**The page (as the scan document has it)**



3.2.5 Controls. The display shall have a brightness control and a contrast control on its front panel. Each control shall be illuminated for night use and shall be operable with gloved hands in all environmental conditions specified herein.

3.2.6 Displays. The display shall show the following information.

Own ship position and heading.

Target range and bearing.

Time to go to the next waypoint.

The display shall be readable in direct sunlight.

3.2.7 Mass. The display shall weigh less than 5 kg. The mass shall include the mounting tray and all cables supplied with the unit.

0.7 to 4.5 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.5 | The display shall have a brightness control and a contrast control on its front panel. | |
| 3.2.5 | Each control shall be illuminated for night use and shall be operable with gloved hands in all environmental conditions specified herein. | |
| 3.2.6 | The display shall show the following information. Own ship position and heading. | |
| 3.2.6 | The display shall show the following information. Target range and bearing. | |
| 3.2.6 | The display shall show the following information. Time to go to the next waypoint. | |
| (empty) | The display shall be readable in direct sunlight. | |
| 3.2.7 | The display shall weigh less than 5 kg. | |
| 3.2.7 | The mass shall include the mounting tray and all cables supplied with the unit. | |

# LEAD-02  No colon; four single-spaced items right under the lead-in

Lead-ins | document: documents/LEAD-02_no_colon_four_single_spaced_items_right_under_the_lead_in.pdf | typewriter 11 pt, 300 dpi

### Situation

A paragraph, then the lead-in without a colon and, right under it with no blank line, four short indented lines ('Own ship position and heading.', 'Target range and bearing.', 'Time of day.', 'Waypoint number.').

### What the tool must do

All four lines are items (round 3: the first line was put into the lead-in's block and silently dropped, so a list that looked complete was not).

### Source

Review round 3, finding B2; real OCR page r7_colonless_single_spaced.

### The page (as the scan document has it)

```
3.2.5 Controls. The display shall have a brightness control and a
contrast control on its front panel. Each control shall be
illuminated for night use and shall be operable with gloved hands
in all environmental conditions specified herein.

3.2.6 Displays. The display shall show the following information.
     Own ship position and heading.
     Target range and bearing.
     Time of day.
     Waypoint number.

3.2.7 Mass. The display shall weigh less than 5 kg. The mass shall
include the mounting tray and all cables supplied with the unit.
```

0.7 to 3.8 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.5 | The display shall have a brightness control and a contrast control on its front panel. | |
| 3.2.5 | Each control shall be illuminated for night use and shall be operable with gloved hands in all environmental conditions specified herein. | |
| 3.2.6 | The display shall show the following information. Own ship position and heading. | |
| 3.2.6 | The display shall show the following information. Target range and bearing. | |
| 3.2.6 | The display shall show the following information. Time of day. | |
| 3.2.6 | The display shall show the following information. Waypoint number. | |
| 3.2.7 | The display shall weigh less than 5 kg. | |
| 3.2.7 | The mass shall include the mounting tray and all cables supplied with the unit. | |

# LEAD-03  No colon; three single-spaced items after a blank line; then a requirement

Lead-ins | document: documents/LEAD-03_no_colon_three_single_spaced_items_after_a_blank_line_then_a.pdf | typewriter 11 pt, 300 dpi

### Situation

A paragraph, the lead-in without a colon, a blank line, three short single-spaced indented lines (one block of 16 words), a blank line, then 'The display shall be readable in direct sunlight.' at the same indent.

### What the tool must do

The three lines are items; the requirement after them keeps its own row.

### Source

Review round 3, findings B2 and F3; real OCR page r2c_colonless_swallow_ctx.

### The page (as the scan document has it)

```
3.2.5 Controls. The display shall have a brightness control and a
contrast control on its front panel. Each control shall be
illuminated for night use and shall be operable with gloved hands
in all environmental conditions specified herein.

3.2.6 Displays. The display shall show the following information.

     Own ship position and heading.
     Target range and bearing.
     Time to go to the next waypoint.

     The display shall be readable in direct sunlight.

3.2.7 Mass. The display shall weigh less than 5 kg. The mass shall
include the mounting tray and all cables supplied with the unit.
```

0.7 to 4.2 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.5 | The display shall have a brightness control and a contrast control on its front panel. | |
| 3.2.5 | Each control shall be illuminated for night use and shall be operable with gloved hands in all environmental conditions specified herein. | |
| 3.2.6 | The display shall show the following information. Own ship position and heading. | |
| 3.2.6 | The display shall show the following information. Target range and bearing. | |
| 3.2.6 | The display shall show the following information. Time to go to the next waypoint. | |
| (empty) | The display shall be readable in direct sunlight. | |
| 3.2.7 | The display shall weigh less than 5 kg. | |
| 3.2.7 | The mass shall include the mounting tray and all cables supplied with the unit. | |

# LEAD-04  '... listed below.' with plain one-line items

Lead-ins | document: documents/LEAD-04_listed_below_with_plain_one_line_items.pdf | typewriter 11 pt, 300 dpi

### Situation

'3.2.4 Modes. The radar shall provide the operating modes listed below.' then three indented lines without markers or full stops ('Search mode', 'Track while scan mode', 'Ground mapping mode').

### What the tool must do

A lead-in without a colon may lead into plain items too: three rows, each with the lead-in in front (round 2: only marked or labelled items were taken after a colon-less lead-in).

### Source

Review round 2, finding contstem/a3.

### The page (as the scan document has it)

```
3.2.4 Modes. The radar shall provide the operating modes listed below.
     Search mode
     Track while scan mode
     Ground mapping mode

3.2.5 Weight. The radar shall weigh less than 45 kg.
```

0.7 to 2.4 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 | The radar shall provide the operating modes listed below. Search mode | |
| 3.2.4 | The radar shall provide the operating modes listed below. Track while scan mode | |
| 3.2.4 | The radar shall provide the operating modes listed below. Ground mapping mode | |
| 3.2.5 | The radar shall weigh less than 45 kg. | |

# LEAD-05 No colon, at the top of the page; three single-spaced items after a blank line

Lead-ins | document: documents/LEAD-05_no_colon_at_the_top_of_the_page_three_single_spaced_items_af.pdf | typewriter 11 pt, 300 dpi

### Situation

LEAD-03 without the paragraph above it: the lead-in is the first line of the page.

### What the tool must do

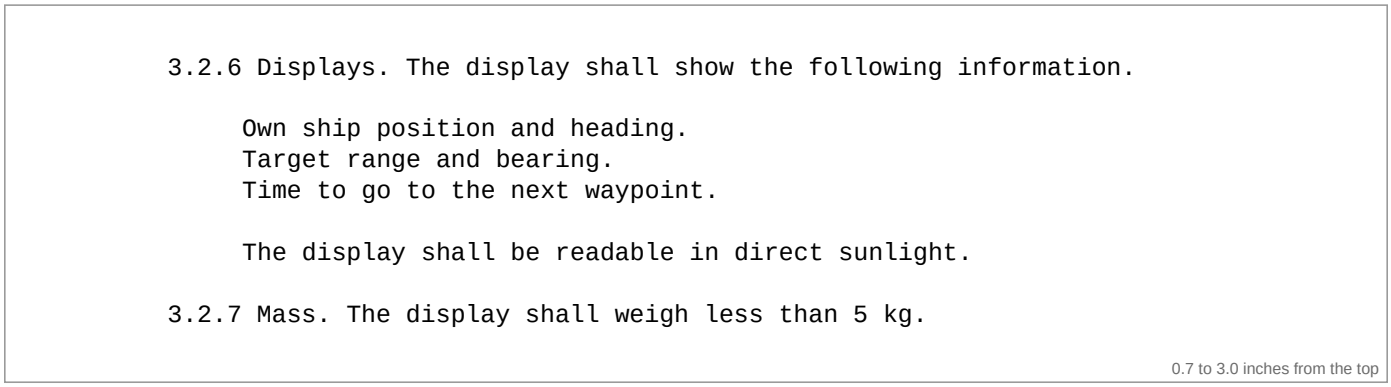
As LEAD-03: three items, then the requirement on its own row.

### Source

Review round 3, finding B2; real OCR page r2_colonless_swallow.

**KNOWN LIMITATION** - the tool still gets this case wrong: With nothing above the lead-in, the three short lines are neither taken as items nor flagged: only the lead-in and the requirement below them are rows (with a paragraph above the lead-in they are found - LEAD-03).

### The page (as the scan document has it)



```
3.2.6 Displays. The display shall show the following information.

     Own ship position and heading.
     Target range and bearing.
     Time to go to the next waypoint.

     The display shall be readable in direct sunlight.

3.2.7 Mass. The display shall weigh less than 5 kg.
```

0.7 to 3.0 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.6 | The display shall show the following information. Own ship position and heading. | |
| 3.2.6 | The display shall show the following information. Target range and bearing. | |
| 3.2.6 | The display shall show the following information. Time to go to the next waypoint. | |
| (empty) | The display shall be readable in direct sunlight. | |
| 3.2.7 | The display shall weigh less than 5 kg. | |

# LEAD-06  No colon, at the top of the page; items separated by blank lines

Lead-ins | document: documents/LEAD-06_no_colon_at_the_top_of_the_page_items_separated_by_blank_lin.pdf | typewriter 11 pt, 300 dpi

### Situation

LEAD-01 without the paragraph above it: the lead-in is the first line of the page.

### What the tool must do

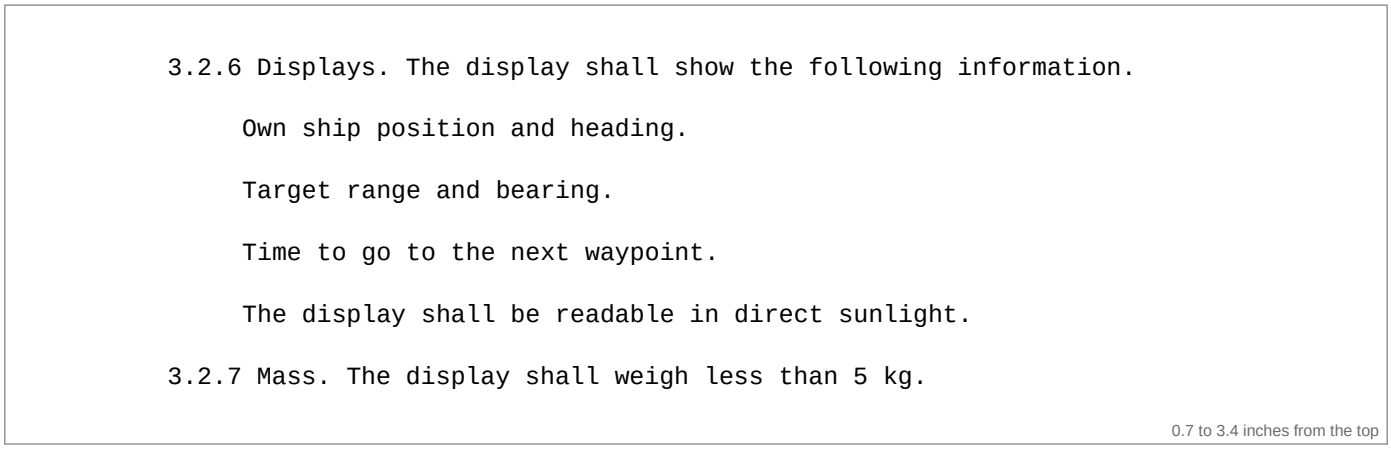
As LEAD-01: three items, then the requirement on its own row.

### Source

Review round 3, finding F3; real OCR page r2b_colonless_swallow_blanks.

**KNOWN LIMITATION** - the tool still gets this case wrong: With nothing above the lead-in, the three short lines are neither taken as items nor flagged: only the lead-in and the requirement below them are rows (with a paragraph above the lead-in they are found - LEAD-01).

### The page (as the scan document has it)



3.2.6 Displays. The display shall show the following information.

Own ship position and heading.

Target range and bearing.

Time to go to the next waypoint.

The display shall be readable in direct sunlight.

3.2.7 Mass. The display shall weigh less than 5 kg.

0.7 to 3.4 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.6 | The display shall show the following information. Own ship position and heading. | |
| 3.2.6 | The display shall show the following information. Target range and bearing. | |
| 3.2.6 | The display shall show the following information. Time to go to the next waypoint. | |
| (empty) | The display shall be readable in direct sunlight. | |
| 3.2.7 | The display shall weigh less than 5 kg. | |

# LEAD-07  Lead-in words in the middle of a paragraph

Lead-ins | document: documents/LEAD-07_lead_in_words_in_the_middle_of_a_paragraph.pdf | typewriter 11 pt, 300 dpi

### Situation

'... shall be repaired and retested on the following day. The results ...' and '... shall be marked as specified below. The marking shall be legible ...' - the words of a lead-in, but inside running text with no list after them.

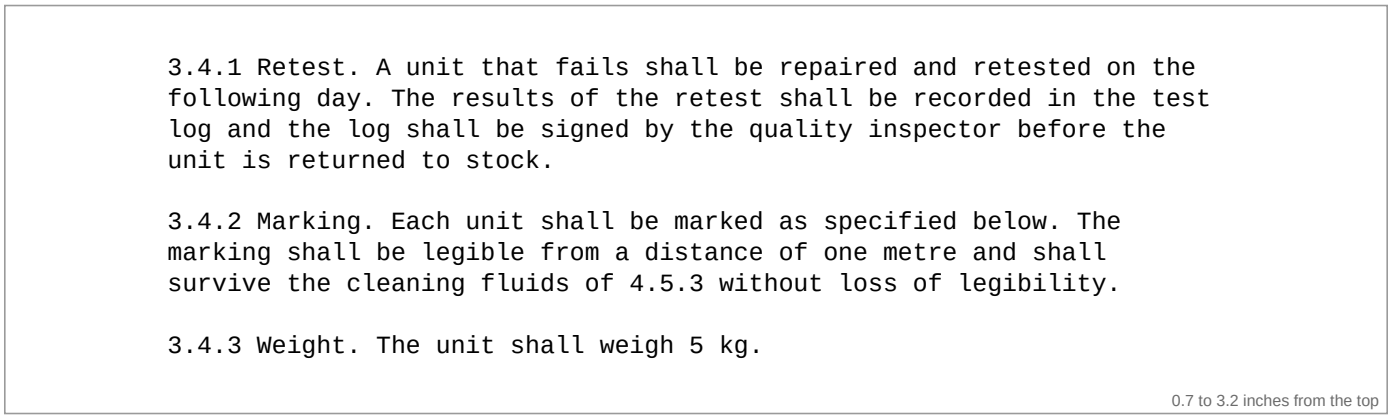
### What the tool must do

No list: each sentence is a row of its own paragraph (round 1: 'on the following day.' was taken as a lead-in and the next sentence became its item).

### Source

Review round 1, finding lists/F1; real OCR page r01_midblock.

### The page (as the scan document has it)



3.4.1 Retest. A unit that fails shall be repaired and retested on the following day. The results of the retest shall be recorded in the test log and the log shall be signed by the quality inspector before the unit is returned to stock.

3.4.2 Marking. Each unit shall be marked as specified below. The marking shall be legible from a distance of one metre and shall survive the cleaning fluids of 4.5.3 without loss of legibility.

3.4.3 Weight. The unit shall weigh 5 kg.

0.7 to 3.2 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.4.1 | A unit that fails shall be repaired and retested on the following day. | |
| 3.4.1 | The results of the retest shall be recorded in the test log and the log shall be signed by the quality inspector before the unit is returned to stock. | |
| 3.4.2 | Each unit shall be marked as specified below. | |
| 3.4.2 | The marking shall be legible from a distance of one metre and shall survive the cleaning fluids of 4.5.3 without loss of legibility. | |
| 3.4.3 | The unit shall weigh 5 kg. | |

# LEAD-08  A colon with nothing below it

Lead-ins | document: documents/LEAD-08_a_colon_with_nothing_below_it.pdf | typewriter 11 pt, 300 dpi

### Situation

'3.2.6 Displays. The display shall show the following:' followed straight away by the next numbered paragraph - the list is missing (lost on the scan, or on another page).

### What the tool must do

The lead-in is a row of its own and Review Reasons says a list was promised but not found; 3.2.7 is not taken as its item.

### Source

Written for the case library (the flag is described in the README).

### The page (as the scan document has it)

```
3.2.6 Displays. The display shall show the following:

3.2.7 Mass. The display shall weigh less than 5 kg.
```

0.7 to 1.9 inches from the top

### The rows the tool must produce

| **Doc Ref** | **Content** (the lead-in in grey) | **Also checked** |
|---|---|---|
| 3.2.6 | The display shall show the following: | review note: Ends with ':' but no list items |
| 3.2.7 | The display shall weigh less than 5 kg. | |

# LEAD-09  Lead-in words, then the next numbered paragraph

Lead-ins | document: documents/LEAD-09_lead_in_words_then_the_next_numbered_paragraph.pdf | typewriter 11 pt, 300 dpi

### Situation

'3.2.6 ... as follows.' (no colon) followed by the next numbered paragraph '3.2.7 Mass. ...'.

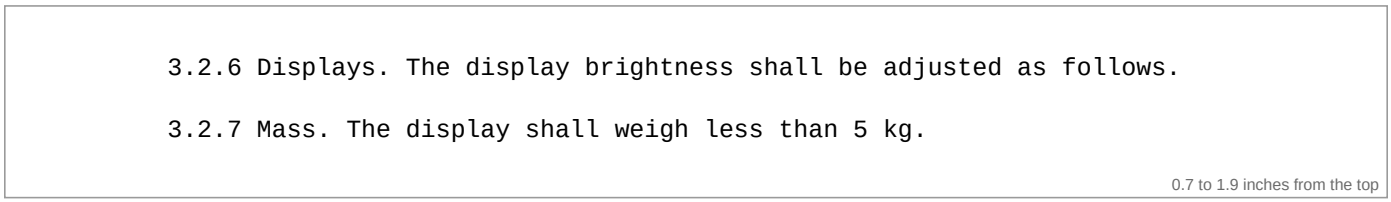
### What the tool must do

A sentence that just ends in a lead-in phrase and is followed by a new paragraph is not a list: two rows, and 3.2.7 keeps its number.

### Source

Written for the case library (round 3 review of rule 7).

### The page (as the scan document has it)




0.7 to 1.9 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.6 | The display brightness shall be adjusted as follows. | |
| 3.2.7 | The display shall weigh less than 5 kg. | |

# Paragraph numbers and headings

Paragraph numbers and headings: which lines start a new numbered paragraph or a section heading, and which numbers are only list items, cross-references or values at the start of a wrapped line. The Section column is checked where a heading must (or must not) become the section of the rows below it.

## NUM-01  Numbered lists in sections 2 and 3, each followed by the next paragraph

Paragraph numbers and headings | document: documents/NUM-01_numbered_lists_in_sections_2_and_3_each_followed_by_the_next.pdf | typewriter 11 pt, 300 dpi

**Situation**

'2. APPLICABLE DOCUMENTS' with 2.1 and its list '1. MIL-STD-461G' / '2. MIL-STD-704F', then 2.2; '3. REQUIREMENTS' with 3.1, 3.2 and its list '1. Search mode.' to '3. Ground mapping mode.', then 3.3.

**What the tool must do**

The numbered items are list items, not section headings: rows 2.1 1, 2.1 2, 3.2 1 to 3.2 3. 2.2 and 3.3 keep their numbers and their sections (round 2: '3.2.5' after the list was taken as the list's own sub-paragraph, and the items became headings).

**Source**

Review round 2, finding contstem/a1 (opens section); real OCR page opens_section.

**The page (as the scan document has it)**

```
2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents form a part of
this specification to the extent specified herein and shall apply:

     1. MIL-STD-461G
     2. MIL-STD-704F

2.2 Order of precedence. In the event of a conflict, the text of
this specification shall take precedence.

3. REQUIREMENTS

3.1 Power. The unit shall operate from 28 VDC.

3.2 Modes. The radar shall provide the following modes:

     1. Search mode.
     2. Track mode.
     3. Ground mapping mode.

3.3 Weight. The radar shall weigh less than 45 kg.
```

0.7 to 5.5 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 2.1 1 | The following documents form a part of this specification to the extent specified herein and shall apply: 1. MIL-STD-461G | |
| 2.1 2 | The following documents form a part of this specification to the extent specified herein and shall apply: 2. MIL-STD-704F | |
| 2.2 | In the event of a conflict, the text of this specification shall take precedence. | Section: 2 APPLICABLE DOCUMENTS |
| 3.1 | The unit shall operate from 28 VDC. | Section: 3 REQUIREMENTS |
| 3.2 1 | The radar shall provide the following modes: 1. Search mode. | |
| 3.2 2 | The radar shall provide the following modes: 2. Track mode. | |
| 3.2 3 | The radar shall provide the following modes: 3. Ground mapping mode. | |
| 3.3 | The radar shall weigh less than 45 kg. | Section: 3 REQUIREMENTS |

# NUM-02  A title-case heading with a full stop after a list at the margin

Paragraph numbers and headings | document: documents/NUM-02_a_title_case_heading_with_a_full_stop_after_a_list_at_the_ma.pdf | typewriter 11 pt, 300 dpi

### Situation

A list 'a. Search mode.' / 'b. Track mode.' at the left margin, a blank line, the heading '4. Verification.' (title case, with a full stop), then 4.1.

### What the tool must do

'4. Verification.' is a heading, not the list's next item: no row for it, and 4.1 is in section '4 Verification' (round 3: it became a false requirement row '... following modes: 4. Verification.').

### Source

Review round 3, finding F2; real OCR page r1b_margin_list_heading_period.

### The page (as the scan document has it)

```
3.9 Modes. The radar shall provide the following modes:

a. Search mode.
b. Track mode.

4. Verification.

4.1 General. The contractor shall verify each requirement of
section 3 by test.
```

0.7 to 3.0 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.9 a | The radar shall provide the following modes: a. Search mode. | |
| 3.9 b | The radar shall provide the following modes: b. Track mode. | |
| 4.1 | The contractor shall verify each requirement of section 3 by test. | Section: 4 Verification |

# NUM-03  A heading with a full stop after a list of documents without full stops

Paragraph numbers and headings | document: documents/NUM-03_a_heading_with_a_full_stop_after_a_list_of_documents_without.pdf | typewriter 11 pt, 300 dpi

### Situation

'a. MIL-STD-461G' / 'b. MIL-STD-704F' at the margin (no full stops), a blank line, then the heading '3. Requirements.' and paragraphs 3.1 and 3.2.

### What the tool must do

The heading ends the list: item b is 'b. MIL-STD-704F' and 3.1 and 3.2 are in section '3 Requirements'.

### Source

Review round 3, finding F2; real OCR page r1_heading_period.

**KNOWN LIMITATION** - the tool still gets this case wrong: Almost every line of this page is followed by a blank line, so a blank line is taken as the page's normal line spacing; item b has no full stop, and the heading one blank line below it is read as its next line: the row is 'b. MIL-STD-704F 3. Requirements.' and 3.1 and 3.2 keep the section '2 APPLICABLE DOCUMENTS'.

### The page (as the scan document has it)

```
2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents form a part of
this specification to the extent specified herein and shall apply:

a. MIL-STD-461G
b. MIL-STD-704F

3. Requirements.

3.1 Power. The unit shall operate from 28 VDC aircraft power.

3.2 Weight. The unit shall weigh less than 5 kg.
```

0.7 to 3.8 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 2.1 a | The following documents form a part of this specification to the extent specified herein and shall apply: a. MIL-STD-461G | |
| 2.1 b | The following documents form a part of this specification to the extent specified herein and shall apply: b. MIL-STD-704F | |
| 3.1 | The unit shall operate from 28 VDC aircraft power. | Section: 3 Requirements |
| 3.2 | The unit shall weigh less than 5 kg. | Section: 3 Requirements |

# NUM-04  Empty sections in a row ('5. Packaging.', '6. Notes.')

Paragraph numbers and headings | document: documents/NUM-04_empty_sections_in_a_row_5_packaging_6_notes.pdf | typewriter 11 pt, 300 dpi

**Situation**

4.3, then the headings '5. Packaging.' and '6. Notes.' each followed only by a blank line, then 6.1.

**What the tool must do**

Both are headings (no rows of their own) and 6.1 is in section '6 Notes' (round 3: a run of empty sections lost '6 Notes').

**Source**

Review round 3, finding F2; real OCR page r1_heading_period.

**The page (as the scan document has it)**

```
4.3 Inspection. The unit shall be inspected.

5. Packaging.

6. Notes.

6.1 Intended use. The unit shall be used in fighter aircraft.
```

0.7 to 2.6 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 4.3 | The unit shall be inspected. | |
| 6.1 | The unit shall be used in fighter aircraft. | Section: 6 Notes |

# NUM-05  Wrapped lines that start with a cross-reference or a quantity

Paragraph numbers and headings | document: documents/NUM-05_wrapped_lines_that_start_with_a_cross_reference_or_a_quantit.pdf | typewriter 11 pt, 300 dpi

### Situation

Paragraph 3.2.4 wraps after '... loads in paragraph' so the next line starts '3.2.5. The weight shall ...'; 3.2.5 has a line starting '3.5 G in each axis ...'; 3.2.6 has one starting '4.2.1 Bonding Test) with ...' after '(see'.

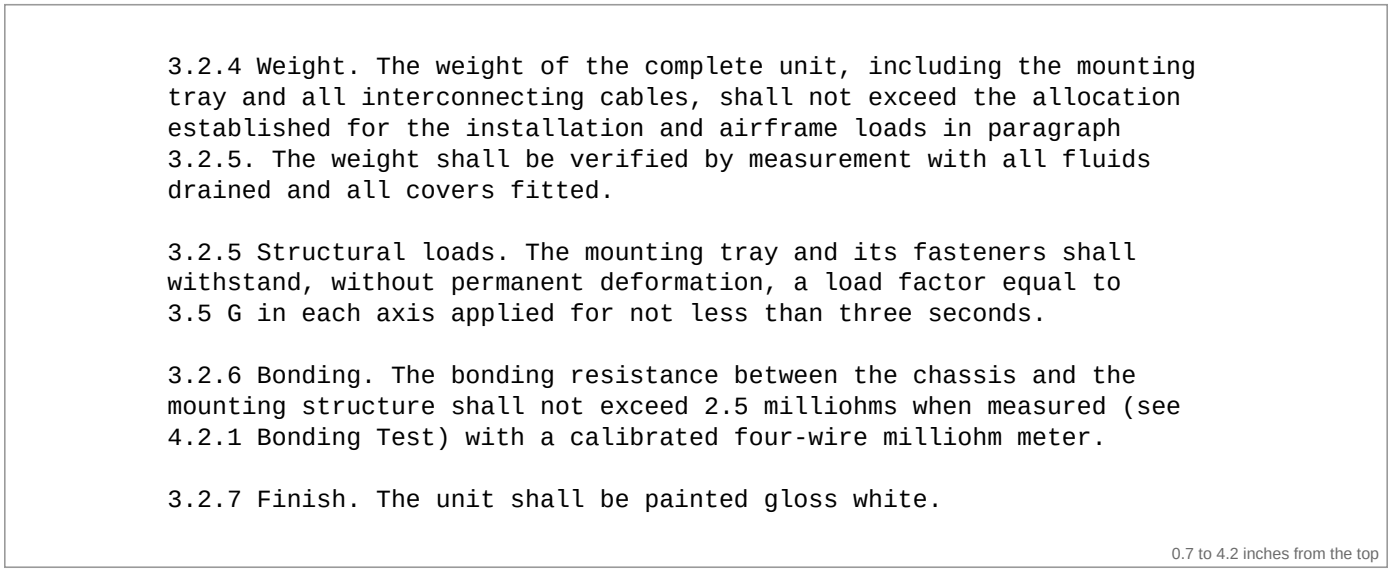
### What the tool must do

None of these numbers starts a new paragraph: 3.2.4 keeps both its sentences, and 3.2.5 and 3.2.6 are each one paragraph (round 1: 3.2.4 was split at '3.2.5.', which is exactly the number that comes next).

### Source

Review round 1, finding numbering/F1; real OCR page r1_crossref.

### The page (as the scan document has it)



3.2.4 Weight. The weight of the complete unit, including the mounting tray and all interconnecting cables, shall not exceed the allocation established for the installation and airframe loads in paragraph 3.2.5. The weight shall be verified by measurement with all fluids drained and all covers fitted.

3.2.5 Structural loads. The mounting tray and its fasteners shall withstand, without permanent deformation, a load factor equal to 3.5 G in each axis applied for not less than three seconds.

3.2.6 Bonding. The bonding resistance between the chassis and the mounting structure shall not exceed 2.5 milliohms when measured (see 4.2.1 Bonding Test) with a calibrated four-wire milliohm meter.

3.2.7 Finish. The unit shall be painted gloss white.

0.7 to 4.2 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 | The weight of the complete unit, including the mounting tray and all interconnecting cables, shall not exceed the allocation established for the installation and airframe loads in paragraph 3.2.5. | |
| 3.2.4 | The weight shall be verified by measurement with all fluids drained and all covers fitted. | |
| 3.2.5 | The mounting tray and its fasteners shall withstand, without permanent deformation, a load factor equal to 3.5 G in each axis applied for not less than three seconds. | |
| 3.2.6 | The bonding resistance between the chassis and the mounting structure shall not exceed 2.5 milliohms when measured (see 4.2.1 Bonding Test) with a calibrated four-wire milliohm meter. | |
| 3.2.7 | The unit shall be painted gloss white. | |

# NUM-06  Paragraph numbers that skip after a list (3.2.4, then 3.2.8)

Paragraph numbers and headings | document: documents/NUM-06_paragraph_numbers_that_skip_after_a_list_3_2_4_then_3_2_8.pdf | typewriter 11 pt, 300 dpi

### Situation

Paragraphs with a first-line indent: 3.2.4 with a list on tab stops, then 3.2.8 and 3.2.9 (a skip of four - or a scan that misread 3.2.5).

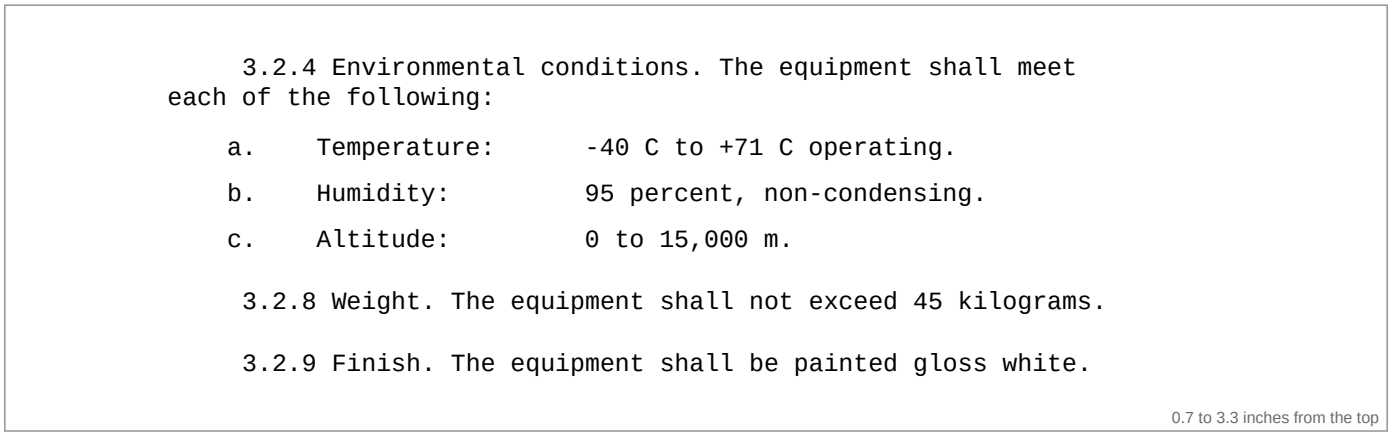
### What the tool must do

3.2.8 and 3.2.9 are paragraphs of their own, not more items of the 3.2.4 list (round 2: both were swallowed as extra items because 3.2.8 does not follow 3.2.4).

### Source

Review round 2, finding numbering/F3; real OCR page r3_swallow.

### The page (as the scan document has it)



3.2.4 Environmental conditions. The equipment shall meet each of the following:

a. Temperature: -40 C to +71 C operating.

b. Humidity: 95 percent, non-condensing.

c. Altitude: 0 to 15,000 m.

3.2.8 Weight. The equipment shall not exceed 45 kilograms.

3.2.9 Finish. The equipment shall be painted gloss white.

0.7 to 3.3 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The equipment shall meet each of the following: a. Temperature: -40 C to +71 C operating. | |
| 3.2.4 b | The equipment shall meet each of the following: b. Humidity: 95 percent, non-condensing. | |
| 3.2.4 c | The equipment shall meet each of the following: c. Altitude: 0 to 15,000 m. | |
| 3.2.8 | The equipment shall not exceed 45 kilograms. | |
| 3.2.9 | The equipment shall be painted gloss white. | |

# NUM-07  A numbered list straight before '4. Verification' and 4.1

Paragraph numbers and headings | document: documents/NUM-07_a_numbered_list_straight_before_4_verification_and_4_1.pdf | typewriter 11 pt, 300 dpi

### Situation

'1. Search mode.' to '3. Ground mapping mode.' under 3.2.4, then the heading '4. Verification' (no full stop) and '4.1 General. ...'.

### What the tool must do

Only '4. Verification' is a heading - the one followed by its own 4.1. Items 1 to 3 stay items (round 2: the whole run 1., 2., 3. and 4. was kept as headings).

### Source

Review round 2, finding contstem/a1c.

### The page (as the scan document has it)

```
3.2.4 Modes. The radar shall provide the following modes:

     1. Search mode.
     2. Track mode.
     3. Ground mapping mode.

4. Verification

4.1 General. The contractor shall verify each requirement of
section 3 by test.
```

0.7 to 3.2 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 1 | The radar shall provide the following modes: 1. Search mode. | |
| 3.2.4 2 | The radar shall provide the following modes: 2. Track mode. | |
| 3.2.4 3 | The radar shall provide the following modes: 3. Ground mapping mode. | |
| 4.1 | The contractor shall verify each requirement of section 3 by test. | Section: 4 Verification |

# NUM-08  A wrapped line that starts with the next paragraph's number

Paragraph numbers and headings | document: documents/NUM-08_a_wrapped_line_that_starts_with_the_next_paragraph_s_number.pdf | typewriter 11 pt, 300 dpi

### Situation

'3.2.1 Power. The unit shall draw its power as defined in paragraph' / '3.2.2 Input Power of MIL-STD-704F and shall not exceed 150 W.' - the line after 'paragraph' starts with 3.2.2, the number that would come next.

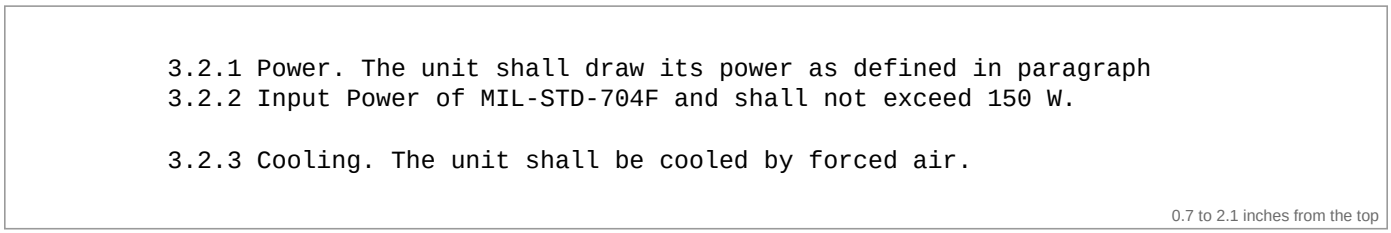
### What the tool must do

After 'paragraph' a number is a cross-reference: 3.2.1 is one paragraph, and there is no paragraph 3.2.2.

### Source

Review round 2 (layout, numbered_next).

### The page (as the scan document has it)



```
3.2.1 Power. The unit shall draw its power as defined in paragraph
3.2.2 Input Power of MIL-STD-704F and shall not exceed 150 W.

3.2.3 Cooling. The unit shall be cooled by forced air.
```

0.7 to 2.1 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.1 | The unit shall draw its power as defined in paragraph 3.2.2 Input Power of MIL-STD-704F and shall not exceed 150 W. | |
| 3.2.3 | The unit shall be cooled by forced air. | |

# NUM-09  Two numbered paragraphs with no blank line between them

Paragraph numbers and headings | document: documents/NUM-09_two_numbered_paragraphs_with_no_blank_line_between_them.pdf | typewriter 11 pt, 300 dpi

### Situation

'3.2.2.1 Weight. ...' and '3.2.2.2 Input power. ...' on consecutive lines, as a scan that lost the blank line between them shows them.

### What the tool must do

Two paragraphs, each with its own number and title - the second sentence is not filed under the first number.

### Source

README (paragraph numbers found inside a block).

### The page (as the scan document has it)

```
3.2.2.1 Weight. The unit shall not exceed 45 kg with its tray.
3.2.2.2 Input power. The unit shall draw less than 150 W.

3.2.2.3 Cooling. The unit shall be cooled by forced air.
```

0.7 to 2.1 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.2.1 | The unit shall not exceed 45 kg with its tray. | |
| 3.2.2.2 | The unit shall draw less than 150 W. | |
| 3.2.2.3 | The unit shall be cooled by forced air. | |

# NUM-10  A numbered list of documents at the margin

Paragraph numbers and headings | document: documents/NUM-10_a_numbered_list_of_documents_at_the_margin.pdf | typewriter 11 pt, 300 dpi

### Situation

'2.1 ... The following documents shall apply:' then '1. MIL-STD-461G' and '2. MIL-STD-704F' at the left margin (all capitals), a blank line, then 2.2.

### What the tool must do

The two lines are items of the list, not section headings '1' and '2' (round 3: all-capital lines in sequence were called headings and the list was lost).

### Source

Review round 3, rule 2 (b12_doclist_margin).

### The page (as the scan document has it)

```
2.1 Government documents. The following documents shall apply:
1. MIL-STD-461G
2. MIL-STD-704F

2.2 Order of precedence. The text of this specification shall take
precedence.
```

0.7 to 2.4 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 2.1 1 | The following documents shall apply: 1. MIL-STD-461G | |
| 2.1 2 | The following documents shall apply: 2. MIL-STD-704F | |
| 2.2 | The text of this specification shall take precedence. | |

# NUM-11  A title page with a document-control block

Paragraph numbers and headings | document: documents/NUM-11_a_title_page_with_a_document_control_block.pdf | typewriter 11 pt, 300 dpi

### Situation

A centred title, a block of 'Document:', 'Revision:', 'Date:', 'Prepared by:', 'Approved by:', 'Distribution:' with their values on a tab stop, then '1. SCOPE' and paragraph 1.1.

### What the tool must do

Only the requirement of 1.1 is a row; the control block is not a list or a requirement.

### Source

Review round 1 (layout, title page); real OCR page title_mono.

### The page (as the scan document has it)

```
                  SYSTEM SPECIFICATION FOR THE
                      AIRBORNE RADAR UNIT


               Document:             SPEC-1234

               Revision:             C

               Date:                 12 March 1998

               Prepared by:          J. Smith
                                     Systems Engineering Department

               Approved by:          R. Jones

               Distribution:         3 copies to the program office,
                                     1 copy to the test site.

      1. SCOPE

      1.1 Scope. This specification establishes the requirements for
      the airborne radar unit. The unit shall be designed for
      installation in the aircraft nose bay.
```

0.7 to 5.2 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 1.1 | The unit shall be designed for installation in the aircraft nose bay. | Section: 1 SCOPE |

# NUM-12 A table of contents with dot leaders

Paragraph numbers and headings | document: documents/NUM-12_a_table_of_contents_with_dot_leaders.pdf | typewriter 11 pt, 300 dpi

**Situation**

'TABLE OF CONTENTS', then numbers, titles with dot leaders and page numbers ('3.2.1 Performance ..... 4').

**What the tool must do**

No requirement rows and no sections come from the contents list; it is given as one Table item.

**Source**

Review round 1 (layout, table of contents); real OCR page toc_mono.

**The page (as the scan document has it)**

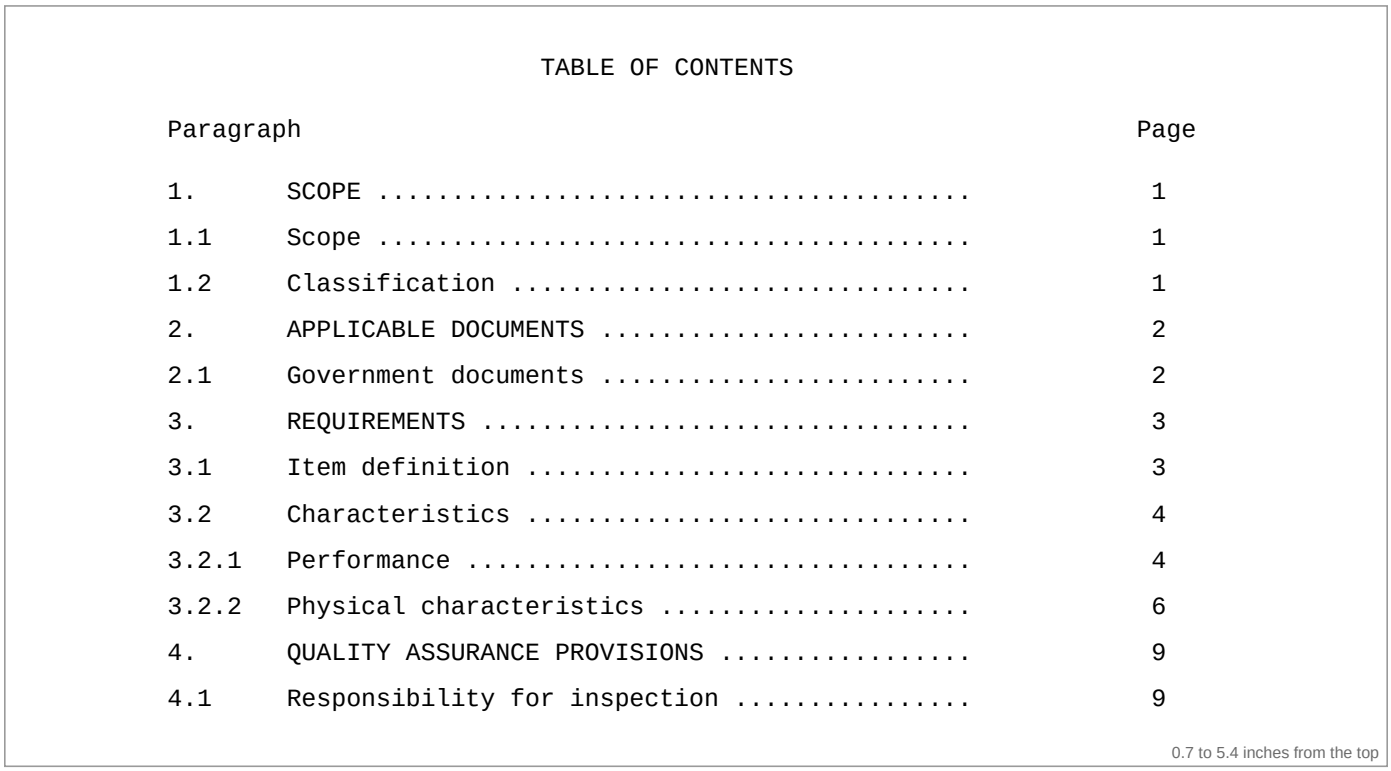


TABLE OF CONTENTS

| Paragraph | | Page |
|---|---|---|
| 1. | SCOPE ..... | 1 |
| 1.1 | Scope ..... | 1 |
| 1.2 | Classification ..... | 1 |
| 2. | APPLICABLE DOCUMENTS ..... | 2 |
| 2.1 | Government documents ..... | 2 |
| 3. | REQUIREMENTS ..... | 3 |
| 3.1 | Item definition ..... | 3 |
| 3.2 | Characteristics ..... | 4 |
| 3.2.1 | Performance ..... | 4 |
| 3.2.2 | Physical characteristics ..... | 6 |
| 4. | QUALITY ASSURANCE PROVISIONS ..... | 9 |
| 4.1 | Responsibility for inspection ..... | 9 |

0.7 to 5.4 inches from the top

**The rows the tool must produce**

*No Requirement rows.*

Also expected: 1 Table item(s), containing 'QUALITY ASSURANCE PROVISIONS'.

# NUM-13  A paragraph number run into its title ('4.2Extended verification.')

Paragraph numbers and headings | document: documents/NUM-13_a_paragraph_number_run_into_its_title_4_2extended_verificati.pdf | typewriter 11 pt, 300 dpi

### Situation

'4.2Extended verification. The contractor shall ...' - the scan lost the space after the number.

### What the tool must do

The number is still split off: Doc Ref '4.2'.

### Source

README (a faint scan can lose the space after a paragraph number).

### The page (as the scan document has it)

```
4.1 General. The contractor shall verify each requirement by test.

4.2Extended verification. The contractor shall repeat each test
at the highest and the lowest operating temperature.
```

0.7 to 2.1 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 4.1 | The contractor shall verify each requirement by test. | |
| 4.2 | The contractor shall repeat each test at the highest and the lowest operating temperature. | |

# Paragraphs around lists

Paragraphs around lists: what comes after the last item - an ordinary paragraph with its first line indented, a sub-paragraph, a sentence in the items' text column, a NOTE - must not be swallowed by the last item, and the last item must keep all of its own lines. Also paragraphs that are numbered or lettered in a hanging indent.

## PARA-01  Paragraphs with a first-line indent after a list at the margin

Paragraphs around lists | document: documents/PARA-01_paragraphs_with_a_first_line_indent_after_a_list_at_the_marg.pdf | typewriter 11 pt, 300 dpi

**Situation**

'(a) Search mode.' to '(c) Ground mapping mode.' at the margin, then two ordinary paragraphs whose first line is indented and whose second line is back at the margin, then 3.2.5.

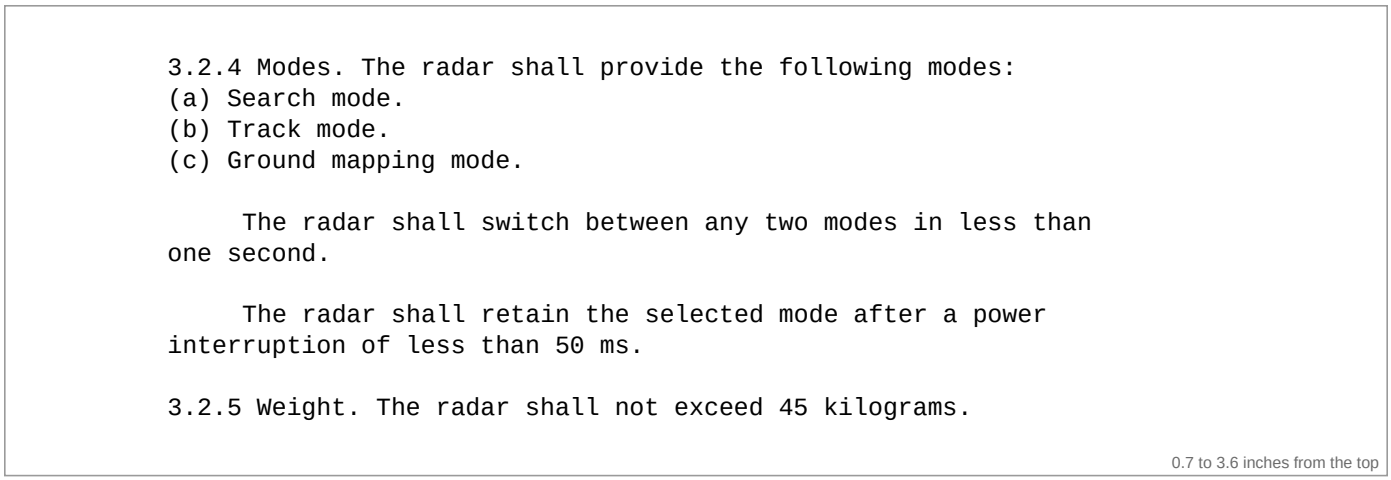
**What the tool must do**

The paragraphs are not more of item (c): each is a row of its own (they have no number of their own, so their Doc Ref is empty).

**Source**

Review round 1 (layout, hanging indents); real OCR page hang_paren.

**The page (as the scan document has it)**



```
3.2.4 Modes. The radar shall provide the following modes:
(a) Search mode.
(b) Track mode.
(c) Ground mapping mode.

     The radar shall switch between any two modes in less than
one second.

     The radar shall retain the selected mode after a power
interruption of less than 50 ms.

3.2.5 Weight. The radar shall not exceed 45 kilograms.
```

0.7 to 3.6 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The radar shall provide the following modes: (a) Search mode. | |
| 3.2.4 b | The radar shall provide the following modes: (b) Track mode. | |
| 3.2.4 c | The radar shall provide the following modes: (c) Ground mapping mode. | |
| (empty) | The radar shall switch between any two modes in less than one second. | |
| (empty) | The radar shall retain the selected mode after a power interruption of less than 50 ms. | |
| 3.2.5 | The radar shall not exceed 45 kilograms. | |

# PARA-02  A numbered sub-paragraph indented under a list

Paragraphs around lists | document: documents/PARA-02_a_numbered_sub_paragraph_indented_under_a_list.pdf | typewriter 11 pt, 300 dpi

**Situation**

'a. Search mode.' / 'b. Track mode.', a blank line, then '3.2.4.1 Mode switching. The radar shall switch ...' indented to the items' text column, then 3.2.5.

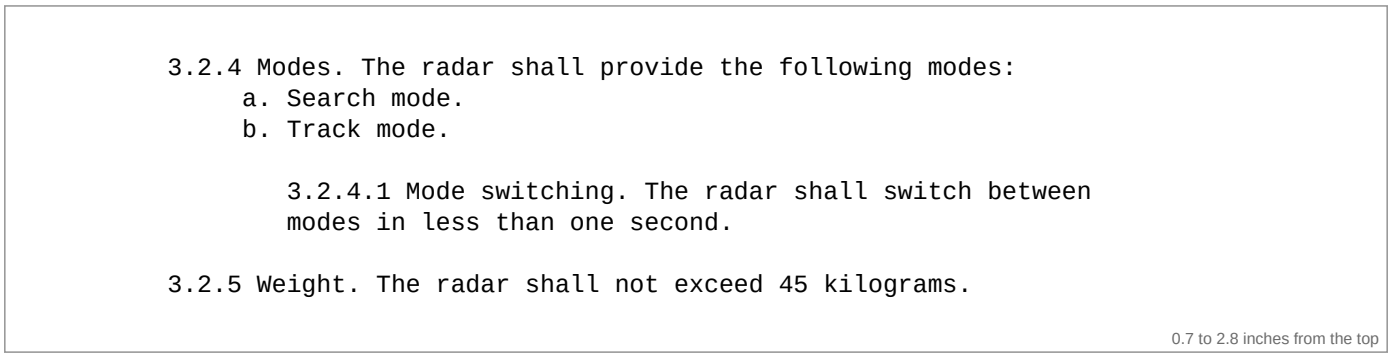
**What the tool must do**

3.2.4.1 is a paragraph of its own with its own number - not more of item b.

**Source**

Review round 1 (layout, hanging indents); real OCR page hang_numbered.

**The page (as the scan document has it)**



```
3.2.4 Modes. The radar shall provide the following modes:
     a. Search mode.
     b. Track mode.

        3.2.4.1 Mode switching. The radar shall switch between
        modes in less than one second.

3.2.5 Weight. The radar shall not exceed 45 kilograms.
```

0.7 to 2.8 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The radar shall provide the following modes: a. Search mode. | |
| 3.2.4 b | The radar shall provide the following modes: b. Track mode. | |
| 3.2.4.1 | The radar shall switch between modes in less than one second. | |
| 3.2.5 | The radar shall not exceed 45 kilograms. | |

# PARA-03  A one-line indented paragraph after a list at the margin

Paragraphs around lists | document: documents/PARA-03_a_one_line_indented_paragraph_after_a_list_at_the_margin.pdf | typewriter 11 pt, 300 dpi

### Situation

'a.', 'b.', 'c.' at the margin, a blank line, then one sentence indented eight characters, then 3.2.5.

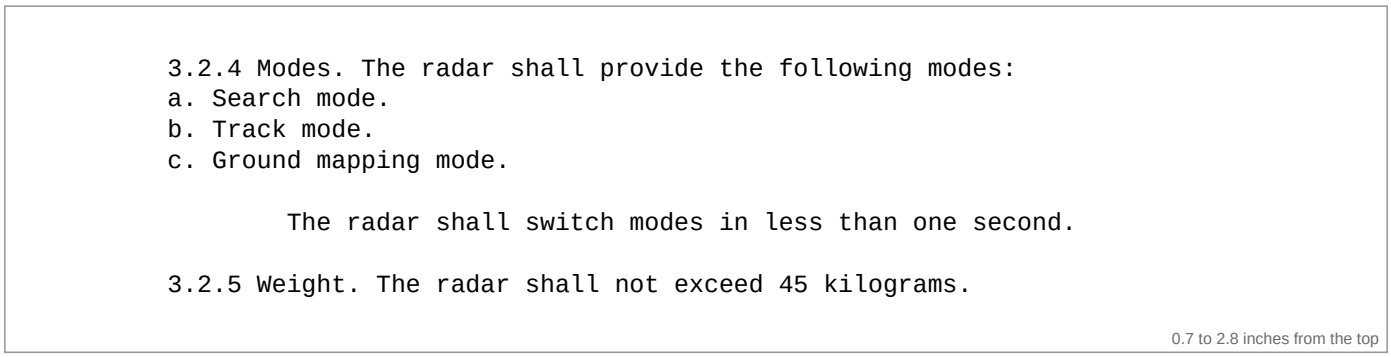
### What the tool must do

The sentence is a requirement of its own, not more of item c. It sits between the list and the next numbered paragraph, so it keeps the list's lead-in and paragraph number (Doc Ref '3.2.4', no letter).

### Source

Review round 1, finding lists/F3; real OCR page indented_para.

### The page (as the scan document has it)



```
3.2.4 Modes. The radar shall provide the following modes:
a. Search mode.
b. Track mode.
c. Ground mapping mode.

        The radar shall switch modes in less than one second.

3.2.5 Weight. The radar shall not exceed 45 kilograms.
```

0.7 to 2.8 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The radar shall provide the following modes: a. Search mode. | |
| 3.2.4 b | The radar shall provide the following modes: b. Track mode. | |
| 3.2.4 c | The radar shall provide the following modes: c. Ground mapping mode. | |
| 3.2.4 | The radar shall provide the following modes: The radar shall switch modes in less than one second. | |
| 3.2.5 | The radar shall not exceed 45 kilograms. | |

# PARA-04  A paragraph after a last item of three lines

Paragraphs around lists | document: documents/PARA-04_a_paragraph_after_a_last_item_of_three_lines.pdf | typewriter 11 pt, 300 dpi

### Situation

Item c. wraps onto three lines; after a blank line, a two-line sentence in the items' text column; then 3.2.5.

### What the tool must do

The blank line ends item c: the sentence keeps its own row (with the lead-in, no letter) and item c keeps its three lines (round 2: the three-line item's line spacing spanned the blank line and the sentence was joined to it).

### Source

Review round 2, finding contstem/a5; real OCR page para_after_long_item.

### The page (as the scan document has it)

```
3.2.4 Modes. The radar shall provide the following modes:
     a. Search mode.
     b. Track mode.
     c. Ground mapping mode, with the antenna tilted down by
        the angle set by the operator and the range scale set
        to 10, 20 or 40 km.

        The radar shall switch between any two modes in less
        than one second.

3.2.5 Weight. The radar shall weigh less than 45 kg.
```

0.7 to 3.4 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The radar shall provide the following modes: a. Search mode. | |
| 3.2.4 b | The radar shall provide the following modes: b. Track mode. | |
| 3.2.4 c | The radar shall provide the following modes: c. Ground mapping mode, with the antenna tilted down by the angle set by the operator and the range scale set to 10, 20 or 40 km. | |
| 3.2.4 | The radar shall provide the following modes: The radar shall switch between any two modes in less than one second. | |
| 3.2.5 | The radar shall weigh less than 45 kg. | |

# PARA-05  A NOTE on tab stops after a list

Paragraphs around lists | document: documents/PARA-05_a_note_on_tab_stops_after_a_list.pdf | typewriter 11 pt, 300 dpi

### Situation

The list, a blank line, then 'NOTE:' at the margin with its text on a tab stop over two lines, then 3.2.5.

### What the tool must do

The NOTE is a row of its own, whole, and is not joined to item c (round 1: the note's text went into the list).

### Source

Review round 1 (layout, notes); real OCR page note_after_list.

### The page (as the scan document has it)

```
3.2.4 Modes. The radar shall provide the following modes:
     a. Search mode.
     b. Track mode.
     c. Ground mapping mode.

NOTE:   The operator shall confirm each mode change on the
        control panel before it takes effect.

3.2.5 Weight. The radar shall not exceed 45 kilograms.
```

0.7 to 3.0 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The radar shall provide the following modes: a. Search mode. | |
| 3.2.4 b | The radar shall provide the following modes: b. Track mode. | |
| 3.2.4 c | The radar shall provide the following modes: c. Ground mapping mode. | |
| (empty) | NOTE: The operator shall confirm each mode change on the control panel before it takes effect. | |
| 3.2.5 | The radar shall not exceed 45 kilograms. | |

# PARA-06  'Note.' level with the list's markers

Paragraphs around lists | document: documents/PARA-06_note_level_with_the_list_s_markers.pdf | typewriter 11 pt, 300 dpi

### Situation

'a.', 'b.', 'c.' at the margin and straight under them, with no blank line, 'Note. The operator shall confirm each mode change.'

### What the tool must do

The note is not item c's continuation and is not given a letter: a row of its own ('Note' is taken as its title).

### Source

Review round 1 (layout, notes); real OCR page note_level.

### The page (as the scan document has it)

```
3.2.4 Modes. The radar shall provide the following modes:
a. Search mode.
b. Track mode.
c. Ground mapping mode.
Note. The operator shall confirm each mode change.

3.2.5 Weight. The radar shall not exceed 45 kilograms.
```

0.7 to 2.6 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The radar shall provide the following modes: a. Search mode. | |
| 3.2.4 b | The radar shall provide the following modes: b. Track mode. | |
| 3.2.4 c | The radar shall provide the following modes: c. Ground mapping mode. | |
| (empty) | The operator shall confirm each mode change. | |
| 3.2.5 | The radar shall not exceed 45 kilograms. | |

# PARA-07  Numbered items without full stops, then a paragraph

Paragraphs around lists | document: documents/PARA-07_numbered_items_without_full_stops_then_a_paragraph.pdf | typewriter 11 pt, 300 dpi

### Situation

A paragraph, then '1. Search mode' / '2. Track mode' (no full stops), a blank line, then 'The operator shall select the mode from the front panel.' in the items' text column.

### What the tool must do

The sentence is a requirement of its own (with the lead-in, no number), not more of item 2 (round 3: an item without a full stop looked unfinished and the sentence was glued to it). Lettered items: LST-25.

### Source

Review round 3, finding B3; real OCR page r8_numbered_noperiod_para.

### The page (as the scan document has it)

```
3.2.5 Controls. The display shall have a brightness control and a
contrast control on its front panel. Each control shall be
illuminated for night use and shall be operable with gloved hands
in all environmental conditions specified herein.

3.2.6 Modes. The radar shall provide the following modes:
     1. Search mode
     2. Track mode

        The operator shall select the mode from the front panel.

3.2.7 Mass. The radar shall weigh less than 45 kg. The mass shall
include the mounting tray and all cables supplied with the unit.
```

0.7 to 3.8 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.5 | The display shall have a brightness control and a contrast control on its front panel. | |
| 3.2.5 | Each control shall be illuminated for night use and shall be operable with gloved hands in all environmental conditions specified herein. | |
| 3.2.6 1 | The radar shall provide the following modes: 1. Search mode | |
| 3.2.6 2 | The radar shall provide the following modes: 2. Track mode | |
| 3.2.6 | The radar shall provide the following modes: The operator shall select the mode from the front panel. | |
| 3.2.7 | The radar shall weigh less than 45 kg. | |
| 3.2.7 | The mass shall include the mounting tray and all cables supplied with the unit. | |

# PARA-08  Paragraph numbers set apart from their text (hanging indent)

Paragraphs around lists | document: documents/PARA-08_paragraph_numbers_set_apart_from_their_text_hanging_indent.pdf | typewriter 11 pt, 300 dpi

### Situation

'3.2.1' at the margin and its text ('Power. The unit shall ...') on a tab stop an inch further in, for 3.2.1 to 3.2.3; then the heading '3.3 Interfaces' and two ordinary paragraphs.

### What the tool must do

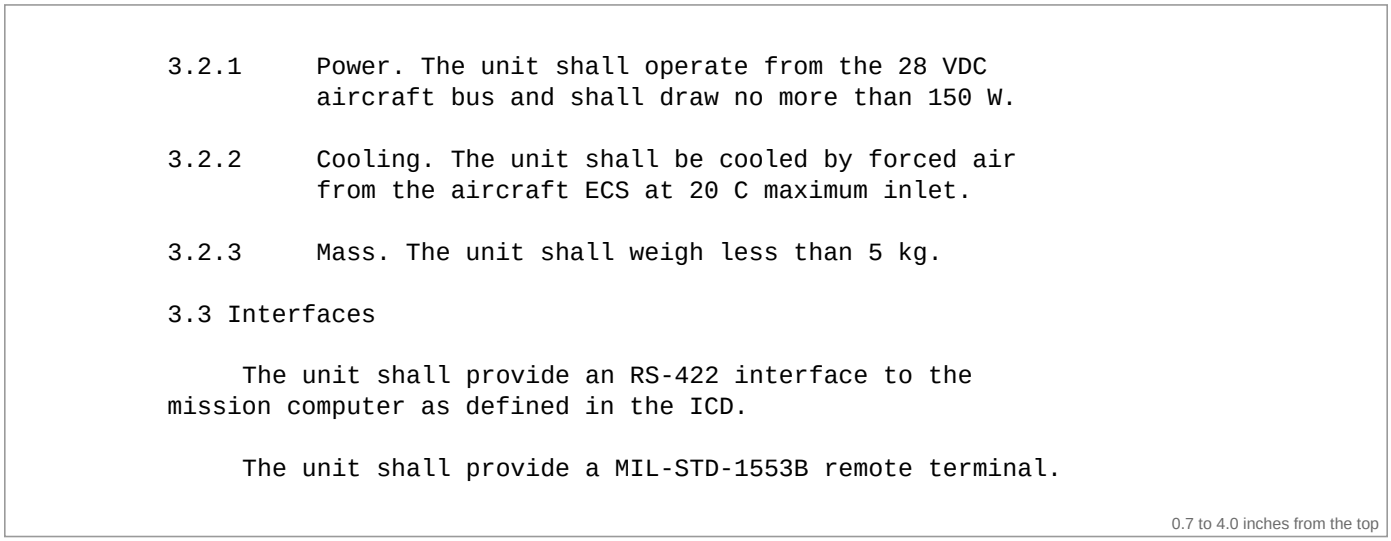
Each numbered paragraph is one row with its number as the Doc Ref and its title split off ('The unit shall operate from ...'); no table; the two paragraphs are in section '3.3 Interfaces'.

### Source

Review round 1 (layout); real OCR page hanging_numbers.

**KNOWN LIMITATION** - the tool still gets this case wrong: The numbers and their text are also read as a two-column table (an extra Table item), and each title stays at the start of the text ('Power. The unit shall operate ...').

### The page (as the scan document has it)



```
3.2.1     Power. The unit shall operate from the 28 VDC
          aircraft bus and shall draw no more than 150 W.

3.2.2     Cooling. The unit shall be cooled by forced air
          from the aircraft ECS at 20 C maximum inlet.

3.2.3     Mass. The unit shall weigh less than 5 kg.

3.3 Interfaces

     The unit shall provide an RS-422 interface to the
mission computer as defined in the ICD.

     The unit shall provide a MIL-STD-1553B remote terminal.
```

0.7 to 4.0 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.1 | The unit shall operate from the 28 VDC aircraft bus and shall draw no more than 150 W. | |
| 3.2.2 | The unit shall be cooled by forced air from the aircraft ECS at 20 C maximum inlet. | |
| 3.2.3 | The unit shall weigh less than 5 kg. | |
| (empty) | The unit shall provide an RS-422 interface to the mission computer as defined in the ICD. | Section: 3.3 Interfaces |
| (empty) | The unit shall provide a MIL-STD-1553B remote terminal. | Section: 3.3 Interfaces |

# PARA-09  Lettered paragraphs 'A. General:' under a heading

Paragraphs around lists | document: documents/PARA-09_lettered_paragraphs_a_general_under_a_heading.pdf | typewriter 11 pt, 300 dpi

### Situation

The heading '3.1 Installation.', then 'A. General: The contractor shall provide all labor, materials' with its next line back at the margin, and 'B. Cabling: ...' the same way; then 3.2.

### What the tool must do

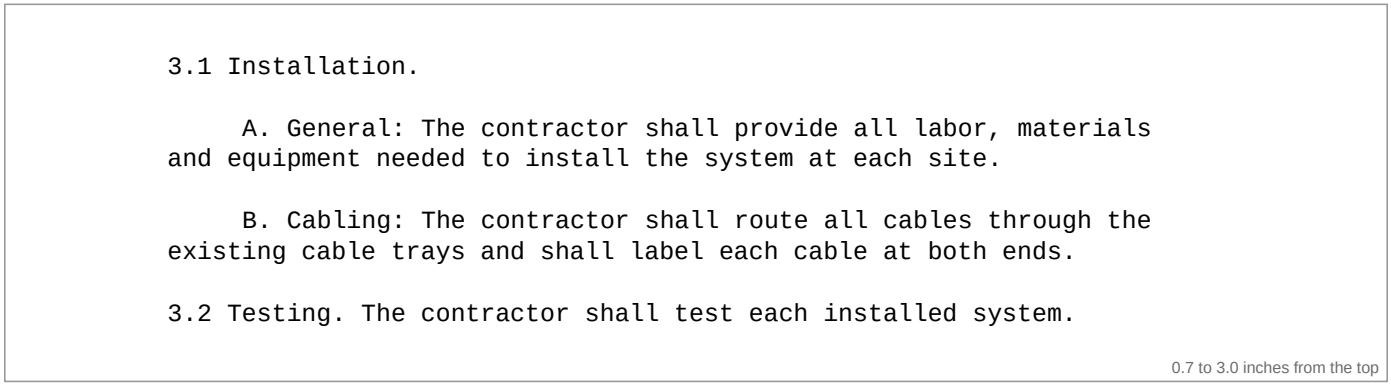
Each lettered paragraph is one whole row (round 3: the sentence was cut after 'materials' and the rest dropped). Its letter is kept: Doc Ref '3.1 A' / '3.1 B', as for the items of a list.

### Source

Review round 3, finding F6; real OCR page lettered_para.

**KNOWN LIMITATION** - the tool still gets this case wrong: The letters are lost: both rows have an empty Doc Ref and their text starts at the label ('General: The contractor shall ...'). The text itself is complete.

### The page (as the scan document has it)



```
3.1 Installation.

     A. General: The contractor shall provide all labor, materials
and equipment needed to install the system at each site.

     B. Cabling: The contractor shall route all cables through the
existing cable trays and shall label each cable at both ends.

3.2 Testing. The contractor shall test each installed system.
```

0.7 to 3.0 inches from the top

### The rows the tool must produce

| **Doc Ref** | **Content** (the lead-in in grey) | **Also checked** |
|---|---|---|
| 3.1 A | A. General: The contractor shall provide all labor, materials and equipment needed to install the system at each site. | |
| 3.1 B | B. Cabling: The contractor shall route all cables through the existing cable trays and shall label each cable at both ends. | |
| 3.2 | The contractor shall test each installed system. | |

# Two-column pages

Two-column pages: each column must be read on its own, top to bottom, and a narrow column of prose or a short list beside it must not be taken for the wide tab stops of a list (or the other way round). Several pages are set in a 10 pt typewriter face, as two-column typed pages usually are.

## COL-01  Two columns of prose with a tight list in the left column (control)

Two-column pages | document: documents/COL-01_two_columns_of_prose_with_a_tight_list_in_the_left_column_co.pdf | proportional 12 pt, 200 dpi

**Situation**

Proportional font, 200 dpi. '3.9 Construction.' across the page, then a left column 3.9.1 to 3.9.3 (with the list a. to c. under 3.9.2) and a right column 3.9.4 to 3.9.6, 0.4 inch apart.

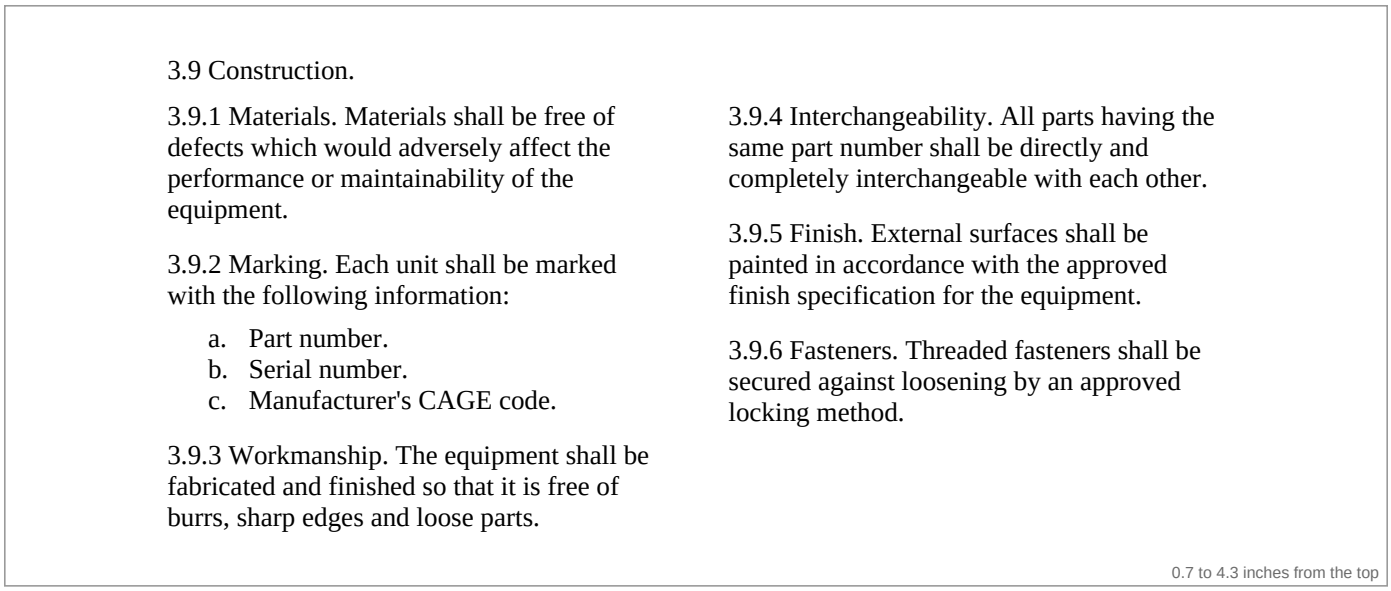
**What the tool must do**

Eight rows in column order: the whole left column, then the right column. Nothing from the right column is joined to the left.

**Source**

List suite p09 (control).

**The page (as the scan document has it)**



3.9 Construction.

3.9.1 Materials. Materials shall be free of defects which would adversely affect the performance or maintainability of the equipment.

3.9.2 Marking. Each unit shall be marked with the following information:

a. Part number.
b. Serial number.
c. Manufacturer's CAGE code.

3.9.3 Workmanship. The equipment shall be fabricated and finished so that it is free of burrs, sharp edges and loose parts.

3.9.4 Interchangeability. All parts having the same part number shall be directly and completely interchangeable with each other.

3.9.5 Finish. External surfaces shall be painted in accordance with the approved finish specification for the equipment.

3.9.6 Fasteners. Threaded fasteners shall be secured against loosening by an approved locking method.

0.7 to 4.3 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.9.1 | Materials shall be free of defects which would adversely affect the performance or maintainability of the equipment. | |
| 3.9.2 a | Each unit shall be marked with the following information: a. Part number. | |
| 3.9.2 b | Each unit shall be marked with the following information: b. Serial number. | |
| 3.9.2 c | Each unit shall be marked with the following information: c. Manufacturer's CAGE code. | |
| 3.9.3 | The equipment shall be fabricated and finished so that it is free of burrs, sharp edges and loose parts. | |
| 3.9.4 | All parts having the same part number shall be directly and completely interchangeable with each other. | |
| 3.9.5 | External surfaces shall be painted in accordance with the approved finish specification for the equipment. | |
| 3.9.6 | Threaded fasteners shall be secured against loosening by an approved locking method. | |

# COL-02 A list of documents in the left column, prose in the right

Two-column pages | document: documents/COL-02_a_list_of_documents_in_the_left_column_prose_in_the_right.pdf | typewriter 10 pt, 300 dpi

### Situation

10 pt typewriter. Left column: '2.1 Government documents. The following documents form a part of this specification:' and ten documents a. to j.; right column: paragraphs 3.1 to 3.3.

### What the tool must do

The right column's paragraphs are whole and in order; nothing from the list is mixed into them (round 1: the gutter was taken for the tab stops of a list and the columns were read across).

### Source

Review round 1 (layout, list-heavy two-column pages); real OCR page twocol_docs_mono.

### The page (as the scan document has it)

```
2.1 Government documents. The
following documents form a part
of this specification:
a. MIL-STD-461G
b. MIL-STD-704F
c. MIL-STD-810H
d. MIL-STD-1275E
e. MIL-STD-1553B
f. MIL-STD-883L
g. MIL-HDBK-217F
h. MIL-PRF-38534
i. MIL-DTL-38999
j. MIL-C-17

3.1 Power. The unit shall operate
from the 28 VDC aircraft bus
defined in MIL-STD-704F and shall
draw no more than 150 W in any
operating mode, including warm-up.

3.2 Cooling. The unit shall be
cooled by forced air from the
aircraft ECS and shall not require
any liquid coolant during ground
running or flight at any altitude.

3.3 Mass. The unit shall weigh
less than 5 kg including its tray.
```

0.7 to 3.7 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.1 | The unit shall operate from the 28 VDC aircraft bus defined in MIL-STD-704F and shall draw no more than 150 W in any operating mode, including warm-up. | |
| 3.2 | The unit shall be cooled by forced air from the aircraft ECS and shall not require any liquid coolant during ground running or flight at any altitude. | |
| 3.3 | The unit shall weigh less than 5 kg including its tray. | |

# COL-03  A list of documents in the left column, prose in the right, proportional font

Two-column pages | document: documents/COL-03_a_list_of_documents_in_the_left_column_prose_in_the_right_pr.pdf | proportional 11 pt, 300 dpi

**Situation**

COL-02 in an 11 pt proportional (serif) font.

**What the tool must do**

As COL-02.

**Source**

Review round 1 (layout); real OCR page twocol_docs_serif.

**The page (as the scan document has it)**

2.1 Government documents. The following documents form a part of this specification:
a. MIL-STD-461G
b. MIL-STD-704F
c. MIL-STD-810H
d. MIL-STD-1275E
e. MIL-STD-1553B
f. MIL-STD-883L
g. MIL-HDBK-217F
h. MIL-PRF-38534
i. MIL-DTL-38999
j. MIL-C-17

3.1 Power. The unit shall operate from the 28 VDC aircraft bus defined in MIL-STD-704F and shall draw no more than 150 W in any operating mode, including warm-up.

3.2 Cooling. The unit shall be cooled by forced air from the aircraft ECS and shall not require any liquid coolant during ground running or flight at any altitude.

3.3 Mass. The unit shall weigh less than 5 kg including its tray.

0.7 to 3.8 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.1 | The unit shall operate from the 28 VDC aircraft bus defined in MIL-STD-704F and shall draw no more than 150 W in any operating mode, including warm-up. | |
| 3.2 | The unit shall be cooled by forced air from the aircraft ECS and shall not require any liquid coolant during ground running or flight at any altitude. | |
| 3.3 | The unit shall weigh less than 5 kg including its tray. | |

# COL-04  A lead-in and short labelled items in the left column, prose in the right

Two-column pages | document: documents/COL-04_a_lead_in_and_short_labelled_items_in_the_left_column_prose_.pdf | typewriter 10 pt, 300 dpi

### Situation

10 pt typewriter. Left: '3.4 Modes. The radar shall provide the following operating modes:' and seven items 'a. Search:' to 'g. Off:'; right: paragraphs 3.5 and 3.6.

### What the tool must do

Seven list rows under 3.4, then 3.5 and 3.6 whole.

### Source

Review round 1 (layout); real OCR page twocol_modes_mono.

### The page (as the scan document has it)

```
3.4 Modes. The radar shall provide        3.5 Power. The unit shall operate
the following operating modes:            from the 28 VDC aircraft bus
a. Search:                                defined in MIL-STD-704F and shall
b. Track:                                 draw no more than 150 W in any
c. Map:                                   operating mode, including warm-up.
d. Standby:
e. Test:                                  3.6 Cooling. The unit shall be
f. Maintenance:                           cooled by forced air from the
g. Off:                                   aircraft ECS.
```

0.7 to 2.8 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.4 a | The radar shall provide the following operating modes: a. Search: | |
| 3.4 b | The radar shall provide the following operating modes: b. Track: | |
| 3.4 c | The radar shall provide the following operating modes: c. Map: | |
| 3.4 d | The radar shall provide the following operating modes: d. Standby: | |
| 3.4 e | The radar shall provide the following operating modes: e. Test: | |
| 3.4 f | The radar shall provide the following operating modes: f. Maintenance: | |
| 3.4 g | The radar shall provide the following operating modes: g. Off: | |
| 3.5 | The unit shall operate from the 28 VDC aircraft bus defined in MIL-STD-704F and shall draw no more than 150 W in any operating mode, including warm-up. | |
| 3.6 | The unit shall be cooled by forced air from the aircraft ECS. | |

# COL-05  A list set in two columns (a-c left, d-f right)

Two-column pages | document: documents/COL-05_a_list_set_in_two_columns_a_c_left_d_f_right.pdf | typewriter 11 pt, 300 dpi

### Situation

A page-wide lead-in, then the items a. to c. in a left column and d. to f. in a right column, then 3.2.8.

### What the tool must do

The six items in order a to f (column by column), each with the lead-in; 3.2.8 after them.

### Source

Review round 2 (layout, column-major list); real OCR page colmajor_mono.

### The page (as the scan document has it)

```
3.2.7 Documents. The unit shall comply with each of the following:

    a. MIL-STD-461G                  d. MIL-STD-1275E
    b. MIL-STD-704F                  e. MIL-STD-1553B
    c. MIL-STD-810H                  f. MIL-STD-883L

3.2.8 Weight. The unit shall not exceed 45 kilograms.
```

0.7 to 2.5 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.7 a | The unit shall comply with each of the following: a. MIL-STD-461G | |
| 3.2.7 b | The unit shall comply with each of the following: b. MIL-STD-704F | |
| 3.2.7 c | The unit shall comply with each of the following: c. MIL-STD-810H | |
| 3.2.7 d | The unit shall comply with each of the following: d. MIL-STD-1275E | |
| 3.2.7 e | The unit shall comply with each of the following: e. MIL-STD-1553B | |
| 3.2.7 f | The unit shall comply with each of the following: f. MIL-STD-883L | |
| 3.2.8 | The unit shall not exceed 45 kilograms. | |

# COL-06  A list set in two columns, proportional font

Two-column pages | document: documents/COL-06_a_list_set_in_two_columns_proportional_font.pdf | proportional 11 pt, 300 dpi

**Situation**

COL-05 in a proportional (serif) font.

**What the tool must do**

As COL-05.

**Source**

Review round 2 (layout); real OCR page colmajor_serif.

**The page (as the scan document has it)**

3.2.7 Documents. The unit shall comply with each of the following:

a. MIL-STD-461G
b. MIL-STD-704F
c. MIL-STD-810H
d. MIL-STD-1275E
e. MIL-STD-1553B
f. MIL-STD-883L

3.2.8 Weight. The unit shall not exceed 45 kilograms.

0.7 to 2.5 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.7 a | The unit shall comply with each of the following: a. MIL-STD-461G | |
| 3.2.7 b | The unit shall comply with each of the following: b. MIL-STD-704F | |
| 3.2.7 c | The unit shall comply with each of the following: c. MIL-STD-810H | |
| 3.2.7 d | The unit shall comply with each of the following: d. MIL-STD-1275E | |
| 3.2.7 e | The unit shall comply with each of the following: e. MIL-STD-1553B | |
| 3.2.7 f | The unit shall comply with each of the following: f. MIL-STD-883L | |
| 3.2.8 | The unit shall not exceed 45 kilograms. | |

# COL-07  A short list beside a narrow column of prose

Two-column pages | document: documents/COL-07_a_short_list_beside_a_narrow_column_of_prose.pdf | typewriter 10 pt, 300 dpi

### Situation

10 pt typewriter. Left: a list of seven documents (no lead-in); right: a column of prose 29 characters wide.

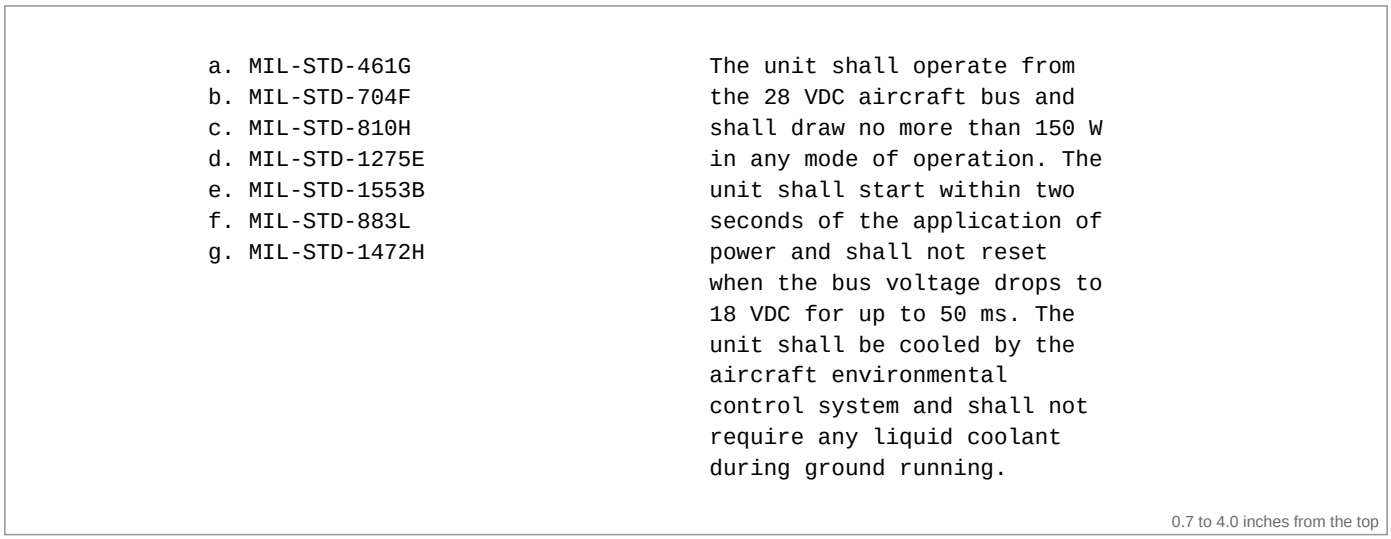
### What the tool must do

The three sentences of the prose, whole; the list is not read across into them (round 3: the narrow prose looked like the value column of a list).

### Source

Review round 3, finding F2; real OCR page twocol_narrow_mono.

### The page (as the scan document has it)



```
a. MIL-STD-461G                  The unit shall operate from
b. MIL-STD-704F                  the 28 VDC aircraft bus and
c. MIL-STD-810H                  shall draw no more than 150 W
d. MIL-STD-1275E                 in any mode of operation. The
e. MIL-STD-1553B                 unit shall start within two
f. MIL-STD-883L                  seconds of the application of
g. MIL-STD-1472H                 power and shall not reset
                                 when the bus voltage drops to
                                 18 VDC for up to 50 ms. The
                                 unit shall be cooled by the
                                 aircraft environmental
                                 control system and shall not
                                 require any liquid coolant
                                 during ground running.
```

0.7 to 4.0 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| (empty) | The unit shall operate from the 28 VDC aircraft bus and shall draw no more than 150 W in any mode of operation. | |
| (empty) | The unit shall start within two seconds of the application of power and shall not reset when the bus voltage drops to 18 VDC for up to 50 ms. | |
| (empty) | The unit shall be cooled by the aircraft environmental control system and shall not require any liquid coolant during ground running. | |

# COL-08  A short list beside a ragged column of typed prose

Two-column pages | document: documents/COL-08_a_short_list_beside_a_ragged_column_of_typed_prose.pdf | typewriter 10 pt, 300 dpi

### Situation

As COL-07, but the prose is typed with a ragged right edge (lines of 27 to 35 characters).

### What the tool must do

As COL-07 (round 3: lines that stopped short were taken as finished rows of a list).

### Source

Review round 3, finding F3; real OCR page twocol_ragged_mono.

### The page (as the scan document has it)

```
a. MIL-STD-461G              The unit shall operate from
b. MIL-STD-704F              the 28 VDC aircraft bus and shall
c. MIL-STD-810H              draw no more than 150 W in
d. MIL-STD-1275E             any mode of operation. The unit
e. MIL-STD-1553B             shall start within two seconds of
f. MIL-STD-883L              the application of power and
g. MIL-STD-1472H             shall not reset when the
                             bus voltage drops to 18 VDC for up
                             to 50 ms. The unit shall be
                             cooled by the aircraft
                             environmental control system
                             and shall not require any
                             liquid coolant during ground
                             running.
```

0.7 to 4.0 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| (empty) | The unit shall operate from the 28 VDC aircraft bus and shall draw no more than 150 W in any mode of operation. | |
| (empty) | The unit shall start within two seconds of the application of power and shall not reset when the bus voltage drops to 18 VDC for up to 50 ms. | |
| (empty) | The unit shall be cooled by the aircraft environmental control system and shall not require any liquid coolant during ground running. | |

# COL-09  Two lists side by side

Two-column pages | document: documents/COL-09_two_lists_side_by_side.pdf | typewriter 10 pt, 300 dpi

### Situation

10 pt typewriter. Left: five documents a. to e.; right: 'a. The radar shall search.' to 'e. The radar shall answer.'

### What the tool must do

The right list's five requirements, each on its own row with its own marker; the left list is not mixed in (round 3: the two lists were read across).

### Source

Review round 3, finding F5; real OCR page twocol_two_lists_mono.

### The page (as the scan document has it)

```
a. MIL-STD-461G                  a. The radar shall search.
b. MIL-STD-704F                  b. The radar shall track.
c. MIL-STD-810H                  c. The radar shall map.
d. MIL-STD-1275E                 d. The radar shall show weather.
e. MIL-STD-1553B                 e. The radar shall answer.
```

0.7 to 2.2 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| a. | The radar shall search. | |
| b. | The radar shall track. | |
| c. | The radar shall map. | |
| d. | The radar shall show weather. | |
| e. | The radar shall answer. | |

# COL-10  A list beside prose that has a small list of its own

Two-column pages | document: documents/COL-10_a_list_beside_prose_that_has_a_small_list_of_its_own.pdf | typewriter 10 pt, 300 dpi

### Situation

10 pt typewriter. Left: seven documents; right: '3.4 Modes. The radar shall have the following operating modes:' with 'a. search mode,' to 'd. weather mode.' and a closing sentence.

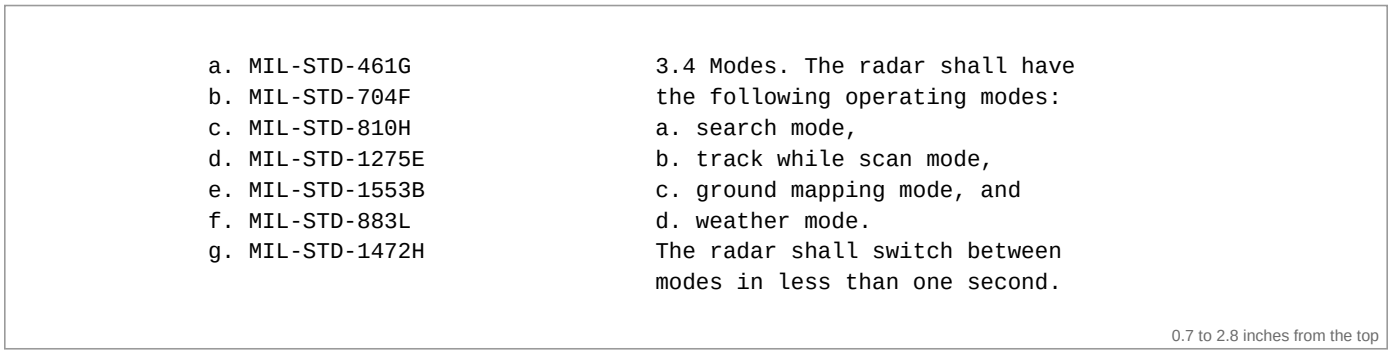
### What the tool must do

The right column's list under 3.4 (a to d) and its closing sentence; the left list is not mixed in.

### Source

Review round 3 (layout); real OCR page twocol_sublist_mono.

### The page (as the scan document has it)



```
a. MIL-STD-461G                3.4 Modes. The radar shall have
b. MIL-STD-704F                the following operating modes:
c. MIL-STD-810H                a. search mode,
d. MIL-STD-1275E               b. track while scan mode,
e. MIL-STD-1553B               c. ground mapping mode, and
f. MIL-STD-883L                d. weather mode.
g. MIL-STD-1472H               The radar shall switch between
                               modes in less than one second.
```

0.7 to 2.8 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.4 a | The radar shall have the following operating modes: a. search mode, | |
| 3.4 b | The radar shall have the following operating modes: b. track while scan mode, | |
| 3.4 c | The radar shall have the following operating modes: c. ground mapping mode, and | |
| 3.4 d | The radar shall have the following operating modes: d. weather mode. | |
| (empty) | The radar shall switch between modes in less than one second. | |

# COL-11  A short right column beside a list and prose

Two-column pages | document: documents/COL-11_a_short_right_column_beside_a_list_and_prose.pdf | typewriter 10 pt, 300 dpi

### Situation

10 pt typewriter. Left: five documents, a blank line, then six lines of prose; right: only six lines of prose, level with the list, ending mid-sentence.

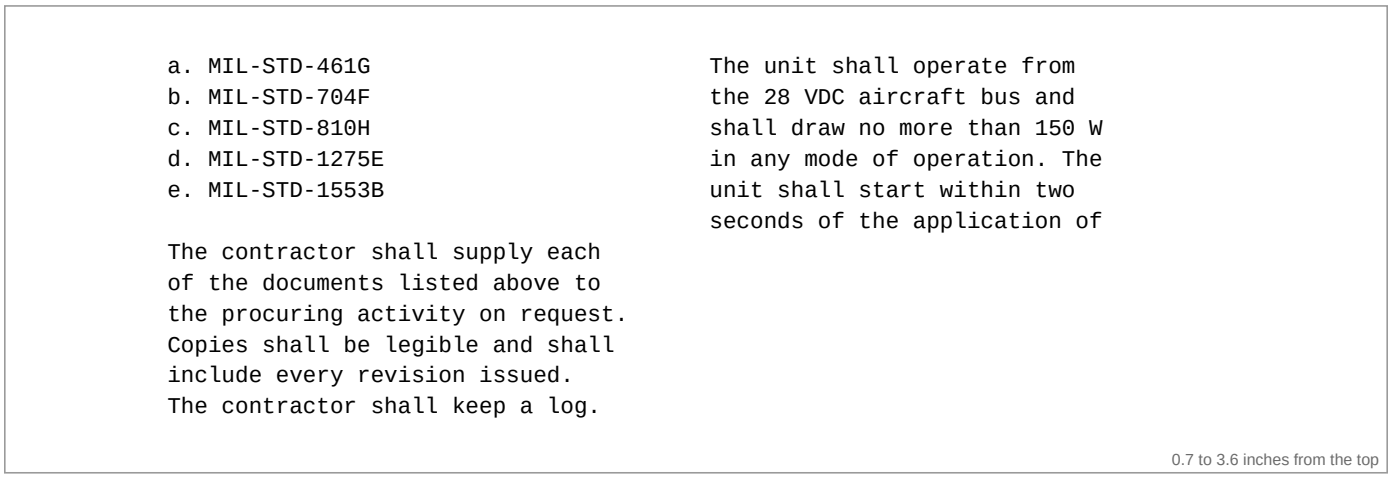
### What the tool must do

Left column first (three sentences), then the right column; the right column's unfinished last sentence is flagged for review.

### Source

Review round 3 (layout); real OCR page twocol_short_right_mono.

### The page (as the scan document has it)




0.7 to 3.6 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| (empty) | The contractor shall supply each of the documents listed above to the procuring activity on request. | |
| (empty) | Copies shall be legible and shall include every revision issued. | |
| (empty) | The contractor shall keep a log. | |
| (empty) | The unit shall operate from the 28 VDC aircraft bus and shall draw no more than 150 W in any mode of operation. | |
| (empty) | The unit shall start within two seconds of the application of | review note: no final punctuation |

# COL-12  A short list beside prose with one numbered paragraph

Two-column pages | document: documents/COL-12_a_short_list_beside_prose_with_one_numbered_paragraph.pdf | typewriter 10 pt, 300 dpi

### Situation

10 pt typewriter. Left: seven documents; right: '3.4 Cooling. The unit shall be ...' - one numbered paragraph whose lines often start with capitals ('MIL-STD-810H', 'Method 507.6', 'Result').

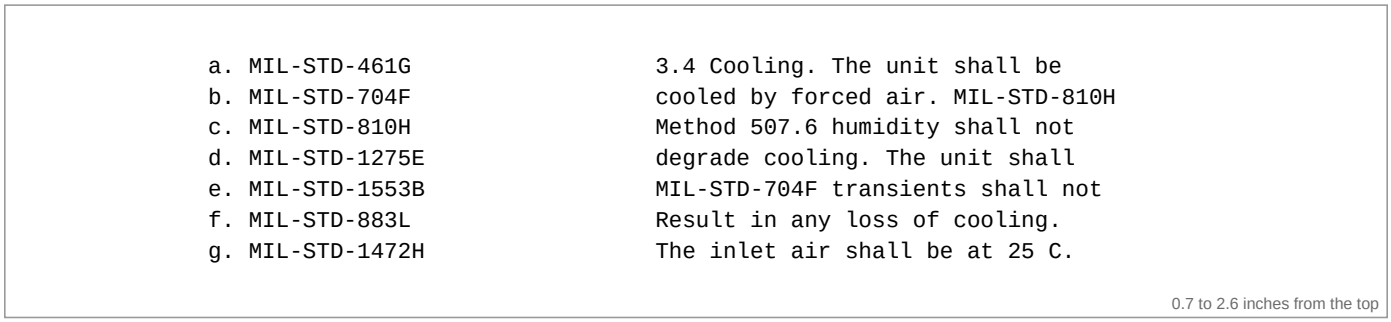
### What the tool must do

The paragraph's four sentences, all under 3.4, in order (round 3: a single titled paragraph made the right column look like the second column of a table).

### Source

Review round 3, finding F4; real OCR page twocol_one_titled_mono.

### The page (as the scan document has it)



```
a. MIL-STD-461G                3.4 Cooling. The unit shall be
b. MIL-STD-704F                cooled by forced air. MIL-STD-810H
c. MIL-STD-810H                Method 507.6 humidity shall not
d. MIL-STD-1275E               degrade cooling. The unit shall
e. MIL-STD-1553B               MIL-STD-704F transients shall not
f. MIL-STD-883L                Result in any loss of cooling.
g. MIL-STD-1472H               The inlet air shall be at 25 C.
```

0.7 to 2.6 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.4 | The unit shall be cooled by forced air. | |
| 3.4 | MIL-STD-810H Method 507.6 humidity shall not degrade cooling. | |
| 3.4 | The unit shall MIL-STD-704F transients shall not Result in any loss of cooling. | |
| 3.4 | The inlet air shall be at 25 C. | |

# COL-13  A list on tab stops straight above a two-column section

Two-column pages | document: documents/COL-13_a_list_on_tab_stops_straight_above_a_two_column_section.pdf | typewriter 10 pt, 300 dpi

### Situation

10 pt typewriter. A lead-in and a list a. to d. whose values sit on a tab stop at 3.3 inches, then, one blank line below, two columns of prose whose right column starts at the same 3.3 inches.

### What the tool must do

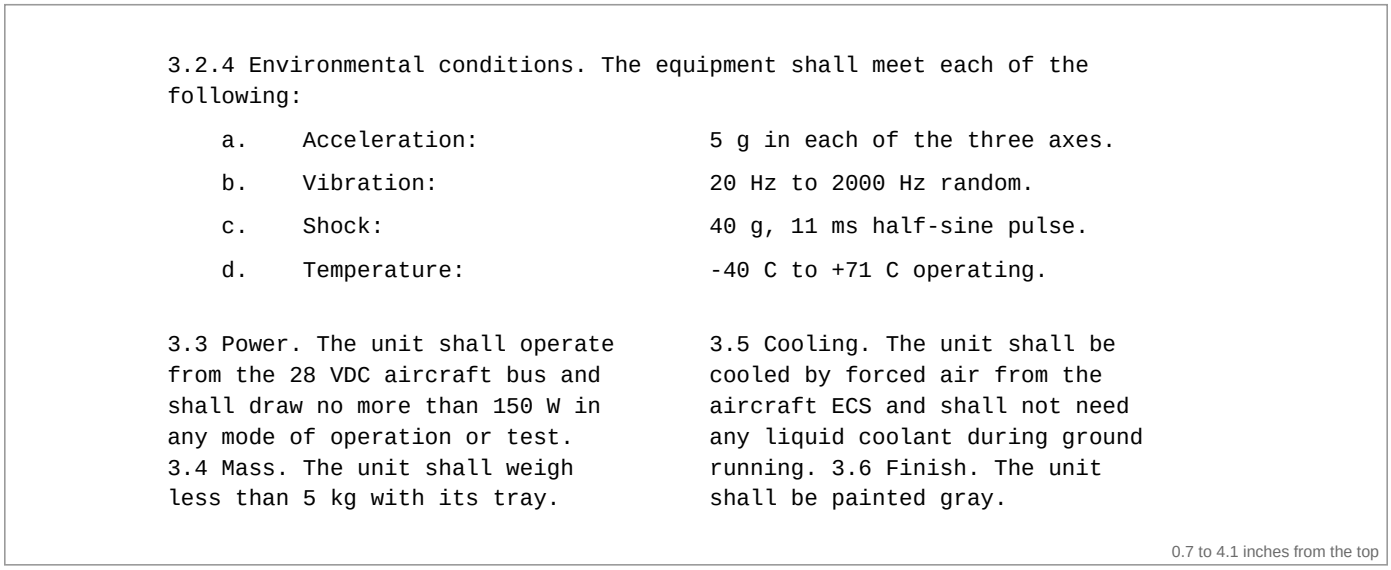
The list's four items with their values, then 3.3 to 3.6 in column order.

### Source

Review round 3, finding B1; known limitation.

**KNOWN LIMITATION** - the tool still gets this case wrong: The two-column section's gutter lines up with the list's value tab stop, so the page is cut into two columns from the top: the values leave their items (the rows keep only 'a. Acceleration:') and come out as a separate Table item.

### The page (as the scan document has it)



3.2.4 Environmental conditions. The equipment shall meet each of the following:

| | | |
|---|---|---|
| a. | Acceleration: | 5 g in each of the three axes. |
| b. | Vibration: | 20 Hz to 2000 Hz random. |
| c. | Shock: | 40 g, 11 ms half-sine pulse. |
| d. | Temperature: | -40 C to +71 C operating. |

3.3 Power. The unit shall operate from the 28 VDC aircraft bus and shall draw no more than 150 W in any mode of operation or test.

3.4 Mass. The unit shall weigh less than 5 kg with its tray.

3.5 Cooling. The unit shall be cooled by forced air from the aircraft ECS and shall not need any liquid coolant during ground running.

3.6 Finish. The unit shall be painted gray.

0.7 to 4.1 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 a | The equipment shall meet each of the following: a. Acceleration: 5 g in each of the three axes. | |
| 3.2.4 b | The equipment shall meet each of the following: b. Vibration: 20 Hz to 2000 Hz random. | |
| 3.2.4 c | The equipment shall meet each of the following: c. Shock: 40 g, 11 ms half-sine pulse. | |
| 3.2.4 d | The equipment shall meet each of the following: d. Temperature: -40 C to +71 C operating. | |
| 3.3 | The unit shall operate from the 28 VDC aircraft bus and shall draw no more than 150 W in any mode of operation or test. | |
| 3.4 | The unit shall weigh less than 5 kg with its tray. | |
| 3.5 | The unit shall be cooled by forced air from the aircraft ECS and shall not need any liquid coolant during ground running. | |
| 3.6 | The unit shall be painted gray. | |

# Tables and lists

Tables and lists: a borderless table under a lead-in must stay a table (one Table item, its rows not made into list items), and a list of labels on tab stops must stay a list, even with a table caption or a sentence about a table right under it. Requirement tables ('Requirement' column) give one requirement row per cell.

## TBL-01  A borderless table with real columns (control)

Tables and lists | document: documents/TBL-01_a_borderless_table_with_real_columns_control.pdf | typewriter 11 pt, 300 dpi

**Situation**

Paragraph 3.8.3, the caption 'TABLE I. Serial interface characteristics.', a three-column table with no rulings (Parameter / Value / Unit and six rows, columns 1.4 to 2 inches apart), then 3.8.4.

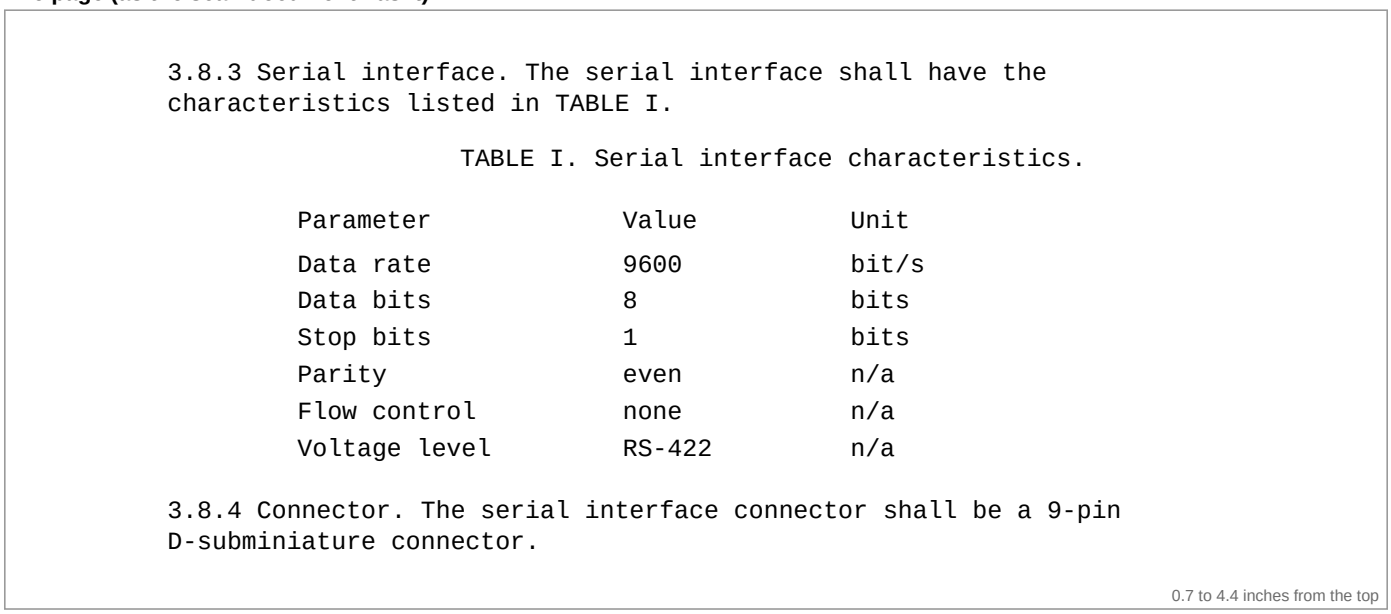
**What the tool must do**

Two paragraph rows and ONE Table item: the table's rows are not list items or requirements.

**Source**

List suite p08 (control).

**The page (as the scan document has it)**



3.8.3 Serial interface. The serial interface shall have the characteristics listed in TABLE I.

TABLE I. Serial interface characteristics.

| Parameter | Value | Unit |
|---|---|---|
| Data rate | 9600 | bit/s |
| Data bits | 8 | bits |
| Stop bits | 1 | bits |
| Parity | even | n/a |
| Flow control | none | n/a |
| Voltage level | RS-422 | n/a |

3.8.4 Connector. The serial interface connector shall be a 9-pin D-subminiature connector.

0.7 to 4.4 inches from the top

**The rows the tool must produce**

| **Doc Ref** | **Content** (the lead-in in grey) | **Also checked** |
|---|---|---|
| 3.8.3 | The serial interface shall have the characteristics listed in TABLE I. | |
| 3.8.4 | The serial interface connector shall be a 9-pin D-subminiature connector. | |

Also expected: 1 Table item(s), containing 'Flow control'; caption(s): TABLE I.

# TBL-02  A requirement table with a header row and labels with colons

Tables and lists | document: documents/TBL-02_a_requirement_table_with_a_header_row_and_labels_with_colons.pdf | typewriter 11 pt, 300 dpi

### Situation

'3.2 ... the requirements of Table I.', the caption 'TABLE I. Characteristics', then a borderless table with the header 'Parameter / Requirement / Method' and four rows ('Input voltage:', 'Power:' ...) whose Requirement cells are sentences, some over two lines; then 3.3.

### What the tool must do

One Table item with its caption, and one requirement row per Requirement cell (their Doc Ref is empty: the table has no ID column). The labels are not a list.

### Source

Review round 1 (layout, requirement tables); real OCR page tbl_hdr_colon.

### The page (as the scan document has it)

```
3.2 Characteristics. The unit shall meet the requirements of Table I.

                     TABLE I. Characteristics

Parameter            Requirement                            Method

Input voltage:       The unit shall operate from 22 to      Test
                     29 VDC.

Power:               The unit shall draw less than 150 W.   Test

Weight:              The unit shall weigh less than 5 kg.   Inspection

Cooling:             The unit shall be cooled by forced     Analysis
                     air at 20 C inlet.

3.3 Marking. Each unit shall carry a nameplate.
```

0.7 to 4.0 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2 | The unit shall meet the requirements of Table I. | |
| (empty) | The unit shall operate from 22 to 29 VDC. | |
| (empty) | The unit shall draw less than 150 W. | |
| (empty) | The unit shall weigh less than 5 kg. | |
| (empty) | The unit shall be cooled by forced air at 20 C inlet. | |
| 3.3 | Each unit shall carry a nameplate. | |

Also expected: 1 Table item(s), containing 'Inspection'; caption(s): TABLE I.

# TBL-03  A requirement table with a header row and labels without colons

Tables and lists | document: documents/TBL-03_a_requirement_table_with_a_header_row_and_labels_without_col.pdf | typewriter 11 pt, 300 dpi

**Situation**

TBL-02 with the labels' colons left out.

**What the tool must do**

As TBL-02.

**Source**

Review round 1 (layout); real OCR page tbl_hdr_nocolon.

**The page (as the scan document has it)**

```
3.2 Characteristics. The unit shall meet the requirements of Table I.

                     TABLE I. Characteristics

Parameter            Requirement                          Method

Input voltage        The unit shall operate from 22 to    Test
                     29 VDC.

Power                The unit shall draw less than 150 W. Test

Weight               The unit shall weigh less than 5 kg. Inspection

Cooling              The unit shall be cooled by forced   Analysis
                     air at 20 C inlet.

3.3 Marking. Each unit shall carry a nameplate.
```

0.7 to 4.0 inches from the top

**The rows the tool must produce**

| **Doc Ref** | **Content** (the lead-in in grey) | **Also checked** |
|---|---|---|
| 3.2 | The unit shall meet the requirements of Table I. | |
| (empty) | The unit shall operate from 22 to 29 VDC. | |
| (empty) | The unit shall draw less than 150 W. | |
| (empty) | The unit shall weigh less than 5 kg. | |
| (empty) | The unit shall be cooled by forced air at 20 C inlet. | |
| 3.3 | Each unit shall carry a nameplate. | |

Also expected: 1 Table item(s), containing 'Inspection'; caption(s): TABLE I.

# TBL-04  A requirement table without a header row, labels with colons

Tables and lists | document: documents/TBL-04_a_requirement_table_without_a_header_row_labels_with_colons.pdf | typewriter 11 pt, 300 dpi

### Situation

TBL-02 without its header row.

### What the tool must do

As TBL-02: the table is recognised from its columns and caption even without a header.

### Source

Review round 1 (layout); real OCR page tbl_nohdr_colon.

### The page (as the scan document has it)

```
3.2 Characteristics. The unit shall meet the requirements of Table I.

                     TABLE I. Characteristics

Input voltage:       The unit shall operate from 22 to      Test
                     29 VDC.

Power:               The unit shall draw less than 150 W.   Test

Weight:              The unit shall weigh less than 5 kg.   Inspection

Cooling:             The unit shall be cooled by forced     Analysis
                     air at 20 C inlet.

3.3 Marking. Each unit shall carry a nameplate.
```

0.7 to 3.7 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2 | The unit shall meet the requirements of Table I. | |
| (empty) | The unit shall operate from 22 to 29 VDC. | |
| (empty) | The unit shall draw less than 150 W. | |
| (empty) | The unit shall weigh less than 5 kg. | |
| (empty) | The unit shall be cooled by forced air at 20 C inlet. | |
| 3.3 | Each unit shall carry a nameplate. | |

Also expected: 1 Table item(s), containing 'Inspection'; caption(s): TABLE I.

# TBL-05  A requirement table without a header row, labels without colons

Tables and lists | document: documents/TBL-05_a_requirement_table_without_a_header_row_labels_without_colo.pdf | typewriter 11 pt, 300 dpi

### Situation

TBL-02 without its header row and without the labels' colons.

### What the tool must do

As TBL-02.

### Source

Review round 1 (layout); real OCR page tbl_nohdr_nocolon.

### The page (as the scan document has it)

```
3.2 Characteristics. The unit shall meet the requirements of Table I.

                     TABLE I. Characteristics

Input voltage        The unit shall operate from 22 to      Test
                     29 VDC.

Power                The unit shall draw less than 150 W.   Test

Weight               The unit shall weigh less than 5 kg.   Inspection

Cooling              The unit shall be cooled by forced     Analysis
                     air at 20 C inlet.

3.3 Marking. Each unit shall carry a nameplate.
```

0.7 to 3.7 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2 | The unit shall meet the requirements of Table I. | |
| (empty) | The unit shall operate from 22 to 29 VDC. | |
| (empty) | The unit shall draw less than 150 W. | |
| (empty) | The unit shall weigh less than 5 kg. | |
| (empty) | The unit shall be cooled by forced air at 20 C inlet. | |
| 3.3 | Each unit shall carry a nameplate. | |

Also expected: 1 Table item(s), containing 'Inspection'; caption(s): TABLE I.

# TBL-06  A three-column table with a header under a lead-in

Tables and lists | document: documents/TBL-06_a_three_column_table_with_a_header_under_a_lead_in.pdf | typewriter 11 pt, 300 dpi

### Situation

'3.2.1 ... shall have the following properties:' then a borderless table 'Characteristic / Requirement / Method' with four rows ('Voltage: 28 VDC nominal Test'), then 3.2.5.

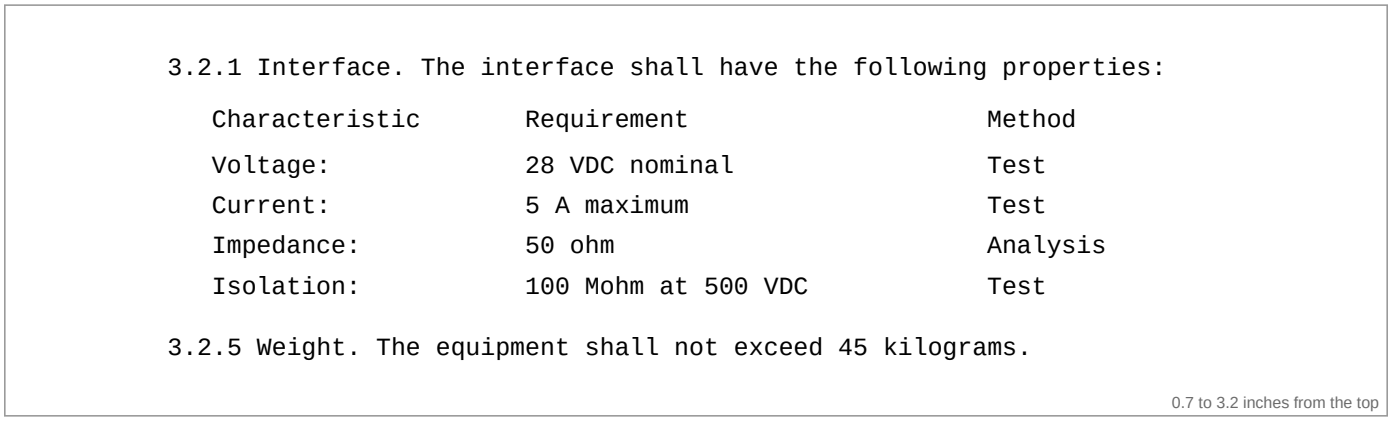
### What the tool must do

A Table item, not four list rows (round 2: the colons made the first column look like list labels); the Requirement cells give one requirement row each; the lead-in says no list items were found.

### Source

Review round 2 (tab-list attack on tables); real OCR page tbl_colon_hdr.

### The page (as the scan document has it)



3.2.1 Interface. The interface shall have the following properties:

| Characteristic | Requirement | Method |
|---|---|---|
| Voltage: | 28 VDC nominal | Test |
| Current: | 5 A maximum | Test |
| Impedance: | 50 ohm | Analysis |
| Isolation: | 100 Mohm at 500 VDC | Test |

3.2.5 Weight. The equipment shall not exceed 45 kilograms.

0.7 to 3.2 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.1 | The interface shall have the following properties: | review note: Ends with ':' but no list items |
| (empty) | 28 VDC nominal | |
| (empty) | 5 A maximum | |
| (empty) | 50 ohm | |
| (empty) | 100 Mohm at 500 VDC | |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

Also expected: 1 Table item(s), containing 'Impedance:', 'Analysis'.

# TBL-07  A three-column table without a header under a lead-in

Tables and lists | document: documents/TBL-07_a_three_column_table_without_a_header_under_a_lead_in.pdf | typewriter 11 pt, 300 dpi

**Situation**

TBL-06 without its header row.

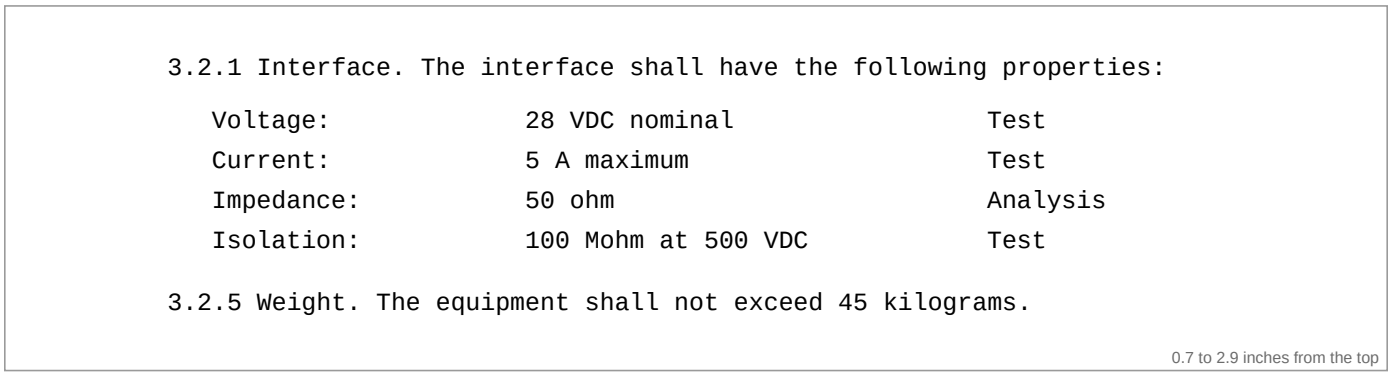
**What the tool must do**

A Table item, not four list rows (no Requirement column, so no requirement rows from it).

**Source**

Review round 2 (tab-list attack on tables); real OCR page tbl_colon_nohdr.

**The page (as the scan document has it)**



3.2.1 Interface. The interface shall have the following properties:

| | | |
|---|---|---|
| Voltage: | 28 VDC nominal | Test |
| Current: | 5 A maximum | Test |
| Impedance: | 50 ohm | Analysis |
| Isolation: | 100 Mohm at 500 VDC | Test |

3.2.5 Weight. The equipment shall not exceed 45 kilograms.

0.7 to 2.9 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.1 | The interface shall have the following properties: | review note: Ends with ':' but no list items |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

Also expected: 1 Table item(s), containing 'Impedance:', 'Analysis'.

# TBL-08  A three-column table with a header, labels without colons

Tables and lists | document: documents/TBL-08_a_three_column_table_with_a_header_labels_without_colons.pdf | typewriter 11 pt, 300 dpi

**Situation**

TBL-06 with the labels' colons left out.

**What the tool must do**

As TBL-06.

**Source**

Review round 2 (tab-list attack on tables); real OCR page tbl_plain_hdr.

**The page (as the scan document has it)**

```
3.2.1 Interface. The interface shall have the following properties:

   Characteristic      Requirement                  Method
   Voltage             28 VDC nominal               Test
   Current             5 A maximum                  Test
   Impedance           50 ohm                       Analysis
   Isolation           100 Mohm at 500 VDC          Test

3.2.5 Weight. The equipment shall not exceed 45 kilograms.
```

0.7 to 3.2 inches from the top

**The rows the tool must produce**

| **Doc Ref** | **Content** (the lead-in in grey) | **Also checked** |
|---|---|---|
| 3.2.1 | The interface shall have the following properties: | review note: Ends with ':' but no list items |
| (empty) | 28 VDC nominal | |
| (empty) | 5 A maximum | |
| (empty) | 50 ohm | |
| (empty) | 100 Mohm at 500 VDC | |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

Also expected: 1 Table item(s), containing 'Impedance', 'Analysis'.

# TBL-09  A three-column table without a header, labels without colons

Tables and lists | document: documents/TBL-09_a_three_column_table_without_a_header_labels_without_colons.pdf | typewriter 11 pt, 300 dpi

**Situation**

TBL-07 with the labels' colons left out.

**What the tool must do**

As TBL-07.

**Source**

Review round 2 (tab-list attack on tables); real OCR page tbl_plain_nohdr.

**The page (as the scan document has it)**

```
3.2.1 Interface. The interface shall have the following properties:

   Voltage             28 VDC nominal               Test
   Current             5 A maximum                  Test
   Impedance           50 ohm                       Analysis
   Isolation           100 Mohm at 500 VDC          Test

3.2.5 Weight. The equipment shall not exceed 45 kilograms.
```

0.7 to 2.9 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.1 | The interface shall have the following properties: | review note: Ends with ':' but no list items |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

Also expected: 1 Table item(s), containing 'Impedance', 'Analysis'.

# TBL-10  A table of documents under a lead-in

Tables and lists | document: documents/TBL-10_a_table_of_documents_under_a_lead_in.pdf | typewriter 11 pt, 300 dpi

### Situation

'2.1 Documents. The following documents form a part of this specification:' then document numbers and titles in two columns (one title over two lines), then 2.2.

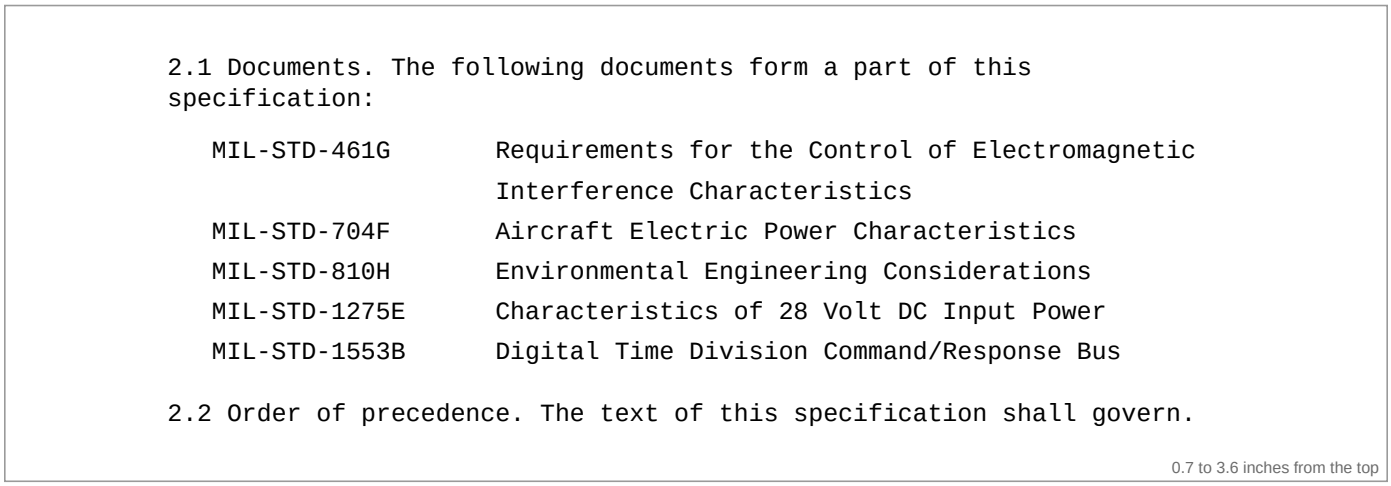
### What the tool must do

One Table item; the documents are not list items; 2.2 keeps its number.

### Source

Review round 2 (tab-list attack on tables); real OCR page tbl_docs.

### The page (as the scan document has it)



```
2.1 Documents. The following documents form a part of this
specification:

   MIL-STD-461G       Requirements for the Control of Electromagnetic
                      Interference Characteristics
   MIL-STD-704F       Aircraft Electric Power Characteristics
   MIL-STD-810H       Environmental Engineering Considerations
   MIL-STD-1275E      Characteristics of 28 Volt DC Input Power
   MIL-STD-1553B      Digital Time Division Command/Response Bus

2.2 Order of precedence. The text of this specification shall govern.
```

0.7 to 3.6 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 2.2 | The text of this specification shall govern. | |

Also expected: 1 Table item(s), containing 'MIL-STD-810H'.

# TBL-11  A table of documents with a header row

Tables and lists | document: documents/TBL-11_a_table_of_documents_with_a_header_row.pdf | typewriter 11 pt, 300 dpi

**Situation**

TBL-10 with the header row 'Document / Title'.

**What the tool must do**

As TBL-10.

**Source**

Review round 2 (tab-list attack on tables); real OCR page tbl_docs_hdr.

**The page (as the scan document has it)**

2.1 Documents. The following documents form a part of this specification:

| Document | Title |
|---|---|
| MIL-STD-461G | Requirements for the Control of Electromagnetic Interference Characteristics |
| MIL-STD-704F | Aircraft Electric Power Characteristics |
| MIL-STD-810H | Environmental Engineering Considerations |
| MIL-STD-1275E | Characteristics of 28 Volt DC Input Power |
| MIL-STD-1553B | Digital Time Division Command/Response Bus |

2.2 Order of precedence. The text of this specification shall govern.

0.7 to 3.9 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 2.2 | The text of this specification shall govern. | |

Also expected: 1 Table item(s), containing 'MIL-STD-810H'.

# TBL-12  Labels with a requirement sentence each, under a lead-in

Tables and lists | document: documents/TBL-12_labels_with_a_requirement_sentence_each_under_a_lead_in.pdf | typewriter 11 pt, 300 dpi

### Situation

'3.2.1 ... the following properties:' then two columns: 'Voltage:' / 'The unit shall operate from 28 VDC.', 'Current:' / 'The unit shall draw less than 5 A.', and so on.

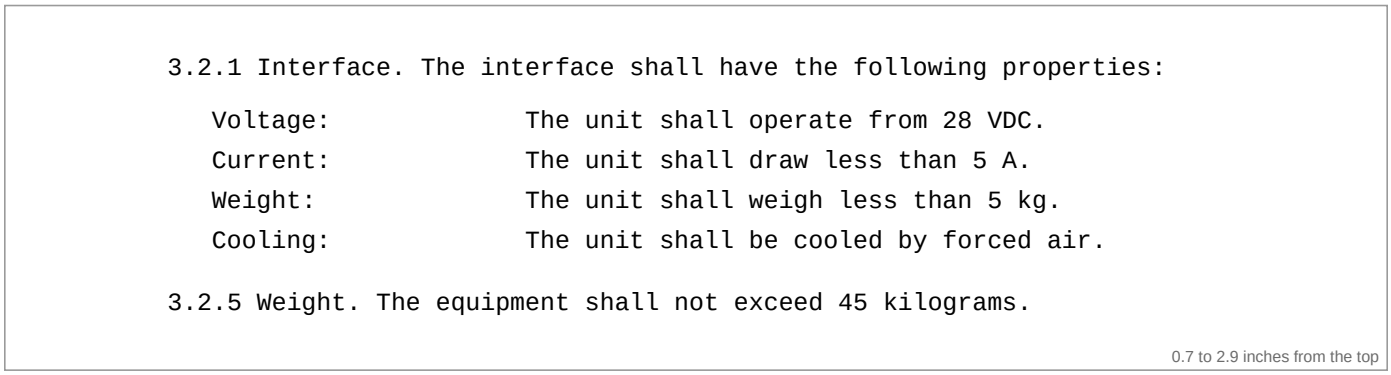
### What the tool must do

A list: one row per label, with the lead-in in front ('... properties: Voltage: The unit shall operate from 28 VDC.'), not a table.

### Source

Review round 2 (tab-list attack on tables); real OCR page tbl_reqs_2col.

### The page (as the scan document has it)



```
3.2.1 Interface. The interface shall have the following properties:

   Voltage:            The unit shall operate from 28 VDC.
   Current:            The unit shall draw less than 5 A.
   Weight:             The unit shall weigh less than 5 kg.
   Cooling:            The unit shall be cooled by forced air.

3.2.5 Weight. The equipment shall not exceed 45 kilograms.
```

0.7 to 2.9 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.1 | The interface shall have the following properties: Voltage: The unit shall operate from 28 VDC. | |
| 3.2.1 | The interface shall have the following properties: Current: The unit shall draw less than 5 A. | |
| 3.2.1 | The interface shall have the following properties: Weight: The unit shall weigh less than 5 kg. | |
| 3.2.1 | The interface shall have the following properties: Cooling: The unit shall be cooled by forced air. | |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

# TBL-13  A table whose first column holds long labels

Tables and lists | document: documents/TBL-13_a_table_whose_first_column_holds_long_labels.pdf | typewriter 11 pt, 300 dpi

**Situation**

'3.2.1 ... the following properties:' then 'Maximum steady state input current: 5 A Test' and three more rows with labels of five to six words.

**What the tool must do**

A Table item, not a list.

**Source**

Review round 2 (tab-list attack on tables); real OCR page tbl_longlabel.

**The page (as the scan document has it)**

```
3.2.1 Interface. The interface shall have the following properties:

   Maximum steady state input current:          5 A         Test
   Minimum insulation resistance at 500 VDC:    100 Mohm    Test
   Maximum inrush current at turn on:           20 A        Test
   Maximum ripple voltage on the bus:           1 V         Analysis

3.2.5 Weight. The equipment shall not exceed 45 kilograms.
```

0.7 to 2.9 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.1 | The interface shall have the following properties: | review note: Ends with ':' but no list items |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

Also expected: 1 Table item(s), containing 'Maximum inrush current'.

# TBL-14  A table under a lead-in with its caption BELOW it

Tables and lists | document: documents/TBL-14_a_table_under_a_lead_in_with_its_caption_below_it.pdf | typewriter 11 pt, 300 dpi

**Situation**

TBL-07's table with the caption 'TABLE I. Interface characteristics.' under it, then 3.2.5.

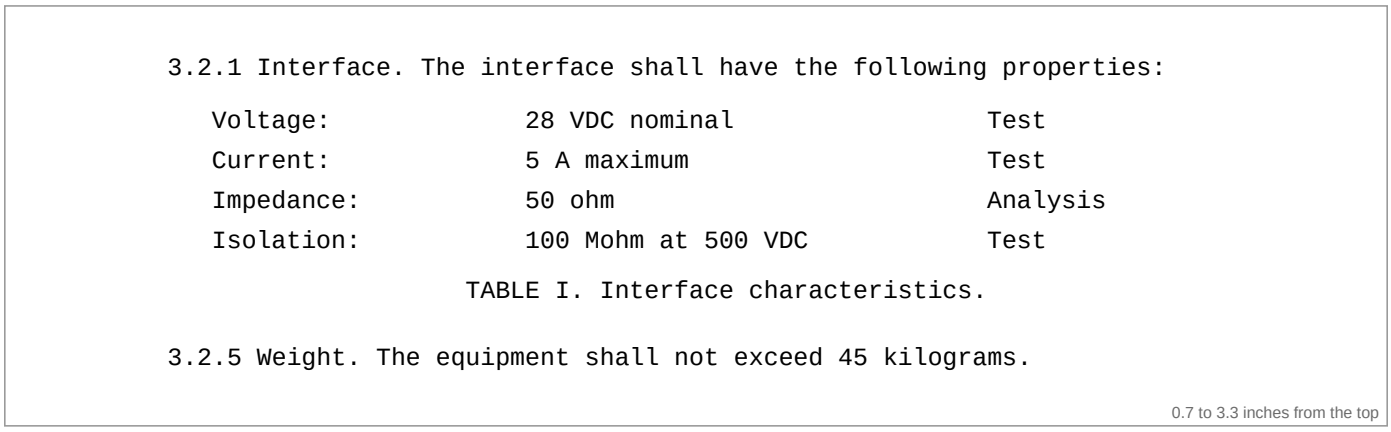
**What the tool must do**

A Table item linked to its caption; not a list.

**Source**

Review round 2 (tab-list attack on tables); real OCR page tbl_colon_caption_below.

**The page (as the scan document has it)**



3.2.1 Interface. The interface shall have the following properties:

| | | |
|---|---|---|
| Voltage: | 28 VDC nominal | Test |
| Current: | 5 A maximum | Test |
| Impedance: | 50 ohm | Analysis |
| Isolation: | 100 Mohm at 500 VDC | Test |

TABLE I. Interface characteristics.

3.2.5 Weight. The equipment shall not exceed 45 kilograms.

0.7 to 3.3 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.1 | The interface shall have the following properties: | review note: Ends with ':' but no list items |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

Also expected: 1 Table item(s), containing 'Impedance:'; caption(s): TABLE I.

# TBL-15  A table whose labels are abbreviations with full stops ('Min.', 'Nom.', 'Max.')

Tables and lists | document: documents/TBL-15_a_table_whose_labels_are_abbreviations_with_full_stops_min_n.pdf | typewriter 11 pt, 300 dpi

### Situation

'3.2.3 ... within the following limits:' then 'Min. 22.0 VDC Test', 'Nom. 28.0 VDC Test', 'Max. 29.0 VDC Test'.

### What the tool must do

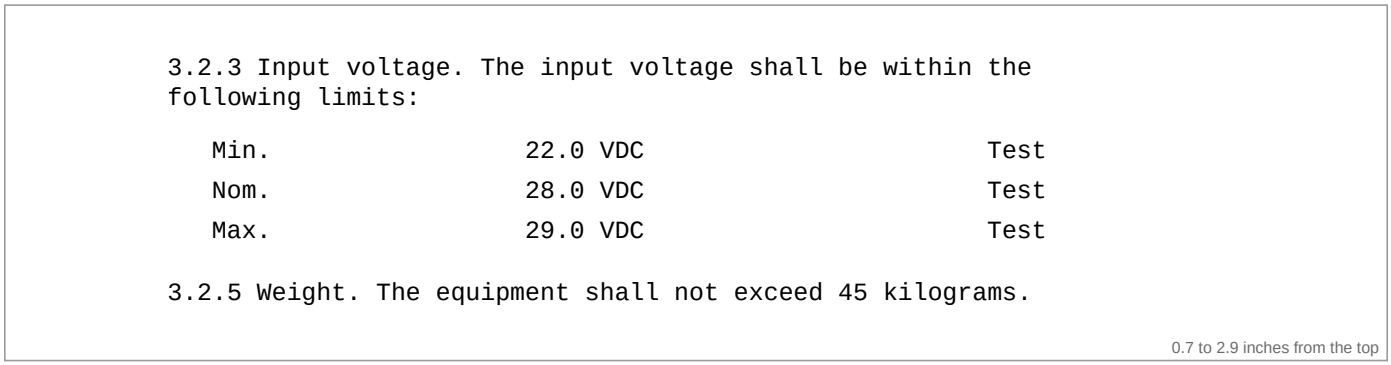
A Table item, not a list.

### Source

Review round 2 (tab-list attack on tables); real OCR page tbl_minmax; known limitation.

**KNOWN LIMITATION** - the tool still gets this case wrong: The rows are taken as the lead-in's list, with the labels and the rest of each row apart: six rows ('... following limits: Min.', 'Nom.', 'Max.', then '22.0 VDC Test' ...), and no Table item.

### The page (as the scan document has it)



3.2.3 Input voltage. The input voltage shall be within the following limits:

| | | |
|---|---|---|
| Min. | 22.0 VDC | Test |
| Nom. | 28.0 VDC | Test |
| Max. | 29.0 VDC | Test |

3.2.5 Weight. The equipment shall not exceed 45 kilograms.

0.7 to 2.9 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.3 | The input voltage shall be within the following limits: | review note: Ends with ':' but no list items |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

Also expected: 1 Table item(s), containing '29.0 VDC'.

# TBL-16  A table whose labels are abbreviations ('Temp.', 'Alt.', 'Vib.', 'Hum.')

Tables and lists | document: documents/TBL-16_a_table_whose_labels_are_abbreviations_temp_alt_vib_hum.pdf | typewriter 11 pt, 300 dpi

### Situation

'3.2.1 ... the following properties:' then 'Temp. -40 C to +71 C Test' and three more such rows.

### What the tool must do

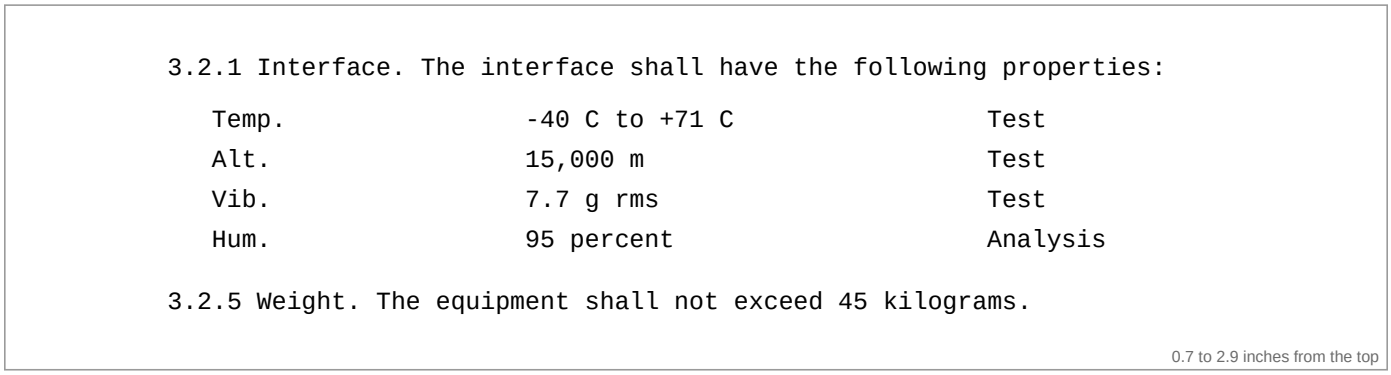
A Table item, not a list.

### Source

Review round 2 (tab-list attack on tables); real OCR page tbl_abbrev; known limitation.

**KNOWN LIMITATION** - the tool still gets this case wrong: The rows are taken as the lead-in's list: four rows such as '... properties: Temp. -40 C to +71 C Test', and no Table item.

### The page (as the scan document has it)



3.2.1 Interface. The interface shall have the following properties:

| | | |
|---|---|---|
| Temp. | -40 C to +71 C | Test |
| Alt. | 15,000 m | Test |
| Vib. | 7.7 g rms | Test |
| Hum. | 95 percent | Analysis |

3.2.5 Weight. The equipment shall not exceed 45 kilograms.

0.7 to 2.9 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.1 | The interface shall have the following properties: | review note: Ends with ':' but no list items |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

Also expected: 1 Table item(s), containing '15,000 m'.

# TBL-17  Requirement IDs with a colon on a tab stop ('SRS-101:')

Tables and lists | document: documents/TBL-17_requirement_ids_with_a_colon_on_a_tab_stop_srs_101.pdf | typewriter 11 pt, 300 dpi

### Situation

'3.2 Performance. The radar shall meet the following requirements:' then 'SRS-101:' at the margin and its requirement on a tab stop (two lines), 'SRS-102:' and 'SRS-103:' likewise; then 3.3.

### What the tool must do

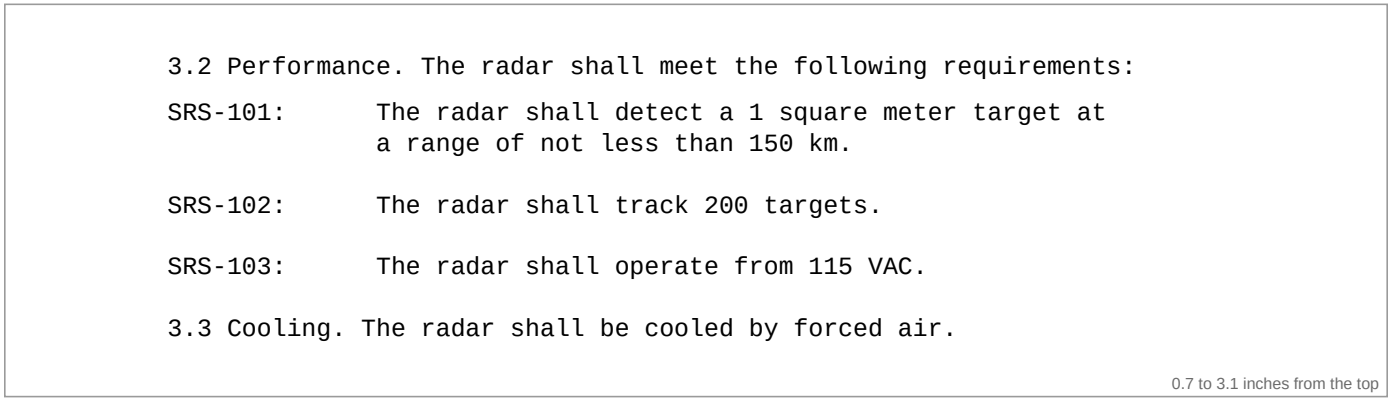
Each requirement is a row whose Doc Ref is its own ID (SRS-101, SRS-102, SRS-103) - the ID is not a list label (round 3: 'SRS-101:' was accepted as a label and the table was lost).

### Source

Review round 3 (idcolon); real OCR page idcolon_mono.

**KNOWN LIMITATION** - the tool still gets this case wrong: The IDs are read as the first column of a table, and the requirement rows taken from it have an empty Doc Ref instead of SRS-101, SRS-102 and SRS-103.

### The page (as the scan document has it)



```
3.2 Performance. The radar shall meet the following requirements:

SRS-101:      The radar shall detect a 1 square meter target at
              a range of not less than 150 km.

SRS-102:      The radar shall track 200 targets.

SRS-103:      The radar shall operate from 115 VAC.

3.3 Cooling. The radar shall be cooled by forced air.
```

0.7 to 3.1 inches from the top

### The rows the tool must produce

| **Doc Ref** | **Content** (the lead-in in grey) | **Also checked** |
|---|---|---|
| 3.2 | The radar shall meet the following requirements: | review note: Ends with ':' but no list items |
| SRS-101 | The radar shall detect a 1 square meter target at a range of not less than 150 km. | |
| SRS-102 | The radar shall track 200 targets. | |
| SRS-103 | The radar shall operate from 115 VAC. | |
| 3.3 | The radar shall be cooled by forced air. | |

Also expected: 1 Table item(s), containing 'SRS-102'.

# TBL-18  A list of labels, then the next table's caption and a ruled table

Tables and lists | document: documents/TBL-18_a_list_of_labels_then_the_next_table_s_caption_and_a_ruled_t.pdf | typewriter 11 pt, 300 dpi

### Situation

'3.2.6 ... shall meet the following:' with five labels on tab stops (no markers), one blank line, the caption 'TABLE II. Connector J1 pin assignments.' and a ruled table Pin / Signal.

### What the tool must do

The labels stay a list (five rows with the lead-in), and TABLE II is a Table item with its caption (round 3: the caption below the labels turned the list into a borderless table).

### Source

Review round 3, finding F1; real OCR page cap_list_mono.

### The page (as the scan document has it)

3.2.6 Input power. The unit shall meet the following:

Input voltage: 22 to 29 VDC.

Ripple: 1 V peak to peak maximum.

Power: 150 W maximum.

Transients: as in MIL-STD-704F.

Hold-up: 50 ms after loss of input.

TABLE II. Connector J1 pin assignments.

| Pin | Signal |
|---|---|
| A | +28 VDC |
| B | Return |
| C | Chassis |

0.7 to 4.4 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.6 | The unit shall meet the following: Input voltage: 22 to 29 VDC. | |
| 3.2.6 | The unit shall meet the following: Ripple: 1 V peak to peak maximum. | |
| 3.2.6 | The unit shall meet the following: Power: 150 W maximum. | |
| 3.2.6 | The unit shall meet the following: Transients: as in MIL-STD-704F. | |
| 3.2.6 | The unit shall meet the following: Hold-up: 50 ms after loss of input. | |

Also expected: 1 Table item(s), containing 'Chassis'; caption(s): TABLE II.

# TBL-19 A list of labels, then the next table's caption and a ruled table, proportional font

Tables and lists | document: documents/TBL-19_a_list_of_labels_then_the_next_table_s_caption_and_a_ruled_t.pdf | proportional 11 pt, 300 dpi

**Situation**

TBL-18 in a proportional (serif) font.

**What the tool must do**

As TBL-18.

**Source**

Review round 3, finding F1; real OCR page cap_list_serif.

**The page (as the scan document has it)**

3.2.6 Input power. The unit shall meet the following:

| | |
|---|---|
| Input voltage: | 22 to 29 VDC. |
| Ripple: | 1 V peak to peak maximum. |
| Power: | 150 W maximum. |
| Transients: | as in MIL-STD-704F. |
| Hold-up: | 50 ms after loss of input. |

TABLE II. Connector J1 pin assignments.

| Pin | Signal |
|---|---|
| A | +28 VDC |
| B | Return |
| C | Chassis |

0.7 to 4.4 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.6 | The unit shall meet the following: Input voltage: 22 to 29 VDC. | |
| 3.2.6 | The unit shall meet the following: Ripple: 1 V peak to peak maximum. | |
| 3.2.6 | The unit shall meet the following: Power: 150 W maximum. | |
| 3.2.6 | The unit shall meet the following: Transients: as in MIL-STD-704F. | |
| 3.2.6 | The unit shall meet the following: Hold-up: 50 ms after loss of input. | |

Also expected: 1 Table item(s), containing 'Chassis'; caption(s): TABLE II.

# TBL-20  A list of labels, then a sentence that starts 'Table 4 ...'

Tables and lists | document: documents/TBL-20_a_list_of_labels_then_a_sentence_that_starts_table_4.pdf | typewriter 11 pt, 300 dpi

### Situation

The five labels of TBL-18, a blank line, then 'Table 4 gives the limits that apply in each mode of operation.'

### What the tool must do

The labels stay a list; the sentence is not a caption (round 3: 'Table 4 gives ...' was taken as a caption and the list became a table).

### Source

Review round 3, finding F1; real OCR page table_ref_after_list_mono.

### The page (as the scan document has it)

```
3.2.6 Input power. The unit shall meet the following:
       Input voltage:       22 to 29 VDC.
       Ripple:              1 V peak to peak maximum.
       Power:               150 W maximum.
       Transients:          as in MIL-STD-704F.
       Hold-up:             50 ms after loss of input.

Table 4 gives the limits that apply in each mode of operation.
```

0.7 to 3.3 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.6 | The unit shall meet the following: Input voltage: 22 to 29 VDC. | |
| 3.2.6 | The unit shall meet the following: Ripple: 1 V peak to peak maximum. | |
| 3.2.6 | The unit shall meet the following: Power: 150 W maximum. | |
| 3.2.6 | The unit shall meet the following: Transients: as in MIL-STD-704F. | |
| 3.2.6 | The unit shall meet the following: Hold-up: 50 ms after loss of input. | |

# TBL-21  A lead-in whose short last line sits right above a table's header

Tables and lists | document: documents/TBL-21_a_lead_in_whose_short_last_line_sits_right_above_a_table_s_h.pdf | typewriter 11 pt, 300 dpi

### Situation

TBL-06 with its lead-in wrapped: the second line 'characteristics:' is short and sits straight above the table's header row 'Characteristic Requirement Method'.

### What the tool must do

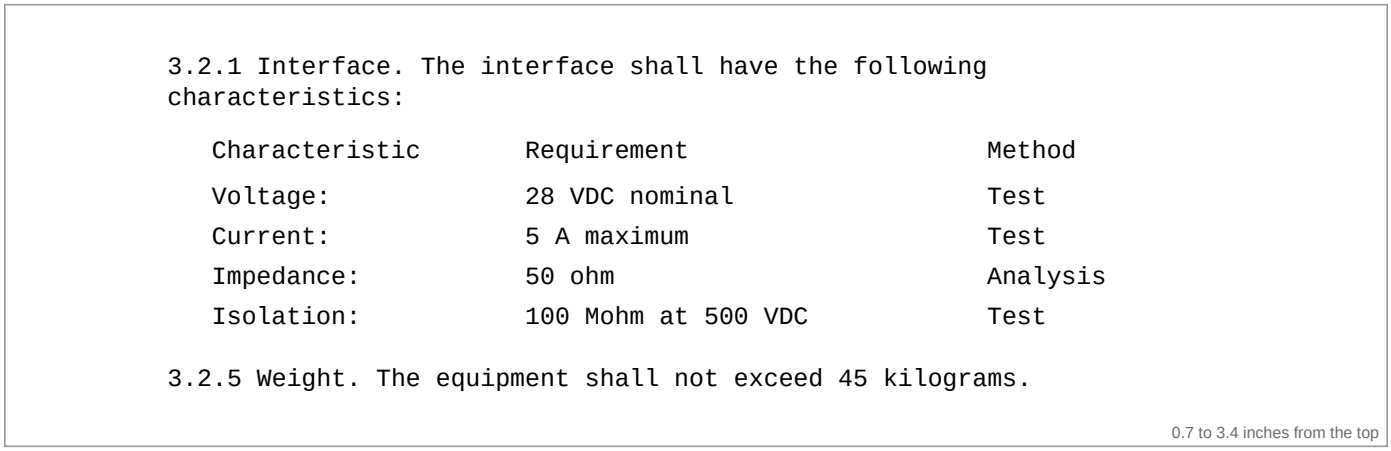
The lead-in keeps its last line ('... have the following characteristics:'), and the table is one Table item that starts at its header row.

### Source

Written for the case library (a variant of TBL-06).

**KNOWN LIMITATION** - the tool still gets this case wrong: The lead-in's short last line is taken into the table as the start of its header row, so the lead-in's row stops at '... have the following' (and is flagged as possibly incomplete).

### The page (as the scan document has it)



3.2.1 Interface. The interface shall have the following characteristics:

| Characteristic | Requirement | Method |
|---|---|---|
| Voltage: | 28 VDC nominal | Test |
| Current: | 5 A maximum | Test |
| Impedance: | 50 ohm | Analysis |
| Isolation: | 100 Mohm at 500 VDC | Test |

3.2.5 Weight. The equipment shall not exceed 45 kilograms.

0.7 to 3.4 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.1 | The interface shall have the following characteristics: | review note: Ends with ':' but no list items |
| (empty) | 28 VDC nominal | |
| (empty) | 5 A maximum | |
| (empty) | 50 ohm | |
| (empty) | 100 Mohm at 500 VDC | |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

Also expected: 1 Table item(s), containing 'Impedance:', 'Analysis'.

# Page breaks

Page breaks: an item or a paragraph that carries on at the top of the next page must be joined back into one row (its Page(s) column then reads '1-2'), and what starts the next page must not be taken for a new section just because it starts with a number.

## PAGE-01  A tab-list value that carries on at the top of the next page

Page breaks | document: documents/PAGE-01_a_tab_list_value_that_carries_on_at_the_top_of_the_next_page.pdf | typewriter 11 pt, 300 dpi, 2 pages

**Situation**

Item b's value starts at the foot of page 1 ('20 Hz to 2000 Hz random, 0.04 g2/Hz,') and carries on at the top of page 2 in the value column ('3 hours per axis, 7.7 g rms overall.'); c. and 3.2.5 follow on page 2.

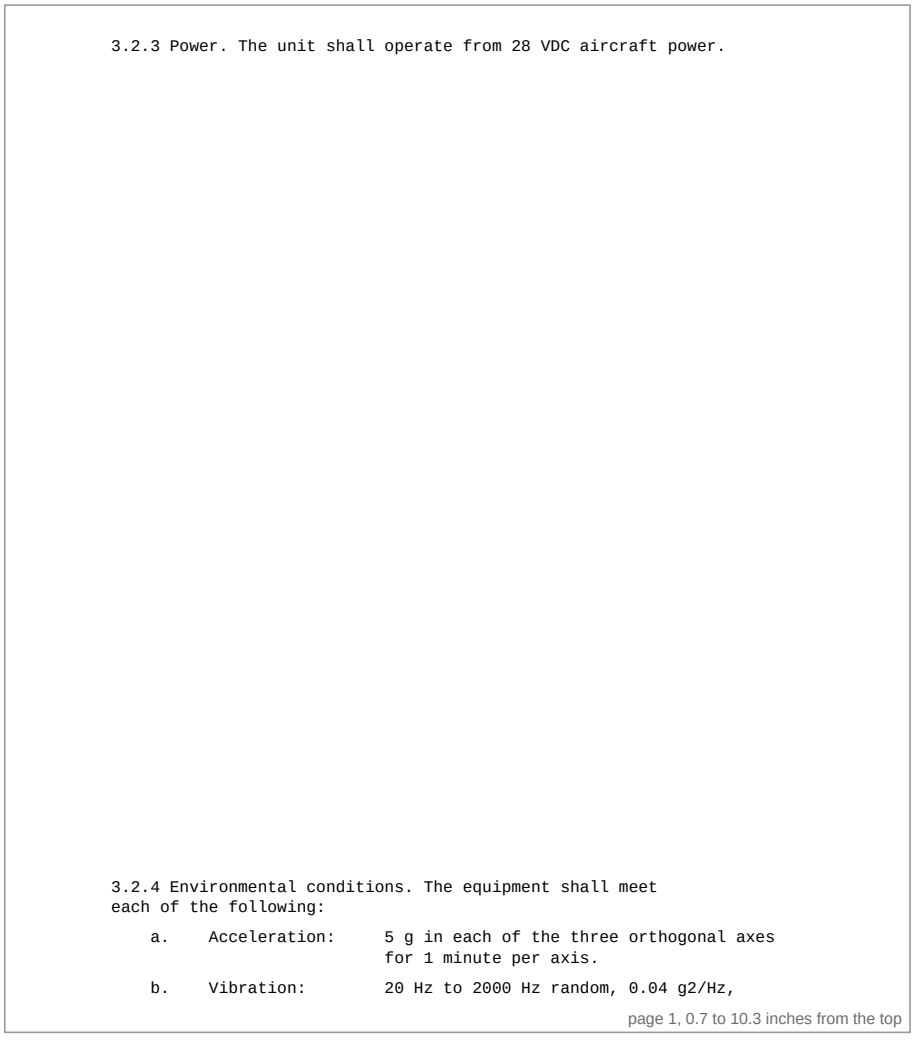
**What the tool must do**

Item b is one row with its whole value, pages '1-2'; c is still item c.

**Source**

Review round 2, finding contstem/a8 (written as a two-page document for the case library).

**The page (as the scan document has it)**



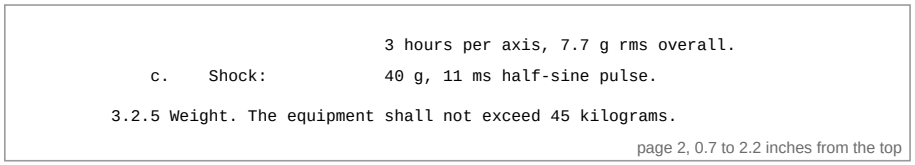
```
3.2.3 Power. The unit shall operate from 28 VDC aircraft power.

3.2.4 Environmental conditions. The equipment shall meet
each of the following:

    a.    Acceleration:     5 g in each of the three orthogonal axes
                            for 1 minute per axis.

    b.    Vibration:        20 Hz to 2000 Hz random, 0.04 g2/Hz,
```

page 1, 0.7 to 10.3 inches from the top



```
                            3 hours per axis, 7.7 g rms overall.

    c.    Shock:            40 g, 11 ms half-sine pulse.

3.2.5 Weight. The equipment shall not exceed 45 kilograms.
```

page 2, 0.7 to 2.2 inches from the top

**The rows the tool must produce**

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.3 | The unit shall operate from 28 VDC aircraft power. | |
| 3.2.4 a | The equipment shall meet each of the following: a. Acceleration: 5 g in each of the three orthogonal axes for 1 minute per axis. | Page(s): 1 |
| 3.2.4 b | The equipment shall meet each of the following: b. Vibration: 20 Hz to 2000 Hz random, 0.04 g2/Hz, 3 hours per axis, 7.7 g rms overall. | Page(s): 1-2 |
| 3.2.4 c | The equipment shall meet each of the following: c. Shock: 40 g, 11 ms half-sine pulse. | Page(s): 1-2 |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

# PAGE-02  A value carried onto the next page that starts with a decimal quantity

Page breaks | document: documents/PAGE-02_a_value_carried_onto_the_next_page_that_starts_with_a_decima.pdf | typewriter 11 pt, 300 dpi, 2 pages

### Situation

As PAGE-01, but the line at the top of page 2 is '2.5 G rms overall. The unit shall be powered during' / 'the test.' - it starts like a paragraph numbered 2.5 with the title 'G rms overall'.

### What the tool must do

The line carries item b's value on (pages '1-2'); there is no section 2.5 (round 2: it became section '2.5' and item c was filed under it).

### Source

Review round 2, finding contstem/a10.

### The page (as the scan document has it)

```
        3.2.3 Power. The unit shall operate from 28 VDC aircraft power.









        3.2.4 Environmental conditions. The equipment shall meet
        each of the following:

            a.    Acceleration:     5 g in each of the three orthogonal axes
                                    for 1 minute per axis.

            b.    Vibration:        20 Hz to 2000 Hz random, 0.04 g2/Hz,
```

page 1, 0.7 to 10.3 inches from the top

```
                                    2.5 G rms overall. The unit shall be
                                    powered during the test.

            c.    Shock:            40 g, 11 ms half-sine pulse.

        3.2.5 Weight. The equipment shall not exceed 45 kilograms.
```

page 2, 0.7 to 2.4 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.3 | The unit shall operate from 28 VDC aircraft power. | |
| 3.2.4 a | The equipment shall meet each of the following: a. Acceleration: 5 g in each of the three orthogonal axes for 1 minute per axis. | Page(s): 1 |
| 3.2.4 b | The equipment shall meet each of the following: b. Vibration: 20 Hz to 2000 Hz random, 0.04 g2/Hz, 2.5 G rms overall. The unit shall be powered during the test. | Page(s): 1-2 |

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 c | The equipment shall meet each of the following: c. Shock: 40 g, 11 ms half-sine pulse. | Page(s): 1-2 |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

# PAGE-03 A value carried onto the next page that starts with a cross-referenced paragraph

Page breaks | document: documents/PAGE-03_a_value_carried_onto_the_next_page_that_starts_with_a_cross_.pdf | typewriter 11 pt, 300 dpi, 2 pages

### Situation

Item b's value ends at the foot of page 1 with '... as specified in'; page 2 starts, in the value column, with '4.5.3 Random vibration. Levels shall be measured at the' / 'mounting points.'

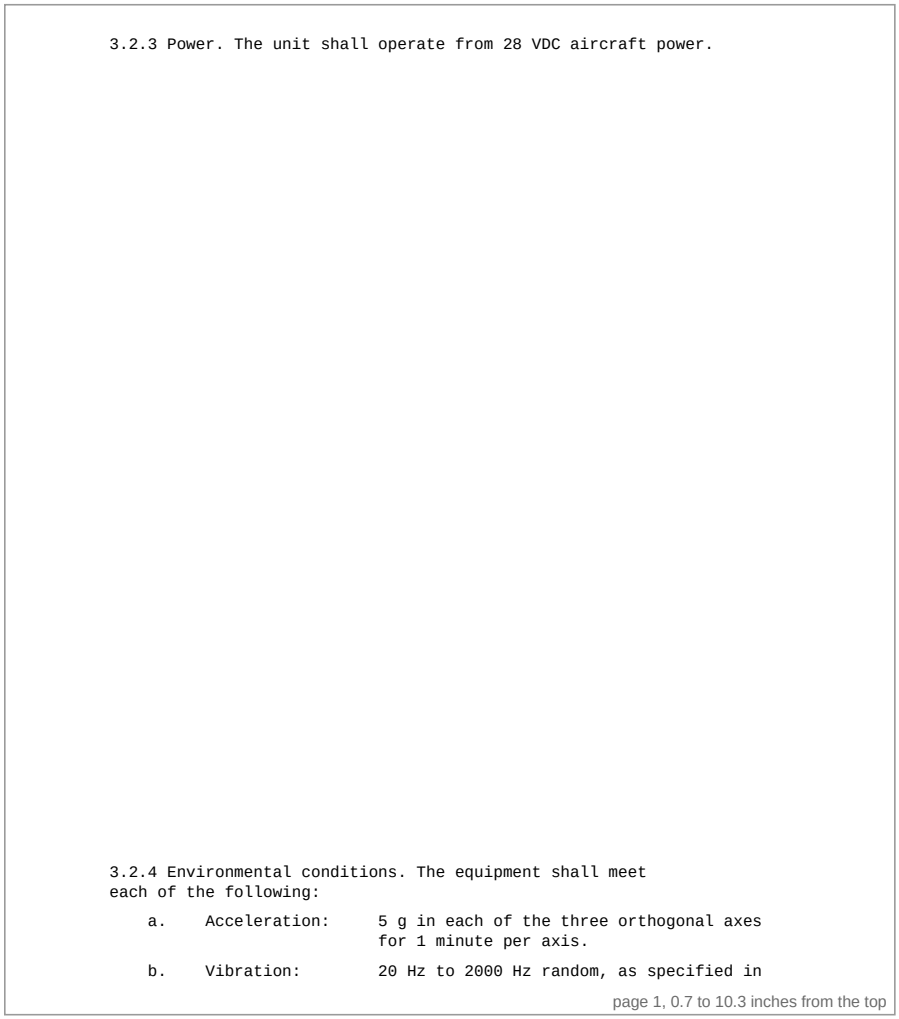
### What the tool must do

The cross-reference carries item b's value on (pages '1-2'): 4.5.3 does not follow 3.2.4 and the value had not ended, so it is not a new section, and c is still item c.

### Source

Review round 2, finding contstem/a12.

### The page (as the scan document has it)

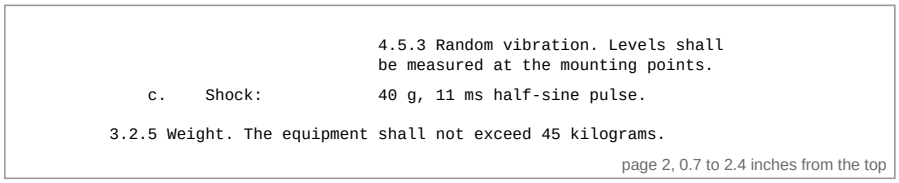


```
3.2.3 Power. The unit shall operate from 28 VDC aircraft power.




3.2.4 Environmental conditions. The equipment shall meet
each of the following:
    a.    Acceleration:     5 g in each of the three orthogonal axes
                            for 1 minute per axis.
    b.    Vibration:        20 Hz to 2000 Hz random, as specified in
```

page 1, 0.7 to 10.3 inches from the top



```
                            4.5.3 Random vibration. Levels shall
                            be measured at the mounting points.
    c.    Shock:            40 g, 11 ms half-sine pulse.

3.2.5 Weight. The equipment shall not exceed 45 kilograms.
```

page 2, 0.7 to 2.4 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.3 | The unit shall operate from 28 VDC aircraft power. | |
| 3.2.4 a | The equipment shall meet each of the following: a. Acceleration: 5 g in each of the three orthogonal axes for 1 minute per axis. | Page(s): 1 |

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.4 b | The equipment shall meet each of the following: b. Vibration: 20 Hz to 2000 Hz random, as specified in 4.5.3 Random vibration. Levels shall be measured at the mounting points. | Page(s): 1-2 |
| 3.2.4 c | The equipment shall meet each of the following: c. Shock: 40 g, 11 ms half-sine pulse. | Page(s): 1-2 |
| 3.2.5 | The equipment shall not exceed 45 kilograms. | |

# PAGE-04  A paragraph that carries on at the top of the next page

Page breaks | document: documents/PAGE-04_a_paragraph_that_carries_on_at_the_top_of_the_next_page.pdf | typewriter 11 pt, 300 dpi, 2 pages

### Situation

Paragraph 3.2.6 ends page 1 in mid-sentence ('... forced air from the'); page 2 starts with the rest ('aircraft environmental control system at an inlet temperature of' / '20 C.'), then 3.2.7.

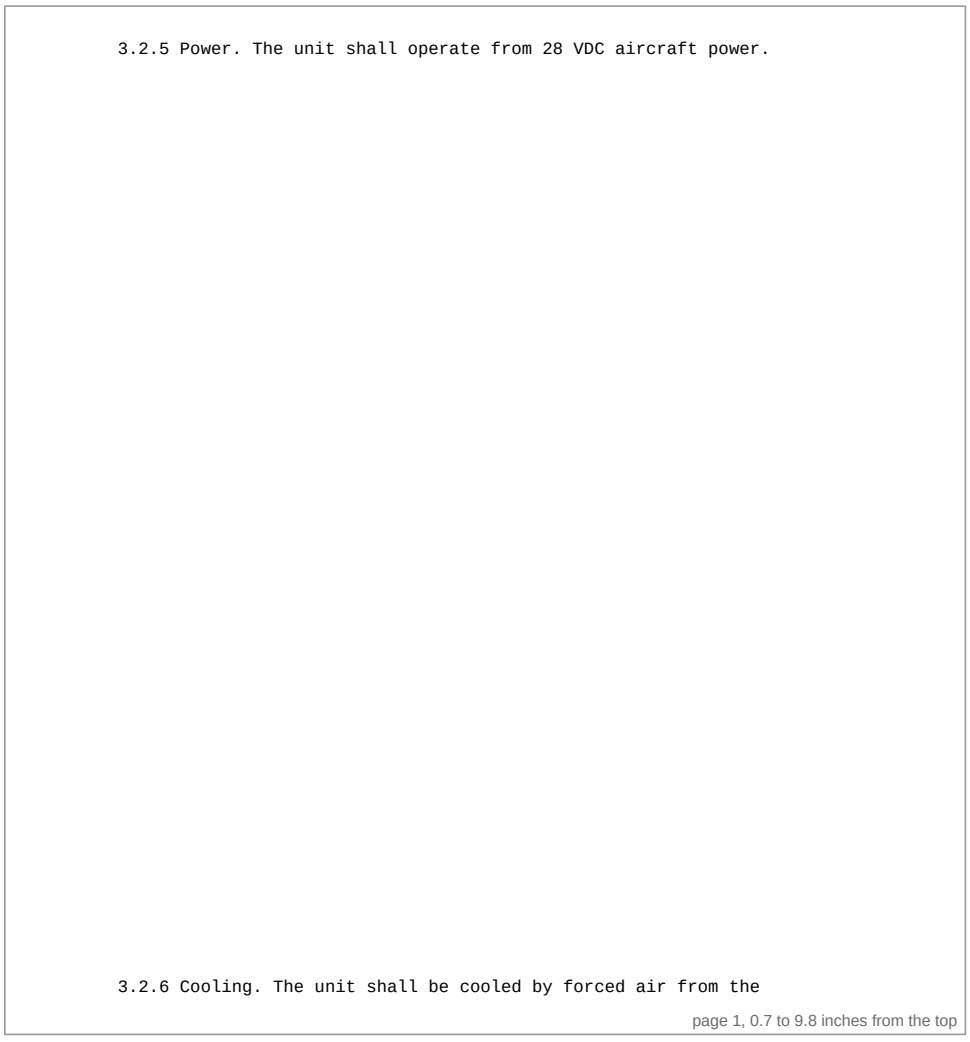
### What the tool must do

One row for 3.2.6 with the whole sentence, pages '1-2', and a review note that the text was joined across the page break.

### Source

Written for the case library (README: text that continues across pages).

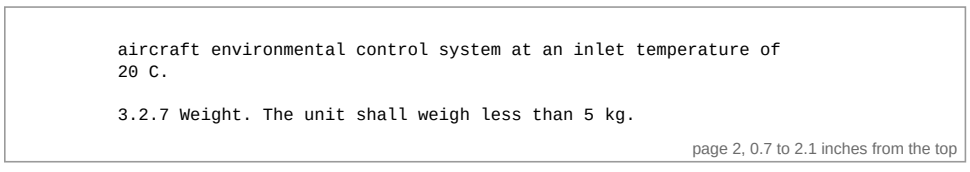
### The page (as the scan document has it)



```
3.2.5 Power. The unit shall operate from 28 VDC aircraft power.

3.2.6 Cooling. The unit shall be cooled by forced air from the
```

page 1, 0.7 to 9.8 inches from the top



```
aircraft environmental control system at an inlet temperature of
20 C.

3.2.7 Weight. The unit shall weigh less than 5 kg.
```

page 2, 0.7 to 2.1 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.5 | The unit shall operate from 28 VDC aircraft power. | |
| 3.2.6 | The unit shall be cooled by forced air from the aircraft environmental control system at an inlet temperature of 20 C. | review note: Text continues across pages; Page(s): 1-2 |
| 3.2.7 | The unit shall weigh less than 5 kg. | |

# PAGE-05  A list that carries on at the top of the next page

Page breaks | document: documents/PAGE-05_a_list_that_carries_on_at_the_top_of_the_next_page.pdf | typewriter 11 pt, 300 dpi, 2 pages

### Situation

The lead-in and items a. and b. at the foot of page 1; items c. and d. at the top of page 2, then 3.2.5.

### What the tool must do

All four items under the same lead-in and paragraph number. The rows of c and d give pages '1-2': each holds the lead-in, which is on page 1.

### Source

Written for the case library.

### The page (as the scan document has it)

```
        3.2.3 Power. The unit shall operate from 28 VDC aircraft power.









        3.2.4 Modes. The radar shall provide the following modes:
             a. Search mode, with a scan of 120 degrees in azimuth.
             b. Track mode, holding up to twenty targets in track.
```

page 1, 0.7 to 9.8 inches from the top

```
             c. Ground mapping mode.
             d. Test mode.

        3.2.5 Weight. The radar shall weigh less than 45 kg.
```

page 2, 0.7 to 2.1 inches from the top

### The rows the tool must produce

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.3 | The unit shall operate from 28 VDC aircraft power. | |
| 3.2.4 a | The radar shall provide the following modes: a. Search mode, with a scan of 120 degrees in azimuth. | Page(s): 1 |
| 3.2.4 b | The radar shall provide the following modes: b. Track mode, holding up to twenty targets in track. | Page(s): 1 |
| 3.2.4 c | The radar shall provide the following modes: c. Ground mapping mode. | Page(s): 1-2 |
| 3.2.4 d | The radar shall provide the following modes: d. Test mode. | Page(s): 1-2 |

| Doc Ref | **Content** (the lead-in in grey) | Also checked |
|---|---|---|
| 3.2.5 | The radar shall weigh less than 45 kg. | |

# Appendix A - the fast rule tests

`python tests/test_rules.py` runs these 236 checks in seconds, without the OCR: each builds OCR lines with realistic page geometry and runs them through the same steps as a scanned page. They test one rule at a time; the cases in this catalog test the whole tool on real scans of whole pages.

| **Rule test** (tests/test_rules.py) - and what it checks, where its description says more |
|---|
| **paragraph continued on next page keeps its order** |
| **page break after complete word does not join paragraphs** |
| **numbered sentence is not a heading** |
| **number at start of wrapped line does not split paragraph** |
| **designator at start of wrapped line stays in text** |
| **unit abbreviation ends a sentence** |
| **real abbreviations do not end a sentence** |
| **stray period before an abbreviation does not cut the requirement** |
| **quantity is not a paragraph number** |
| **a title whose capital was read in lower case keeps its number** - '3.2.7 output. The ...': the scan read the capital 'O' of the title as 'o' (letters such as c, o, s, v, w have the same shape in both cases). |
| **designators are not requirement ids** |
| **bracketed and repeated ids are recognised** |
| **ocr garbled keyword is repaired and flagged** |
| **all caps list items are not headings** |
| **unmarked list under a stem** |
| **list item in footer zone is not page furniture** |
| **repeated header is page furniture** |
| **designator in parentheses at start of wrapped line stays in sentence** |
| **heading without period before a sentence is split** |
| **full width title like line is a wrapped sentence** |
| **keywords in heading rule follow settings** |
| **short text in footer zone is not page furniture** |
| **keyword notice repeated in every footer is reported once** |
| **keyword notice with page number or ocr noise is reported once** |
| **running header with ocr noise is page furniture** |
| **two column page keeps wrapped sentences together** |
| **double spaced paragraph next to single spaced notes** |
| **id led requirement after full line without period** |
| **continuation on next page starting with a number is joined** |
| **numbered caps items under a stem are list items** |
| **last line without final period does not swallow next paragraph** |
| **section headings after a list or an empty section stay headings** |
| **bracketed citation at start of wrapped line stays in sentence** |
| **heading without period on a list heavy page and next to id columns** |
| **heading before a short sentence on a single column page** |
| **different requirements in footer zone are not grouped** |
| **custom keywords protect short requirements** |
| **heading rules still find real headings** |
| **caption row inside table is given back and merges shift** |
| **a caption under the last row is given back not read into it** - 'TABLE I. Interface characteristics.' printed close under a table's last row, where the reader stretched that row down over it: it is the table's caption, not more text of the row's cell (case TBL-14). |
| **a caption close under an aligned table is not a wrapped line of its last row** - A borderless table found from its aligned text, with 'TABLE I. Interface characteristics.' tight under its last row: the caption ends the table - it is not the last row's text wrapping on (case TBL-14). |
| **caption split over cells is given back** |
| **list of tables keeps its rows** |
| **two line caption row is given back whole** |
| **requirement table with requirement id header** |
| **word report handles large tables quickly** |
| **framed figure above a table stays a figure** |
| **framed caption cell above a table is the caption** |
| **requirement table with title and description columns** |
| **merges without row** |

| **Rule test** (tests/test_rules.py) - and what it checks, where its description says more |
|---|
| **captions below tables link to their own table** |
| **captions above tables link to the table below** |
| **caption inside table frame and captions below on one page** |
| **min cost assignment** |
| **sequential ids are not flagged as misreads** |
| **header row may name a keyword** |
| **table cells with suspect text are flagged** |
| **verification column and references** |
| **table with column rules but no row lines is found** - The layout that broke the tool on real specifications: an outer frame and column rules, but no line between the rows. |
| **fully ruled table is unchanged by the row inference** - Inferring the missing row lines must never disturb a table that draws all of them. |
| **row inference does not split a cell that wraps over two lines** - A cell whose text wraps must stay one row: the row pitch, not every line of text, decides. |
| **sparse verification matrix is not rejected as too empty** - A verification cross reference matrix is mostly empty by design - one or two marks per row. |
| **ordinary sparse grid is still rejected** - The exemption above must not open the door to any mostly empty grid (forms, drawings). |
| **verification matrix read from its headers** |
| **verification matrix works when the sideways headers could not be read** - Sideways headers often come back from OCR as nothing at all. |
| **verification matrix without headers or legend is not guessed** - With neither readable headers nor a legend the methods are unknown: say so, never invent. |
| **verification matrix taken from the legend is flagged for review** - A method read from a column order rather than from a header is a deduction, so the row says so. |
| **verification matrix reads a two row header** - Matrices often group the method columns under a merged 'VERIFICATION METHOD' banner. |
| **header rows counts a merged banner above the real headers** |
| **borderless requirement table with long sentences is kept** - An ID column beside a column of whole requirement sentences is a table, not prose. |
| **wordy columns without a requirement header are still prose** - The relaxed limit is earned by the header row alone. |
| **a table whose rows were never ruled is recognised** - Every column of the one data row holds several lines at once: the rows were never ruled. |
| **a sparse matrix whose rows were never ruled is recognised** - A verification matrix is mostly ticks, so only its left-hand column shows the crowding. |
| **a ruled table whose cells wrap is left alone** - The case that must NOT be touched: a properly ruled requirement table whose wide column wraps over two or three lines. |
| **a checklist with tick columns and a deep cell is left alone** - A deeply wrapped cell beside tick columns is still an ordinary table while the table has more rows than any one cell has lines. |
| **rows are recut from the text when the rules do not describe them** - The last resort: cut the rows ourselves, half-way down the white gap between the lines of text, keeping the columns exactly as the printed rules give them. |
| **a header band split by the row inference is put back together** - A sideways header needs a tall band, and the fragments read in it ("DEM" above "ON") can look like two rows of text. |
| **evenly sized rows are never collapsed** - The rejoining above must only ever touch a genuinely tall band, not an ordinary table whose first row happens to have a single filled cell. |
| **a page with little text has its orientation checked by reading it** - The orientation classifier can take a page with a few lines at its top and foot for an upside-down one, and the other way round (case PAGE-05). |
| **unrotate box maps sideways text back to the page** - The inverse of a 90 degree rotation, used to put a sideways header back on its own column. |
| **sideways column headers are read and land on their own columns** - The point of the sideways pass: "INSPECTION" printed vertically above the third column has to become the THIRD column's header, not the second's or the fourth's. |
| **sideways headers are only read where the headers are missing** - A header band that OCR already read normally must be left exactly as it is. |
| **gap threshold splits a tables two spacings** - Wrapped lines sit tight together, records are set apart; the dividing line is measured from the table itself because the leading differs from one document to the next. |
| **evenly spaced lines have no threshold** - A matrix of one-line entries has one spacing only: there is nothing to split, and saying so is what keeps every one of its rows a record of its own. |
| **record breaks follow the key column** - The strongest signal: a value in the left-hand column starts a record, and a line without one continues the record above, however many such lines there are. |
| **record breaks fall back to the leading** - With no key column to go by - every row fills every column - the spacing decides. |
| **record breaks keep every row when nothing distinguishes them** |
| **continuation rows are merged back into their record** - One row per line of text goes in, one row per record comes out, and the merged cell boxes cover the whole record so its text is gathered from all of its lines. |
| **a grid of real rows is left alone by the merge** - A matrix whose entries are one line each must come through untouched. |
| **a candidate that starts inside a record is rejected** - Aligned-text detection can lock on to the lower half of a table, taking its wrapped lines for rows and missing the header and first record above them. |
| **a cell wrapping over two lines does not end a borderless table** - The bug behind a whole table being lost: the run stopped at the first cell that wrapped over two lines, because it demanded that the very next row have cells again. |
| **two borderless tables on one page are both found** - Also a guard on the scan itself: the loop that walks the page must move on after each table. |
| **boxed table with only a header rule gives one row per record** - The layout the supplied pages showed most often: a box and a line under the headers, with nothing between the rows, and requirement cells that wrap over three lines. |
| **a page saved at screen size is enlarged enough to read** - A page exported at screen size is about a third of the resolution the recogniser wants. |

| **Rule test** (tests/test_rules.py) - and what it checks, where its description says more |
|---|
| **readme states the same version as the code** - The version at the top of the README is what a user compares against the line a run prints, so the two must never drift apart. |
| **build id fingerprints the code that is running** - The build id has to follow the FILE, not the version number, so that a folder carrying a half-applied patch or an unreleased edit cannot claim to be a released version. |
| **every ordinary list format yields one row per item** - The formats a specification actually uses, each of which must give one row per item. |
| **list spacing variations all yield one row per item** - The same list set with different spacing: tight, a blank line between items, a wide tab between the marker and its text, and an item indented differently from its siblings. |
| **a misread marker does not end the list** - The heart of it: the last marker comes back as the wrong letter, the wrong case, a digit where letters were used, or glued to its text. |
| **a marker lost altogether does not end the list** - The marker is gone, not merely wrong. |
| **a misread digit does not end a numbered list** |
| **a wrapped item is still one item** - The guard on all of the above: a line that merely continues the sentence of the item above must NOT become an item of its own, however it is indented. |
| **a heading after a list is still a heading** - The counter-case to treating a numbered line level with a list as the next item: a section heading in capitals after a list stays a heading and does not join the list. |
| **a wide tab list that also loses a marker is a known limitation** - Documented, not fixed. |
| **the split is taken at the lowest clear step** - Measured off such a matrix. |
| **the lowest step rule still groups a wrapped requirement table** - The same rule on the gaps of a requirement table whose cells wrap over three lines. |
| **a matrix cell holding a column number and a tick is not a heading** - "1 X" - a column header run together with the tick below it - used to read as the heading "1. |
| **row keys tell a record from a wrapped header** - A legend band of two lines, then a header whose label wraps onto a second line, then the records. |
| **what counts as a row key** - A paragraph number or an identifier names a record; a word, a band of legend text, a tick or an empty cell does not. "2.5" counts, because paragraph 2.5 is a perfectly ordinary key - it reads as an AQL percentage only in the AQL column, ... |
| **a legend matrix with a wrapped header keeps one row per requirement** - End to end on a drawn page: legend band, a header whose label wraps, eleven columns, column rules and no line between the rows. |
| **a lost space after a paragraph number is repaired** - A photocopy loses the space after a paragraph number, or reads it as a speck of dirt. |
| **two numbered paragraphs in one block keep their own numbers** - A faint scan can lose the blank line between two numbered paragraphs, leaving both in one block. |
| **a second numbered paragraph without a title is not lost** - Both paragraphs on ONE line, so they are one block. |
| **a list under the second numbered paragraph of a block carries that number** - A stem in the second of two merged numbered paragraphs: its list items belong to THAT paragraph, so they must not be filed under the first paragraph's number. |
| **a cross reference is not mistaken for a new paragraph** - '... as defined in 3.2.1. Testing shall ...' is one paragraph: the number is a reference in running text, not a heading, so the second sentence keeps the block's own number. |
| **a quantity inside a paragraph is not a new paragraph** - '3.3 V shall' after a full stop is a voltage, not paragraph 3.3. |
| **an ocr misread repeated on every page is still flagged** - The OCR repeats its mistakes, so 'repeated in the document' cannot mean 'a real term' on its own. 'systen' printed beside 'system' is reported; a project term of its own is not. |
| **a project term is not flagged just because english has a near word** - A term of the program's own ('hardpoint') is one edit from nothing the document uses, so there is no evidence of a misread and it is left alone however often it appears. |
| **a list on wide tab stops keeps each label with its value** - The wide tab between 'Acceleration:' and its value is not the gutter between two columns: each row carries its marker, label and the whole multi-line value. |
| **numbered items on tab stops are items not headings** - '2. Vibration: 20 Hz ...' looks like a numbered heading ('2. SCOPE'), but a heading never holds a label and a value. |
| **a lead in without a colon still leads into its list** - '... each of the following.' / 'within the following limits.' introduce a list as surely as a colon does; without one the items used to be attached to nothing. |
| **a lead in phrase with no list below is an ordinary requirement** - Without a colon, 'the following' followed by no list is just a sentence: no 'no list items' warning (a colon promised a list; the phrase alone did not). |
| **value lines that start with numbers stay in their item** - A value line may end with a full stop and the next may start with a number - '2.5 G peak', '20 Hz', '1.5 V' - which used to be taken for a new section or a heading. |
| **a misread list marker is taken from the sequence and reported** - 'c.' read as 'C.' - the letters count a, b, ?, d, so it is c. |
| **a lost list marker is taken from the sequence** - The photocopy lost 'a.' and 'c.' altogether: each of those rows starts with its label, in the label column. |
| **a list that really skips a letter keeps its letters** - a, b, d: a valid marker is only replaced when the NEXT item confirms the expected one, so a list that genuinely skips c is left as printed. |
| **section numbers must follow in sequence** - Paragraph numbers count up: after 3.2.4 come 3.2.4.1, 3.2.5, 3.3 or 4 (a few may be skipped, since the scan can miss a heading). '1.5' or '20' after 3.2.4 is a value, not a section. |
| **a value inside a paragraph is not a new paragraph** - '... per axis. 2.5 G peak ...' inside paragraph 3.2.4: 2.5 does not follow 3.2.4, so it is a value and the paragraph is not split there (3.2.5 would have been). |
| **the next paragraph number starts a new paragraph even without a full stop** - A speck turned '3.5 INTERFACE CONDITIONS' into '3.5 INTERFACE. CONDITIONS', which no longer looks like a heading; '3.5.1' on the next line is the number that should come next, so it still starts its own paragraph and its requirements carry ... |
| **a list on tab stops is not a borderless table** - Items on tab stops line up exactly like a table without ruling lines. |
| **a label that wraps is read before its value** - 'Storage' / 'temperature:' wraps in the label column beside a two-line value: the row reads 'Storage temperature: -55 C to +85 C for up to 5 years', not the two columns interleaved. |
| **a long value line is not taken for a page wide line** - A value line can be long, but it starts far to the right; only a line that starts near the left edge (a running header, a title, a full line of a paragraph) divides the page before the column check. |

| **Rule test** (tests/test_rules.py) - and what it checks, where its description says more |
|---|
| **a short list in one column of a two column page is not read across the page** - Left column: a lead-in and short items 'a. MIL-STD-461G'; right column: numbered prose on the same line grid. |
| **a list set in two columns gives one row per item** - a-c on the left, d-f on the right: a list set in columns, read column by column. |
| **paragraphs after a list are not pulled into its last item** - A blank line after a finished item ends the item, even when the next paragraph is indented to where the item's text starts; a numbered sub-paragraph after the list keeps its own number. |
| **a value that wraps back under its label keeps its order** - Only a LABEL that wraps beside its value is read label-first. |
| **a real table with label like first cells stays a table** - 'Input voltage: \| The unit shall ... \| Test' is a table - its first cells end with ':' but a caption (above or below it), a header row or a real third column (Method: 'Test', 'Inspection') says so, even under a lead-in. |
| **notes are not list items** - 'NOTE: text' after a list, 'Note.' level with the items and a 'Note:' line inside an item are not items of the list: none of them gets a letter, and no letter is used twice. |
| **a note inside an item stays in that item** |
| **the following day is not a lead in and a mid paragraph phrase splits nothing** - '... retested on the following day.' and '... as described below.' inside ordinary paragraphs: every sentence stays a whole requirement of its own. |
| **a wrapped cross reference does not split a paragraph** - '... established in paragraph' / '3.2.5. The weight shall ...' - the line above ran to the margin, so the number is a cross-reference in the sentence, not the start of paragraph 3.2.5. |
| **a titled paragraph ends a list whatever its number** - '3.2.8 Weight. The ...' after a list: a skip, or 3.2.5 misread as 3.2.8 - a paragraph with a run-in title is a paragraph, not more of the list. |
| **a titled inner paragraph is split whatever its number** |
| **a value carried onto the next page stays with its item** - Item d's value goes on at the top of page 2 with '2.5 C per minute ...' - not a heading, not a new section; and the items after it are still found. |
| **an item that opens a sub list keeps its sub items** - 'a. Search mode, which shall include:' / 'a. Vibration: as follows:' - the lines below are the item's sub-items (one row each, under the item), not more of its text and not new letters. |
| **requirement ids under a lead in keep their ids** - 'SRS-101: The software shall ...' on tab stops under a lead-in: |
| **several lost markers in a row are counted forward** |
| **a note between items keeps its own row and the letters go on after it** - a, b, NOTE, C. (misread), d: the NOTE is neither numbered nor glued to b, and 'C.' is still read as c - the letter is expected from the last item that has one. |
| **a misread next letter starts the next item without a full stop** - Items without final punctuation ('b. MIL-STD-704F') never end a sentence, so a misread 'C.' level with them used to be folded into the item above. |
| **a blank line after a finished item ends it** - 'c. Ground mapping mode.', a blank line, then an indented sentence: a paragraph of its own, not more of item c - even on a page whose paragraph spacing is wide. |
| **a note heading or long label near a tab list does not cut it into columns** - Only the rows of the list itself are judged: a NOTE or a heading under the list, or the lead-in's last line above it, sits in the same band but is not part of it. |
| **values in lower case without a full stop stay with their labels** - 'within 22 to 29 VDC' / 'less than 1 V peak to peak,' - values that start in lower case and do not end a sentence are not prose running from line to line: each line breaks where there was still room. |
| **a quantity at the start of a value is not a titled paragraph** - '5.0 V. Ripple shall ...', '1.5 V peak to peak. Measured ...', '2.5 G rms overall. Duration ...' start with a quantity and its unit; '3.1 Power. The unit ...' is a numbered paragraph with a run-in title. |
| **a sub list in the value column stays under its own item** - 'a. Search mode: (1) Range ...' / '(2) Azimuth ...' on the value tab stop: a sub-list that starts again under each item, not a second column of the list. |
| **a sub list printed without its own indent is nested by its markers** - 'b. Vibration:' then '(1) Random: ...', '(2) Sine: ...' at the same indent, then 'c. Shock:': the marker style changes after a line ending ':' and comes back, so (1) and (2) belong to b. |
| **a label that wraps is read before its value even when the value is shorter** - The label's last line may have no value beside it ('temperature:' under a one-line value), have 4 or more words, or follow a first line that ends in an abbreviation ('Max. oper.'). |
| **a table under a lead in with a method column or a caption below stays a table** - 'Voltage: \| 28 VDC nominal \| Test' under '... the following characteristics:' is a table: its third column is short and repeated ('Test', 'Analysis'). |
| **a label list with one colon lost stays a list with tables on** - Labels on tab stops without markers, one of them without its colon ('Power 150 W maximum.'): with table detection on, still one list row per label under the lead-in. |
| **a numbered list is not taken for section headings by the paragraph after it** - '1. Search mode.' ... '3. Ground mapping mode.' followed by '3.2.5 Weight. ...' (or a section-2 document list followed by '2.2 ...') are list items. |
| **a cross reference with a title carries a value onto the next page** - 'b. Vibration: 20 Hz to 2000 Hz random, as specified in' at the foot of page 1, then '4.5.3 Random vibration. Levels ...' at the top of page 2 in the value column: the number does not follow the section numbering and the value had not ... |
| **a blank line under a long item still ends it** - 'Right under an item' is measured with the item's own line spacing, so a blank line under an item of three lines ends it too: the paragraph after it, or the next item whose marker was lost, is not glued on. |
| **an unmarked block between items is joined only when it is a piece of text** - An unmarked block whose would-be letter a later item carries is not that item. |
| **labels under a bare label item are its sub items** - 'a. Operating:' with nothing after it, then 'Temperature: ...' and 'Humidity: ...' in its label column: they belong to a (no invented letters). |
| **a lead in without a colon takes a run of short plain items** - '... shall show the following information.' followed by short indented lines without markers - a list whose lead-in lost its colon. |
| **lost markers are restored from the neighbours** - 'Track mode.' between 'a.' and 'c.' - in the markers' column or in the label column - is item b. Two lost markers at the start of a list are counted back from the first letter the scan kept, even when that letter was read as a capital ... |
| **an item after a figure caption inside a list is kept when its marker was lost** - a., b., a caption, then 'Shock: 40 g ...' with its marker lost and 'd. Temperature:': the caption is exported and the list goes on after it. |
| **an item that lost its marker in a single spaced list is split off by the sequence** - 'a. Search mode.' / 'Track mode.' / 'c. Ground mapping mode.' single-spaced: 'Track mode.' starts where a wrapped line of a would, but the markers skip b, so a is split where the new sentence starts in its text column, and the piece gets 'b.' from the sequence (reported). |

| **Rule test** (tests/test_rules.py) - and what it checks, where its description says more |
|---|
| **a label list stays a list when a caption or table sentence follows it** - Labels on tab stops under a lead-in, then a blank line and the NEXT table's caption, or a sentence that begins 'Table 4 ...': still one list row per label, not a table. |
| **a short list beside a column of prose is not read across** - Two columns: a short list on the left, prose on the right - narrow (27 characters), ragged, with one numbered paragraph, or a second list of its own. |
| **a lettered paragraph with its first line indented is read whole** - 'A. General: The contractor shall provide all labor, materials' with its next line back at the margin: the sentence wraps, it does not end the item. |
| **an item broken early with its next line back at the margin is read whole** - 'b. Track mode, holding up to twenty targets in' / 'track at the same time.' back at the margin, although the next word would have fitted on the line above (a typist's early break): a line that ends in the middle of a phrase ('in', 'at', ... |
| **a lost marker is split off before a marker read as a capital** - 'a. Search mode.' / ' Track mode.' / 'C. Ground mapping mode.' single-spaced: the scan lost 'b.' and read 'c.' as 'C.'. |
| **a requirement after a blank line does not hang on an item without a full stop** - 'b. Track mode' (no full stop), a blank line, then 'The operator shall select ...' in the item's text column: a requirement of its own, not more of item b. |
| **a page of three lines separated by blank lines is not one block** - The top of a page that carries a list on: a value line, a blank line, 'C. Shock: 40 g ...' (the marker read as a capital), a blank line, '3.2.5 weight. The ...' (the title's capital read as a small letter). |
| **a list row whose marker was read as a capital starts a new block** - The top of a page that carries a tab list on: its first line is the value of the item above, then 'C. Shock: 40 g ...' - a marker and a label, although the scan read 'c.' as 'C.' - then 3.2.5. |
| **a label whose capital was read as a small letter still starts an item** - Labels on tab stops under a lead-in: 'Voltage:', then 'current:' (the scan read 'C' as 'c'), 'Weight:', 'Cooling:'. |
| **a value that wraps back under its label is not read as the label** - 'a. Vibration 20 Hz to 2000 Hz random, applied in' / 'each of three axes as follows:' under the label, then (1) and (2): the value stays in order and the sub-items stay under a. |
| **capital letters are judged per list level** - 'A. Operating conditions:' with 'a. Temperature ...', 'b. Humidity ...' below it: the capitals are the list's own letters, not misreads - a sub-list in small letters does not change that. |
| **a numbered heading with a full stop still opens its section** - '4. Verification.' followed by '4.1 General. ...' is a heading even after a list; '5. Packaging.' / '6. Notes.' / '6.1 Intended use. ...' are empty sections, not a list. |
| **a requirement after a colon less list keeps its own row** - '... the following information.' with short plain items, then 'The display shall be readable in direct sunlight.' at the same indent: a requirement of its own, not one more item. |
| **a list that counts in twos or repeats a misread letter is not split** - Tests numbered 1., 3., 5. skip on purpose: no item is invented and nothing is flagged. 'a.', 'c.', 'c.', 'd.' is a misread 'b.' - corrected, not split. |
| **the next numbered item after a split off sentence stays in the list** - '2. Track mode, ...' whose second sentence starts on its own line, then '3. Ground mapping mode.': item 3 is the next item (its number follows 2), not a section heading that would swallow the rest of the page. |
| **short values on a real scan stay with their labels** - Line boxes measured on two rendered, OCR'd pages (typewriter and serif). |
| **a grid of wrapped labels beside list markers is a list** - 'Storage' / 'temperature:' in the first column of a borderless grid is one label wrapped over two rows; with the markers a., b., c. standing left of the grid it is a list on tab stops, not a table. |
| **page selection** |
| **keyword option accepts semicolons** |
| **output names never collide** |
| **cache key changes with keywords and layout code** |
| **image file that cannot be written is not fatal** |
| **unreadable image page does not stop the file** |
| **the case library is well formed and its catalog is up to date** - tests/cases: every case has its own ID shaped like 'TAB-01', a page, and the rows (or tables) the tool must produce; and CASEBOOK.pdf was written from the current cases and rule tests. |
| **the word report has the tools own look without a style file or with the shipped one** - Without a style file the report looks as it always did (Calibri 10, tables 8.5 pt with a shaded heading row, 8 pt blue tags, Letter paper with 0.7/0.6 in margins); the shipped style file changes nothing. |
| **the report text carries no formatting of its own so a template decides its look** - Only the review marks (highlight, colour, bold of a likely misread) and a few labels are put on the words; sizes and fonts come from the styles, so a template or Word itself can change them. |
| **the shipped style file lists every part with its settings and the tools own values** - report_style.ini names every part of the report, every setting in it is valid, every look value is empty (so a template decides), the switches hold the tool's own values, and the (bracketed) sizes match the code. |
| **style file values change the styles not the text** - A size, colour, spacing or font from the style file goes into the style definition (so Word's Styles pane stays truthful), and a font replaces Word's theme fonts, which would otherwise win over it. |
| **a template gives the report its styles page headers and footers** - With a template the report has the template's fonts and sizes, page, header and footer (with its picture and page number), and the styles the template does not have are added to it; the template's text and its author never reach the report. |
| **a style file value wins over the template** |
| **nothing in a template that points outside the computer reaches the report** - The attached template on a network share, the intranet link in the header and a field that fetches a picture from a web address (its name split over runs, with a page-number field inside it) are all left out: the link keeps its text, the ... |
| **a template that could reach outside or run code is refused with what to do** |
| **a dotx template gives a docx report that word opens** |
| **template heading numbering is switched off in the report** - A company heading style that numbers itself ("1.1") would print "1.1 3.2 Electrical requirements": the report's copy of the style has its numbering switched off. |
| **page headings start a new page without blank paragraphs** |
| **page images fit inside the page a template sets** - The page image is sized from the report's real page (here A4 landscape with 2 cm margins and a header with a logo), not from US Letter. |

| **Rule test** (tests/test_rules.py) - and what it checks, where its description says more |
|---|
| **requirement tags can go in front of the text** |
| **requirement keywords can be marked and keep their review highlight** - mark_keywords = yes puts 'shall' in the Requirement Keyword style; a 'shall' the OCR was unsure of keeps its review highlight as well. |
| **the requirement style is used for requirement paragraphs only** |
| **table heading rows repeat on each page and cells to check stay shaded** |
| **style file mistakes are all reported with their line and what to write** |
| **a style file saved by notepad is read** - Notepad's byte-order mark and Windows line ends, an 'ANSI' (cp1252) file, a quoted file name, a % sign, a decimal comma, a section name in capitals and a setting written with a space are all read correctly. |
| **style file layout mistakes are explained** |
| **a style that does not exist or is of the wrong kind is reported before any page is read** |
| **the template is looked for next to the style file** |
| **a mistake in the style file stops the run before any page is read** - The run stops before any document is read, and the message is in the log file as well as on the screen. |
| **the starter template has every report style and works as a template** - make_word_template.bat writes a .docx holding every style with a sample of each; it never overwrites a file, and used as the template it gives reports the same look. |
| **the starter template starts from the template in the style file** |
| **the report and the run info sheet name the look that was used** |
| **a report that would point outside the computer is never saved** - The last check before saving: an outside link, a field that fetches something, or a linked picture anywhere in the file stops the save, whatever put it there. |
| **a field code outside any field never reaches the report** - A field instruction with no field around it ('INCLUDEPICTURE http://...' in a plain run) does nothing in Word, but it is removed all the same, and the last check before saving would catch one. |
| **tracked changes in a template are accepted not copied** - A marking changed from SECRET to UNCLASSIFIED with Track Changes on, and never accepted, must not put the struck-out SECRET (and the reviser's name) into every report; a page layout changed the same way gives the report the NEW layout, not ... |
| **styles added to a template from a non english word follow its own normal style** - German Word stores Normal under the id 'Standard'. |
| **tags and keywords keep their look inside italic or bold paragraphs** - Bold and italic are 'toggles' in Word: a bold tag style inside a bold paragraph would print NOT bold, and the tag's 'not italic' could not undo an italic caption. |
| **an empty table alignment leaves it to the template and tables fit beside a gutter** |
| **margins too wide for the paper stop the run at start up** |
| **the look checked at start up is used for the whole run** - Editing the style file while a batch runs must not make the remaining reports fail: main() keeps the look it checked at start-up in Settings.word_look. |
| **page images leave room for the heading and tall header pictures** - The page image leaves room for the 'Page N' heading and a line of notes (as before), and a tall logo in the header - DrawingML or the older VML, in the default or the even-page header - takes room from it. |
| **the starter template keeps the heading row shading when used as the template** |
| **a damaged template gives a message not a crash** |
| **a template header with a table or a text box is not taken for a tall one** - A header laid out as a table (logo \| title \| marking) or holding a floating text box takes the room of its rows, not of every paragraph in it; the start-up check judges the margins only, and a damaged picture size in a header is ignored ... |
| **damaged or hostile bits of a template are cleaned without breaking the rest** - A stray field code sharing a run with a page number's field mark goes and the page number stays; a deleted field code (delInstrText) left outside its deletion goes; a deleted paragraph and a deleted table row go, as with Word's Accept All ... |
| **a template whose normal style is not marked as the default still has one normal** |
| **tables are centred by the table style and the starter keeps it with its shading** - The tool's own centring goes into the table style (Table Grid), like its heading-row shading in the starter template, so a template that lacks Table Grid gets centred tables and a report made from the unchanged starter looks like the ... |
| **word on off settings written as off are off** |
| **a dark highlight for misread words gets white text and a damaged style file a message** |
| **python version check** |
| **constraints file pins every package for every tested python** |
| **configuration report names every difference** |

# Appendix B - the regression documents

`python tests/run_regression.py` builds these synthetic specifications with `tests/make_test_corpus.py`, ages them into office scans, photocopies and faxes, runs the whole tool and scores it against exact ground truth (README, section 6): specA_source_digital.pdf, specA_clean.pdf, specA_scan.pdf, specA_photocopy.tif, specB_scan.pdf, specB_photocopy.tif, specC_fax.tif, specD_source_digital.pdf, specD_scan.pdf, specE_source_digital.pdf, specE_scan.pdf, specE_photocopy.tif, specG_pages.tif, specH_source_digital.pdf, specH_scan.pdf, specH_photocopy.tif.